Exploration of Lunar Lava Pits for Human Habitation and In-Situ Resource Utilization

Preliminary Design Review
Mission Concept Academy
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Team 3
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Table of Acronyms

Acronym	Description
ADV	Advisory
CDH	Command and Data Handling
CDR	Critical Design Review
CMB(R)	Cosmic Microwave Background (Radiation)
CTE	Critical Technology Elements
ESDMD	(NASA) Exploration Systems Development Mission Directorate
GPR	Ground Penetrating Radar
GRC	Glenn Research Center
ISRU	In-situ Resources
JPL	Jet Propulsion Laboratory
LIBS	Laser-Induced Breakdown Spectroscopy
LID	Light Induced Degradation
LiDAR	Light Detection And Ranging
LEAG	Lunar Exploration Analysis Group
LPNS	Lunar Prospector Neutron Spectrometer
LRO	(NASA) Lunar Reconnaissance Orbiter
MCR	Mission Concept Review
MDR	Mission Definition Review
MRR	Mission Readiness Review
NASA	National Aeronautics Space Administration
NIR	Near Infrared
NIRS	Near-Infrared Spectrometer
ORR	Operational Readiness Review
PCDU	Power Control and Distribution Unit

PDR	Preliminary Design Review
PIR	Program Implementation Reviews
PIXL	Planetary Instrument for X-ray Lithochemistry
RFA	Request for Action
RTD(s)	Resistance Temperature Detector(s)
SFCW	Stepped Frequency Continuous Wave
SIR	System Integration Review
SMD	(NASA) Science Mission Directorate
SQRLi	Space Qualified Rover Lidar
SRR	System Requirements Review
STM	Science Traceability Matrix
TBD	To Be Determined
TBR	To Be Reviewed
TGA	Thermogravimetric Analysis
TIR	Thermal Infrared
TRL	Technology Readiness Level
XRF	X-ray fluorescence

1. Mission Overview

1.1. Mission Statement

The mission aims to explore and evaluate lunar lava pits and caves to determine their potential for long-term human habitation and in-situ resource utilization (ISRU). These geological features, believed to offer natural protection from extreme lunar conditions such as radiation, micrometeorite impacts, and temperature fluctuations, are promising candidates for sustainable lunar base development. The mission seeks to analyze their structural stability, mineral compositions, environmental conditions, and resource availability (e.g., water ice and volatiles) to assess their suitability for supporting future human exploration and settlement.

To achieve these objectives, the mission will deploy a dual-platform system consisting of a stationary lander and a mobile rover. The lander will conduct surface-level assessments, including monitoring radiation levels, temperature fluctuations, and dust dynamics, and will act as a communication hub, relaying data from the rover to Earth. Meanwhile, the rover will perform in-depth exploration of the caves and surrounding areas, using advanced scientific instruments such as ground-penetrating radar to assess subsurface structures, laser-induced breakdown spectroscopy to analyze regolith composition, and thermal sensors to study environmental conditions. The rover will generate high-resolution 3D models of cave interiors and their geological features, offering insights into their geometry, stability, and potential for habitation.

The mission is guided by several key science objectives. These include assessing the structural integrity of lunar caves to determine their habitability, identifying in-situ resources like water ice and hydrogen to support ISRU strategies, and analyzing regolith and rock composition to evaluate the feasibility of extracting oxygen and other materials for construction. Additionally, the mission will monitor seismic activity to understand geological risks, evaluate environmental hazards to human habitation, and map the thermal and radiation environment of the lunar surface and subsurface.

This mission directly aligns with NASA’s broader goals of establishing a sustainable human presence on the Moon. By delivering comprehensive data on the geology, resources, and environmental conditions of lunar lava pits and caves, the mission will inform the design and development of future lunar habitats. Furthermore, the insights gained will contribute to advancing humanity’s understanding of planetary processes, resource utilization, and long-term habitation beyond Earth, serving as a stepping stone for future exploration of the solar system.

1.2. Science Traceability Matrix

The STM (Table 1) outlines major science objectives and detailed measurement requirements for both SMD and ESDMD goals for our mission. It focuses on detecting in-situ resources (ISRU) like water ice and hydrogen within and around the pit, using advanced instruments designed to handle the harsh environment. By setting clear performance goals and technical needs, the STM ensures the mission will produce valuable scientific data.

The primary science objectives of the Science Mission Directorate (SMD) include characterizing the distribution and composition of water ice, investigating volcanic and tectonic activities, analyzing lunar soil composition for potential resources, and assessing lunar dust properties and dynamics. These goals aim to enhance our understanding of the lunar pit's surface and its potential to support future exploration and resource utilization. The Exploration Systems Development Mission Directorate (ESDMD) objectives focus on evaluating the structural integrity of the lunar pit to assess its suitability as a habitat, inspecting in-situ resources (ISRU) within and nearby the lunar pit for sustainable habitation, and analyzing environmental risks and dangers to potential habitability within the lunar pit cave. These efforts align with the overarching goal of enabling sustainable human presence in the lunar pit.

The performance of the instruments is key to meeting these goals. Requirements include a wavelength range of 1-3 μm and an integration time of one second to gather accurate data quickly. Sensitivity is also important, as the instruments need to detect very small traces of elements in the lunar regolith. Additionally, we need instruments to work even under poor lighting and not take much power along with it being lightweight. Therefore, after doing a Trade study, instruments like the Lunar Prospector Neutron Spectrometer (LPNS), WISDOM, SQRLi, etc. are chosen to meet these demands and survive tough conditions like lunar dust and extreme temperatures. These features ensure the mission can succeed while delivering the required scientific results.

Goal	Science Objectives	Science Measurement Requirements		Instrument Performance Requirements		Predicted Instrument Performance	Instrument	Mission Requirements
		Physical Parameters	Observables					
SMD: Develop precursor lunar robotic missions and define those scientific activities that astronauts will conduct on the Moon - (*Origins, Worlds, and Life: A Decadal Strategy for Planetary Science and Astrobiology 2023-2032)*	Characterize the distribution and composition of water ice	Identify water ice content (H2O) and hydrogen concentration at surface and subsurface levels (up to 1 meter)	Measure neutron emissions and hydrogen abundance using neutron spectrometry; detect water ice using near-infrared (NIR) reflectance spectra in the 1-3 μm range	Wavelength range:	1-3 μm	1-3 μm	Lunar Prospector Neutron Spectrometer (LPNS) and Near-Infrared Spectrometer (NIRS)	Instruments should be resistant to lunar dust and operate reliably in extreme temperatures, as well as endure the high-impact vibrations from micrometeorite strikes.
				Integration time	1 second	1 second		
				Sensitivity:	Detect hydrogen down to 10^(-3) by weight; water ice content sensitivity around 1% by weight	Detect hydrogen down to 10^(-3) by weight; water ice content sensitivity around 1% by weight		
				Spatial resolution:	~10-30 meters for surface NIR, ~20 km for neutron spectrometry subsurface	~10-30 meters for surface NIR, ~20 km for neutron spectrometry subsurface depth resolution		

					depth resolution			
		Detect the physical state of water (solid, vapor) and volatiles in permanently shadowed regions (PSRs)	Use passive microwave radiometry to determine the phase of water; analyze volatile sublimation signatures using thermal infrared (TIR) sensors in the 10-15 μm range	Wavelength range:	10-15 μm	10-15 μm	Passive Microwave Radiometer and Thermal Infrared Spectrometer.	
				Integration time	0.5 - 1 second	0.5 - 1 second		
				Sensitivity:	Phase detection sensitivity of 0.1 K; volatile concentration detection down to ~1 ppm	Phase detection sensitivity of 0.1 K; volatile concentration detection down to ~1 ppm		
				Spatial resolution:	100 meters in PSRs	100 meters in PSRs		
	Investigate the volcanic and tectonic activities	Record ground vibrations and detect moonquakes to assess tectonic	Detect ground motion with seismometers capable of measuring displacements	Wavelength range:	1 - 2 Hz	1.5 Hz	CP ZM500 vertical seismometer	
				Integration time	1 second	1 second		

		activity and subsurface structure	as small as 10^-9 m/s² over the frequency range of 0.01 to 50 Hz.	Sensitivity:	9000 V/(m.s -1)	9000 V/(m.s -1)		Install imaging and tracking sensors on the lander, positioned to monitor the maximum ground area, ideally with a wide field of view.
				Spatial resolution:	Placement in areas with suspected tectonic activity for maximum coverage and accuracy	Placement in areas with suspected tectonic activity for maximum coverage and accuracy		
	Analyze lunar soil composition for potential resources	Assess the concentration of metal ores and other extractable resources in lunar soil.	Use Laser-Induced Breakdown Spectroscopy (LIBS) in the 200-900 nm range and X-ray fluorescence (XRF) to detect key elements like silicon, iron, and titanium in surface rock samples.	Wavelength range:	200-900 nm	200-900 nm	ChemCam (LIBS) and Planetary Instrument for X-ray Lithochemistry (PIXL - XRF Analyzer)	High-speed imaging and electrostatic measurements should be stored locally with the capability for real-time or batch transmission to Earth for analysis.
				Integration time	1-10 ms	1-10 ms		
				Sensitivity:	10^(-4) - 10^(-3) w/w	10^(-4) - 10^(-3) w/w		
				Spatial resolution:	≤ 1 mm	≤ 1 mm		

		Evaluate the concentration of volatiles, such as hydrogen, water, and helium-3, in the lunar soil.	Detect and quantify volatiles using mass spectrometry and neutron spectrometry to assess hydrogen and helium-3 abundance.	Wavelength range:	Not applicable	Not applicable	Mass Spectrometer and/or Neutron Spectrometer	
				Integration time	1 seconds	1 seconds		
				Sensitivity:	10^(−6) w/w [10^(−4) w/w for Hydrogen]	10^(−6) w/w [10^(−4) w/w for Hydrogen]		
				Spatial resolution:	100 meters	100 meters		
		Analyze particle size distribution and mechanical properties of the lunar regolith to assess feasibility for excavation and resource extraction.	Measure grain size using laser diffraction and conduct mechanical tests (shear, penetration resistance) to evaluate regolith strength.	Wavelength range:	600-900 nm	600-900 nm	Laser diffraction particle size analyzers	
				Integration time	5 seconds	5 seconds		
				Sensitivity:	0.1 µm	0.1 µm		
				Spatial resolution:	~10-50 cm per sample site	~10-50 cm per sample site		
		Assess the mechanical	Analyze regolith for	Wavelength range:	200-900 nm	200-900 nm	LIBS	

		and chemical properties of lunar soil for potential in-situ resource utilization (ISRU), including construction and oxygen extraction.	oxygen-bearing minerals (e.g., oxides, silicates) using Thermogravimetric Analysis (TGA) and elemental spectroscopy.	Integration time	1 second	1 second		
				Sensitivity:	10^(-4) w/w	10^(-4) w/w		
				Spatial resolution:	~10-50 cm between sample sites	~10-50 cm between sample sites		
	Assess lunar dust properties and dynamics on the lunar surface	Examine blast ejecta behavior during lander descent and landing.	Use high-speed imaging (frame rate ≥1000 fps) and particle tracking sensors to observe the dynamics of dust ejecta caused by lander engine blasts; measure ejecta velocity and dispersion patterns to assess the impact of	Wavelength range:	400-900 nm	400-900 nm	Particle Tracking Sensor	Sensors and Testers should be mounted on a stable surface near the rover, ideally in areas that causes less noise in data.
				Integration time	≤1 ms per frame	≤1 ms per frame		
				Sensitivity:	Detects particles down to 50 µm	Detects particles down to 50 µm		
				Spatial resolution:	1-5 mm per particle tracking point	1-5 mm per particle tracking point		

			lander-induced dust movement.					
		Analyze the mechanical properties of lunar dust to assess its adhesion to surfaces.	Measure dust adhesion forces using a surface adhesion tester with a resolution of ±0.01 N/m² to evaluate the bonding strength between dust particles and different surface materials.	Wavelength range:	Not applicable	Not applicable	Surface Adhesion Tester	
				Integration time	1-10 seconds per test	1-10 seconds per test		
				Sensitivity:	Resolution of ±0.01 N/m²	Resolution of ±0.01 N/m²		
				Spatial resolution:	~1 mm² measurement area	~1 mm² measurement area		
		Assess the electrostatic charge on dust particles and its effect on dust lofting.	Measure dust particle charge using electrostatic sensors with a sensitivity of ±1 nC; monitor dust	Wavelength range:	Not applicable	Not applicable	Electrostatic Sensors	
				Integration time	5 seconds	5 seconds		
				Sensitivity:	±1 nC	±1 nC		

			lofting events through high-speed optical imaging during lunar day/night transitions.	Spatial resolution:	Measures particles within 1-5 mm	Measures particles within 1-5 mm		
		Monitor dust dynamics in response to solar wind and micrometeorite impacts.	Use plasma sensors to detect solar wind interactions (particle flux and energy range 10 eV to 10 keV) and accelerometers to detect dust motion caused by micrometeorite impacts.	Energy range:	10 eV - 10 keV	10 eV - 10 keV	Plasma Sensors	
				Integration time	10 seconds	10 seconds		
				Sensitivity:	Detects down to 10^(4) particles/cm^(2)/s	Detects down to 10^(4) particles/cm^(2)/s		
				Spatial resolution:	10 cm around the sensor area	10 cm around the sensor area		
ESDMD: Provide safe and enduring habitation systems to protect individuals,	Evaluate the structural integrity of the lunar pit to assess its	Assess the geological structure, shape, and potential structural	Use ground-penetrating radar (GPR) to map subsurface layers, and perform	Wavelength range:	GPR: 30 MHz to 3 GHz; LiDAR: 10-30 cm horizontal resolution	GPR: 30 MHz to 3 GHz; LiDAR: 10-30 cm horizontal resolution	WISDOM for Stepped Frequency Continuous Wave (SFCW) Ground-Penetrating Radar (GPR) & Space	All instruments should be designed to withstand extreme lunar temperatures, from below -

equipment, and associated infrastructure - *(Lunar Exploration Analysis Group (LEAG) Habitation: HAB-SAT Report, Priority Objectives)*	suitability as a habitat.	weaknesses of the pit cave.	LiDAR scans to create 3D models of the cave geometry.	Integration time	GPR: Milliseconds per scan; LiDAR: 100,000-1,000,000 pulses per second	GPR: Milliseconds per scan; LiDAR: 100,000-1,000,000 pulses per second	Qualified Rover Lidar (SQRLi) for Light Detection and Ranging (LiDAR)	150°C during lunar night to above 100°C during lunar day, and be resistant to lunar dust.
				Sensitivity:	GPR: 10 cm; LiDAR: Surface feature accuracy up to 1 cm	GPR: 10 cm; LiDAR: Surface feature accuracy up to 1 cm		
				Spatial resolution:	GPR: 10-30 cm horizontal resolution; LiDAR: ~1 cm point spacing	GPR: 10-30 cm horizontal resolution; LiDAR: ~1 cm point spacing		Place instruments at various points on the rover so that it properly monitor within the spatial resolution and that creates the least amount of noise when taking measurements.
	Inspect in-situ resources (ISRU) within and near the lunar pit	Measure regolith thickness and distribution, isotopic content, and	Detect mineral signatures using X-ray fluorescence spectroscopy, measure	Energy/Mass range:	XRF: 1-20 keV; QMS: 1-100 amu	XRF: 1-20 keV; QMS: 1-100 amu	PIXL and Quadrupole Mass Spectrometer (QMS)	
				Integration time	XRF: 0.5-1 second per sample;	XRF: 0.5-1 second per sample; QMS: 1-5		

	for sustainable habitation.	mineral composition within and outside the pit cave over a 100 m² area.	isotopic concentrations via mass spectrometry, and map regolith thickness using ground-penetrating radar.		QMS: 1-5 seconds per sample	seconds per sample		
				Sensitivity:	XRF: 10^(-4) w/w; QMS: Concentrations down to ppm	XRF: 10^(-4) w/w; QMS: Concentrations down to ppm		They must maintain operational integrity in vacuum conditions and perform reliably over extended periods to gather data essential for assessing habitability.
				Spatial resolution:	Both: 1-5 cm² per sample site	Both: 1-5 cm² per sample site		
	Analyze the environmental risks and dangers to potential habitability within the lunar pit cave.	Assess heat retention/reflection properties of the pit's walls and soil, and measure radiation and temperature fluctuation	Monitor temperature fluctuations with thermal sensors, and assess radiation and cosmic microwave background	Wavelength range:	TS: 8-14 µm; SR: 0.1-2.5 µm	TS: 8-14 µm; SR: 0.1-2.5 µm	Thermal Sensors (TS); Radiation Dosimeters (RD); Spectroradiometer (SR)	
				Integration time	TS: 1 hr (for 48-72 hrs); SR: 1 minutes	Both: 1 minute		Data should be logged locally with the capability for periodic transmission to Earth or a lunar relay satellite.
				Sensitivity:	TS: ±0.1°C;	TS: ±0.1°C; SR: 10^-6 W/m²/sr		

		over a 24-hour cycle.	(CMB) intensity.		SR: 10^-6 W/m²/sr			The system should have onboard storage capacity to ensure no data is lost during transmission downtimes, particularly for high-frequency radiation and temperature data.
				Spatial resolution:	Both: 5 meters	Both: 5 meters		

Table 1: Science Traceability Matrix

1.3. Summary of Mission Location

We have finalized Mare Tranquillitatis Pit as our final Mission Location. The Mare Tranquillitatis Pit was selected as the prime location for this mission due to its promising characteristics and alignment with key mission objectives (SMD and ESDMD Goal Objects from Table 1). It is located within the Tranquillitatis basin on the Moon's near side. It also offers unique geological features that make it an excellent candidate for further exploration of lunar pit caves and potential lava tube structures. It is currently the deepest known lunar pit i.e. 140m wide, with an inner pit depth of 105m, and ~20m funnel depth. The average elevation is about 2,500m below the Moon's mean radius. The pit was first discovered by SELENE orbiter cameras in 2009 and was further studied using radar imaging by NASA Lunar Reconnaissance Orbiter (Steigerwald 2023). During further studies by NASA's LRO, researchers were able to discover that the pit floor extends underground beyond the opening indicating the presence of a cave conduit (Steigerwald 2023). The temperature in the permanently shadowed areas is around 63°F (17°C) and fluctuates slightly throughout the lunar day (Junichi Haruyama, 2009). As shown in *Figure 1* using simulations researchers were able to discover that the stone ceiling that protects the cave rocks from intense radiation could also protect future human explorers from intense radiation thereby forming a lunar base camp (Leah Crane, 2024).

The criteria that were applied to assess the suitability of the Mare Tranquillitatis Pit in the trade study, included its overhangs, depth, proximity to the equator, inner diameter, and accessibility. The overhangs observed at the pit's entrance strongly suggest an internal cave structure, which may indicate the presence of lava tubes. This structural evidence, combined with a significant depth exceeding 100 meters, increases the likelihood of stable temperature and radiation levels within, creating a potentially hospitable environment for future missions. Furthermore, the pit's moderately large diameter supports the passage of materials and equipment, which is important in achieving the goals. Also, the location's closeness to the equatorial region further supports mission viability, as it provides both stable temperatures and consistent solar energy availability, essential for powering equipment.

We selected the Mare Tranquillitatis Pit because it has features like overhangs and a significant depth, which suggest that there is a higher chance for water ice to be present and possibly a stable environment that could support future missions. By using instruments such as the Lunar Prospector Neutron Spectrometer (LPNS) and Near-Infrared Spectrometer (NIRS), we can also detect and measure hydrogen and water ice content, which are critical for assessing in-situ resources. Ground-penetrating radar (GPR) and thermal sensors will help us analyze the pit's structure, temperature

variations, and possible resources. Overall, this mission at Mare Tranquillitatis will provide essential data for future sustainable lunar exploration and habitation planning.

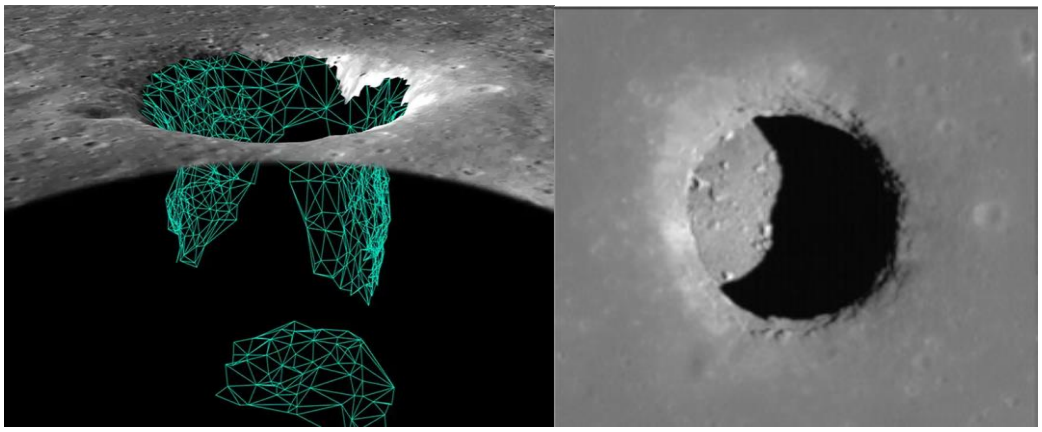

Figure 1: Mare Transquillitaris Pit - Illustration below the Mare Transquillitaris Pit (Left) and Overhead Satellite Image (Right)

The selected 5km radius around the Mare Tranquillitatis Pit offers an optimal landing location for several reasons, key among them being its safety profile and strategic scientific value. This area is characterized by rock-free regolith, as indicated by the white contour lines in the lunar surface map. This smooth, rock-free terrain minimizes the risks associated with landing, such as potential damage from boulder impacts or uneven surfaces that could compromise the landing gear of spacecraft. Additionally, the proximity to the Mare Tranquillitatis Pit—just within a 5km distance—ensures that scientific exploration can commence with minimal delay post-landing. The area allows easy deployment of rovers and equipment to investigate the pit, which is a site of significant geological interest, potentially housing key insights into lunar volcanic activity and subsurface structures such as lava tubes. This proximity is crucial not only for the efficiency of mission operations but also for the safety of overall, as it reduces the need for long-distance traverses on the lunar surface, thus conserving resources and energy.

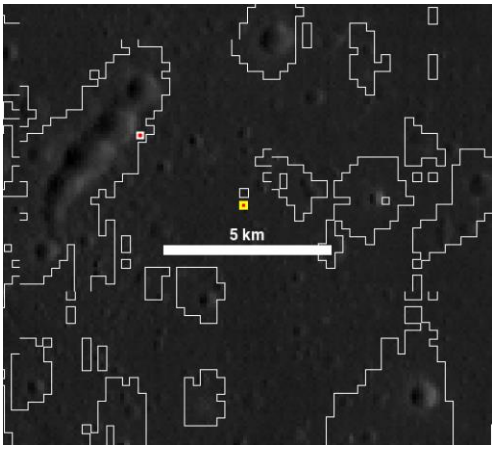

Figure 2: Possible Landing Location - Overview of the Mare Tranquillitatis Pit (Location: 8.335°N, 33.222°E) and the surrounding 5km radius considered for landing.

1.4. Mission Requirements

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Relevant Subsystem	Met/Not Met
MG-1	The mission shall conduct scientific exploration of selected lunar pits/caves to gather data on geological and potential resource characteristics.	The mission must gather scientific data to check the habitability of the lunar caves for human exploration.			Demonstration	Science	Met
PM-1	The mission shall be ready for launch by March 1, 2030.	The mission must align with NASA's schedule to ensure timely execution	Mission Objective		Inspection	Project Management	Met
SC-1	The mission shall utilize a combination of a mobile rover and a stationary lander to achieve science objectives.	The mission requires both surface and subsurface exploration	Mission Objective	Rover and Lander Design	Inspection	Mechanical Design	Met
SC-2	The spacecraft shall establish a communication link with an orbiting spacecraft to relay data back to Earth.	The mission must transfer data back to Earth for analysis	Mission Objective		Test	Communications	Met
SC-3	The mobile rover shall be capable of traversing the lunar surface and entering selected lunar caves.	The rover must navigate the lunar surface and explore caves to gather important scientific data.	Mission Configuration	Distance to Science Site	Demonstration	Mechanical and Drive System	Met

SCI-1	The lander and rover shall be equipped with scientific instruments to collect geological samples and environmental data.	The mission must collect scientific data for analysis, including geological and environmental characteristics.	Mission Objective	Instrument Functionality and Power Supply	Test	Payload	Met
SC-4	The spacecraft shall provide sufficient power to all instruments during operational phases.	All instruments must have sufficient power during operation to ensure maximum efficiency	Science Data Collection		Analysis	Power Systems	Met
SC-5	The mobile rover shall have a compact design that expands after deployment, adhering to the dimension constraints of 2 m x 1.25 m x 1.25 m.	The rover must be compact during launch and expandable for further operations	Mission Configuration		Inspection	Mechanical System	Met
SC-6	The stationary lander shall provide a stable platform for scientific operations, including sample analysis.	The lander must provide a stable platform for conducting scientific operations	Mission Configuration		Test	Mechanical System	Met
SCI-2	The instruments onboard the rover and lander shall operate within specified temperature ranges to ensure proper functioning.	Instruments must operate effectively in the lunar environment	Science Data Collection		Test	Payload	Met
SC-7	The total mass of the spacecraft, including both the rover and lander, shall not exceed 350 kg.	The mission’s total mass must not exceed launch vehicle limitations.	Mission Configuration		Analysis	Mechanical System	Met

PM-2	The total mission cost shall not exceed $425 million.	The mission must remain within budget to ensure cost-effective execution	Mission Objective		Inspection	Project Management	Met
SC-8	The landing site shall not exceed a slope of 10 degrees within a 100m diameter landing zone.	The lander must be deployed on a stable surface to ensure safety and effectiveness	Mission Configuration		Analysis	Navigation	Met
SC-9	The rover shall be capable of navigating to the selected science site within 5 km of the landing zone.	The rover must reach the selected science site within the constraints	Mobility		Demonstration	Navigation	Met
SC-10	The mission shall prioritize essential capabilities that align with the budget constraints, ensuring that only necessary functions are included.	The mission must balance costs to ensure all capabilities are included within the budget	Budget Constraint		Analysis	Project Management	Met
SCI-3	The team shall identify and prioritize science objectives that are achievable within the budget and timeline constraints.	The team must focus on achievable science objectives to maximize mission success	Mission Objective		Analysis	Payload	Met
PM-3	The mission shall conduct analog testing in a comparable volcanic environment to reduce mission risk.	Testing in various environments will reduce mission risk and validate performance	Mission Objective		Demonstration	Validation	Met

SCI-4	The spacecraft shall include systems to assess the structural integrity of selected caves before exploration.	The spacecraft must evaluate the cave's structural integrity to ensure safe exploration and potential habitation	Mission Objective		Test	Payload	Met
SCI-5	The mission shall successfully collect and relay scientific data to Earth, verified by successful communication with the orbiting spacecraft.	The mission's success depends on the complete collection and transmission of scientific data to Earth.	Mission Objective		Demonstration	Communications	Met
SC-11	All design and functional tests shall confirm that the spacecraft can operate within the specified performance constraints.	All systems must function as intended within performance constraints to ensure mission success	Mission Objective		Test	Systems	Met
SC-12	The Black Point Lava Flow in Arizona shall be used for testing the rover's capabilities.	Black Point Lava Flow in Arizona provides a suitable environment for testing rover capabilities	Environmental Testing		Demonstration	Validation	Met
SC-13	The spacecraft shall not use an RTG or any radioactive materials exceeding 5g cumulative mass.	RTGs pose potential safety risks and are prohibited for this mission, limiting the use of radioactive materials to 5g cumulative mass.	Mission Objective		Inspection	Power Systems	Met

Table 2: Mission Requirement

1.5. Concept of Operations (ConOps)

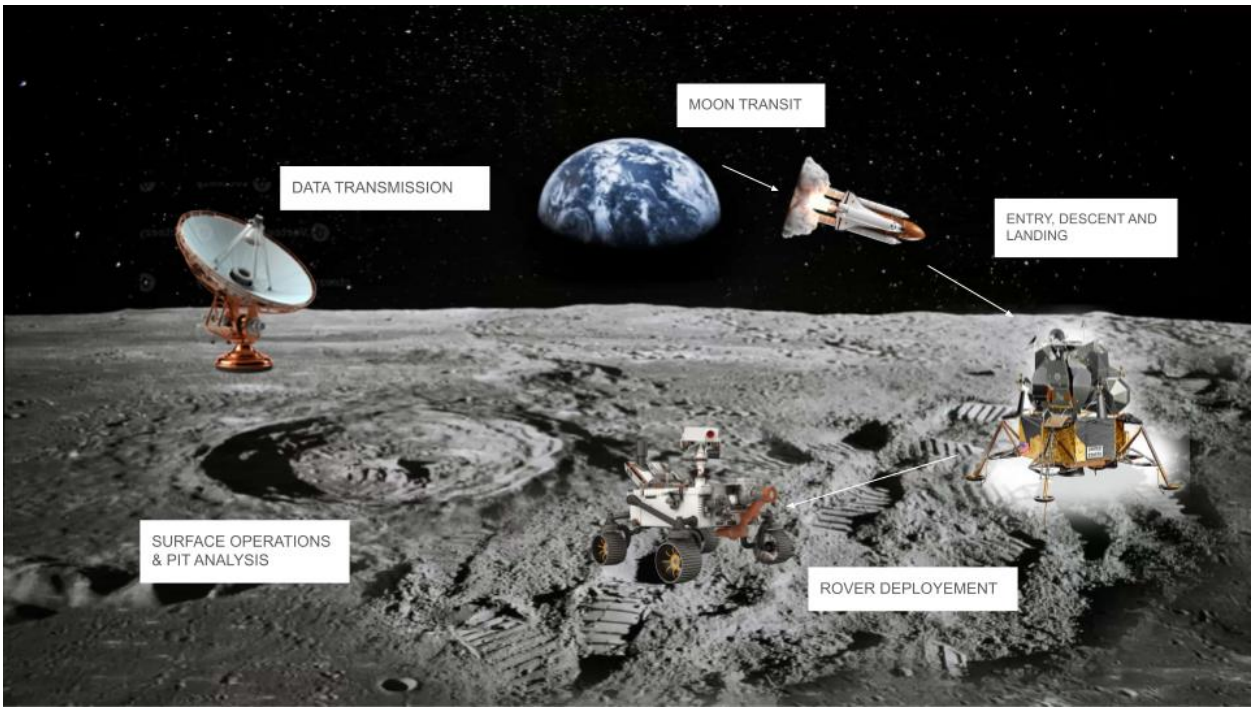

Figure 3: ConOps Illustration Image

System Architecture

Lander:
A stationary lander equipped with scientific instruments for surface analysis, sample collection, and environmental monitoring. The lander will serve as the mission's base of operations, conducting surface science tasks, supporting rover operations, and transmitting collected data back to Earth via a lunar orbiting relay.

Rover:
A mobile rover capable of navigating the lunar surface and exploring lava pits and caves. Equipped with advanced tools and instruments, the rover will analyze geological structures, assess resource availability, and collect data from the lunar surface and subsurface. The rover will also perform environmental measurements, including thermal and radiation analysis, and mechanical property assessments of regolith.

Phase Durations:

Launch and Transit:

- The spacecraft will be launched from Cape Canaveral aboard a designated launch vehicle.
- During transit, the spacecraft systems will remain in a dormant state, with periodic health checks but no science data collection.

Lunar Arrival and Descent:

- The spacecraft will enter lunar orbit and prepare for descent.
- Upon descent initiation, the lander will execute a controlled landing at the pre-selected site within a 5 km radius of the Mare Tranquillitatis Pit.

Landing and Initial Operations:

- The lander will stabilize and activate its systems, conducting health checks and deploying its surface science instruments.
- Environmental monitoring (e.g., temperature, radiation, and dust dynamics) and initial surface analysis will begin.

Rover Deployment and Calibration:

- The rover will detach from the lander and expand into its operational configuration.
- Initial system checks and calibrations will ensure readiness for exploration tasks.

Surface Exploration:

- The rover will traverse the lunar terrain, heading toward the Mare Tranquillitatis Pit or identified cave entrance.
- Using LiDAR and cameras, it will map the pit entrance and surrounding features. Ground-penetrating radar (GPR) will assess structural integrity, and initial thermal and radiation measurements will be conducted.

Subsurface Operations:

- The rover will carefully enter the pit or cave and perform detailed geological and environmental surveys.
- Instruments such as laser-induced breakdown spectroscopy (LIBS), neutron spectrometry, and thermal sensors will analyze resource availability (e.g., volatiles, water ice) and regolith properties.
- High-resolution 3D mapping of the subsurface will be generated using LiDAR, aiding in the assessment of stability and habitability.

Data Collection and Transmission):

- Both the lander and rover will continuously transmit data to Earth via the lunar orbiting relay.
- The lander will act as a relay hub, ensuring uninterrupted communication even when the rover is exploring areas with limited line-of-sight connectivity.

Mission Conclusion and Data Transmission:

- Once the exploration objectives are achieved, the rover and lander will enter a low-power state.
- Final data packets will be transmitted to Earth until all collected information has been relayed.
- The mission will formally conclude with all systems powered down.

GRAPHIC SPECIFICS

1. **Launch:** The spacecraft departing Earth on its designated trajectory.
2. **Lunar Arrival and Descent:** The spacecraft reaching lunar orbit and descending toward the Mare Tranquillitatis Pit.
3. **Landing and Deployment:** The lander stabilizing on the lunar surface and deploying the rover.
4. **Surface Exploration:** The rover mapping and collecting data around the pit's entrance.
5. **Subsurface Operations:** The rover enters the cave and conducts in-depth analysis, shown as an inset of the cave's interior exploration.
6. **Data Transmission:** Arrows indicating data flow from the rover to the lander and from the lander to Earth via the lunar relay satellite.
7. **Mission Conclusion:** The lander and rover in a powered-down state with all data successfully transmitted to Earth.

1.6. Vehicle Design Summary

The spacecraft comprises two primary components: Main Lunar Lander and Exploration Rover, tailored for exploring lunar lava pits at Mare Tranquillitatis. The integrated design supports surface analysis and subsurface exploration, prioritizing safety, efficiency, and scientific value. Below is an overview of the design elements:

Main Lunar Lander

- Purpose: Serves as the mission's stationary base for operations, communications, and scientific data collection.
- Dimensions: 1.5 m x 1.5 m x 1.2 m.
- Mass: ~120 kg, including scientific instruments and structural framework.
- Power: Equipped with high-efficiency solar panels and batteries, ensuring continuous operation during lunar nights. Peak power draw: 1.6 kW.
- Key Features:
    - Stable platform with shock-absorbing landing legs.
    - Scientific instruments for surface-level analysis, including neutron spectrometry and seismic monitoring.
    - Communication hub with relay capability via orbiting satellites.

Exploration Rover

- Purpose: Mobile unit designed to explore the pit's interior and surrounding areas for geological and resource studies.
- Dimensions: 1.9 m x 1.2 m x 1.1 m (deployed), compact during launch.
- Mass: ~95 kg, including chassis, payload, and mobility systems.
- Power: Powered by foldable solar panels and rechargeable batteries; peak power draw: ~545 W.
- Key Features:
    - Six motorized wheels with independent suspension for rough terrain.
    - Instruments like ground-penetrating radar (GPR), laser-induced breakdown spectroscopy (LIBS), and LiDAR for mapping and analysis.
    - Thermal management for extreme lunar conditions, including Multi-Layer Insulation (MLI) and heaters.
    - Semi-autonomous operation with real-time navigation adjustments.

Overall System Metrics

- Total Mass: ≤ 350 kg, adhering to launch constraints.
- Power Consumption: Efficient allocation across subsystems, ensuring operation during lunar nights and in shadowed regions.
- Thermal Control: Systems optimized for temperature swings (-150°C to 100°C) to maintain functionality.

This robust vehicle design ensures the mission meets scientific objectives while addressing challenges like uneven terrain, power scarcity, and harsh lunar environments.

1.7. Science Instrumentation Summary

This mission employs many instruments to comprehensively address our lunar science objectives, spanning resource characterization, geological analysis, and habitability evaluation.

- **Lunar Prospector Neutron Spectrometer (LPNS)**:
  - Detects hydrogen and maps water ice by measuring neutron emissions, essential for understanding the distribution of volatiles.
- **Near-Infrared Spectrometer (NIRS)**:
  - Measures surface reflectance in the 1–3 μm range, enabling the identification of water ice and other volatiles.
- **Passive Microwave Radiometer and Thermal Infrared Spectrometer**:
  - Detects the phase of water (solid or vapor) and analyzes thermal properties to study volatiles in permanently shadowed regions.
- **High-Sensitivity Seismometer**:
  - Captures seismic activity to evaluate lunar tectonics and subsurface structures, critical for assessing habitat stability.
- **ChemCam (LIBS) and Planetary Instrument for X-ray Lithochemistry (PIXL)**:
  - Analyzes elemental composition with laser spectroscopy and X-ray fluorescence for resource mapping and mineral identification.
- **Mass Spectrometer and/or Neutron Spectrometer**:
  - Detects and quantifies volatiles like hydrogen and helium-3 for ISRU planning.
- **Laser Diffraction Particle Size Analyzers**:
  - Measures regolith grain size to assess mechanical properties for excavation and construction potential.
- **LIBS (Laser-Induced Breakdown Spectroscopy)**:
  - Provides detailed elemental analysis of regolith and rocks to identify oxygen-bearing minerals for ISRU.
- **Particle Tracking Sensor**:
  - Tracks the behavior of dust ejecta during lander descent, helping to evaluate lunar dust dynamics.
- **Surface Adhesion Tester**:
  - Measures adhesion forces of lunar dust to assess its impact on equipment and infrastructure.
- **Electrostatic Sensors**:
  - Monitors the charge on dust particles and its effect on lofting, addressing potential challenges to long-term operations.
- **Plasma Sensors**:
  - Measures solar wind particle flux and energy to study interactions with lunar dust and surface properties.
- **WISDOM GPR and Space Qualified Rover LiDAR (SQRLi)**:

    - Provides high-resolution subsurface mapping and detailed 3D surface models to identify geological structures and evaluate habitat potential.
- **PIXL and Quadrupole Mass Spectrometer (QMS)**:
    - Combines high-resolution chemical mapping with volatile detection, crucial for analyzing regolith composition.
- **Thermal Sensors (TS); Radiation Dosimeters (RD); Spectroradiometer (SR)**:
    - Monitors temperature, radiation levels, and reflectance to evaluate environmental risks for habitation.

These instruments work together to enable a deeper understanding of the Moon's surface and subsurface, supporting scientific exploration and resource utilization to pave the way for sustained lunar operations not to mention achieve the science objectives of this mission.

1.8. Programmatic Summary

1.8.1. Team Introduction

- **Ashi Poorey (Project Manager):**
  - University of Illinois Urbana Champaign
  - Astrophysics & Data Science minor
  - EXPERIENCE:
    - Astrophysical and scientific knowledge (coursework)
    - writing expertise (content writer at student journal)
    - Organisation and planning (logistics team for TEDxUIUC)

- **Len Johns Shaji (Deputy Project Manager of Resources):**
  - **University**:
    - *Name*: University of Houston
    - *Location*: Houston, Texas, USA
    - *Major*: Computer Engineering
    - *Minor*: Computer Science and Mathematics
    - *Year*: Senior
  - **Experience brought to the Team:**
    - Team lead for NASA JSC HACK 2024
    - Machine learning Engineer for NASA Airport Throughput Prediction Challenge.
    - Coordinator of projects and events for IEEE-UH and UH Robotics
    - Team lead and Researcher for South Pole Safety: Designing the NASA Lunar Rescue System

- **Jesus Garcia Garcia (Chief Scientist):**
  - University of Minnesota Twin Cities
  - Computational Biology PhD Student
  - Experience:
    - High-performance computing engineer at CU, Boulder
    - Single-cell genomics analysis
    - Moving toward space genomics

- **Jay Kamdar (Chief Systems Engineer)**
  - Missouri University of Science and Technology
  - Aerospace Engineering, MS
  - Topics of Interest:
    - Space Debris Mitigation and Traffic Management
    - Space Solar Power and its Applications
    - Space Systems Design

- **Urva Vashi (Scientist - Astrophysicist):**
  - **University:**

        - *Name*: University of Illinois Urbana Champaign
        - *Location*: Urbana-Champaign, Illinois, USA
        - *Major*: Astrophysics
        - *Minor*: Computer Science and Statistics Minor
        - *Year*: Senior
    - **Experience brought to the Team:**
        - Astrophysical and Scientific knowledge (coursework)
        - Computational and Hardware Knowledge (Student IT Employee and CS Minor coursework)
        - Statistical Manipulation and Data Acquisition Knowledge (STAT Minor coursework)
        - Research & Data Analysis (Scientific Lab Reports coursework)
        - Team Player

- **Vanina Joyce NGUEMSOP FEUTIE (Electrical Engineer)**
    - University:
        - Lone Star College-CyFair
        - Houston, Texas
        - *Major*: Electrical Engineering Undergraduate Student
        - *Year*: Sophomore
    - Experience:
        - Engineering Dynamics (Coursework)
        - Programming for Engineers (Coursework)
        - Organization and planning (ASME-LSCCyFair President, Math Club Event Coordinator)
        - Professional Writing Expertise (Scientific Lab Reports from coursework, Deputy Assistant Project Manager for Lunabotics 2024-2025 LSCCyFair)
        - Research & Data Analysis (Scientific Lab Reports coursework)

- **Benita Benjamin (Computer Hardware Engineer)**

  University: University of Texas at Austin
  - Location: Austin, Texas
  - Major: Computer Science
  - Year: Freshman

  Experience: Computer Science experience from coursework
  Software and programming experience
  Leadership and collaboration from school and college organizations
  Research experience from internships
- Raghav Ramesh

1.8.2. Team Management Overview

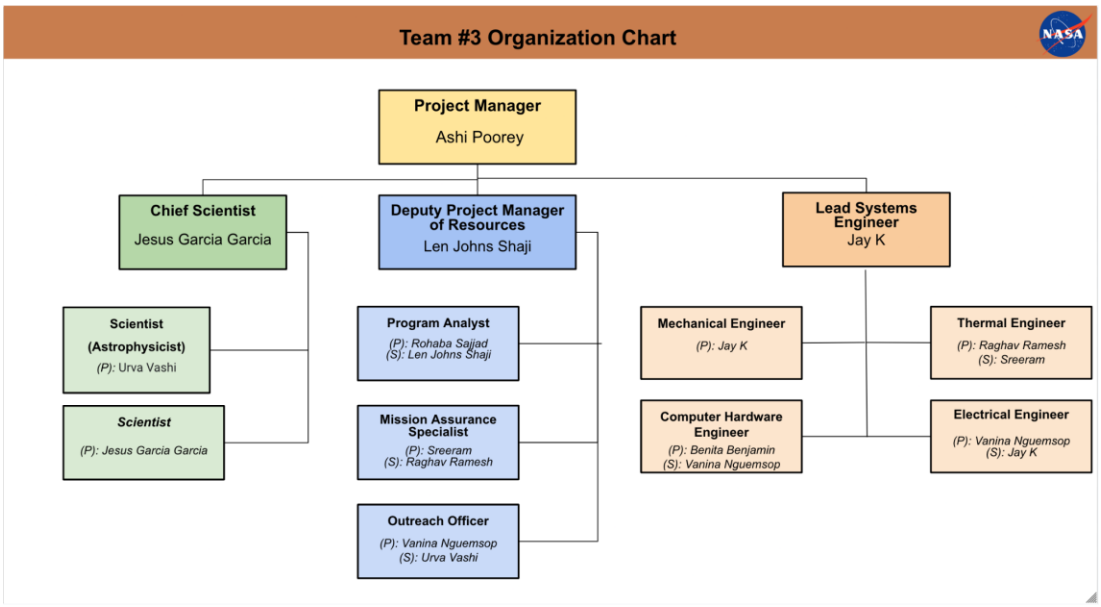

Figure 4: Team #3 Organization Chart

The team is, by now, somewhat well-equipped with each other's work etiquette and ways. The team members can identify their responsibilities and roles effectively. During a deliverable time, the team distributes work according to their assigned roles and respective interests to ensure maximum efficiency, along with understanding everyone's other commitments and workload. Time-sensitive decisions are taken by the leadership team; nonetheless, everyone's take on the situation is considered before the final decision.

For the future, the team would like to be more prompt with tasks and the assigned work, to not get overloaded with work at the last moment. There will be better time management and a more efficient method of assigning work taken into account with further upcoming deliverables. Furthermore, to ensure all the tasks are done, leaders will be assigning tasks to the members of their respective teams through a Google sheet and sharing it with the team which will then be updated by the team members as they finish their tasks. Additionally, leaders will regularly check with the team members through Discord.

The main challenge the team regularly faces is being able to match meeting timings. Not every team member is available all at the same time, and hence, being in sync is something we struggle with. However, to overcome its effects, the team keeps updating all the members on matters discussed and done and ensures that everyone's participation and opinion are taken into consideration by being active in the discord group channel. Furthermore, we started with a team of 15-17 people, but now it is down to 8-9 active members. To compensate for this, we are having some members take two primary roles or other members help out on the people who left. For example, our primary Mechanical Engineer left so we have a member as a secondary and all the other members help out for that role.

1.8.3. Major Milestones Schedule

The mission life cycle is planned in such a way that it ensures that each phase contributes to achieving the overall mission objectives. This structured timeline is built around major milestones that serve as checkpoints, ensuring progress aligns with technical, operational, and scientific goals. From the initial design stages to the final dissemination of findings, every step is designed to maximize efficiency and mission success.

Phase C: Final Design and Fabrication marks the beginning, starting on January 25, 2025, with the completion of subsystem designs and the selection of mission-critical materials by August 2025. This is followed by simulations, fabrication planning, and the Critical Design Review (CDR), concluding by June 24, 2027, to finalize the mission's design and ensure readiness for the next phase.

Phase D: System Assembly, Integration & Test, Launch & Checkout, from January 25, 2027, to March 4, 2030, focuses on assembling and testing the spacecraft. Highlights include instrumental acquisitions by August 2027, integration testing by February 2029, and rover assembly by June 2029. The Operational Readiness Review (ORR) in July 2029 will ensure mission readiness, leading to the spacecraft's launch in March 2030 and subsequent rover deployment on March 4, 2030.

Phase E: Mission Operations, spanning from March 5, 2030, to January 24, 2032, involves the rover's traversal and entry into the targeted pit by January 2030, followed by two years of scientific data collection. This phase concludes with preparations for the mission's end-of-life.

Finally, Phase F: Closeout, from January 25, 2032, to July 31, 2032, wraps up the mission with data archiving, system decommissioning, and the preparation and dissemination of final reports. These milestones collectively ensure the mission's success and provide valuable insights for future endeavors over a total duration of approximately 7.5 years.

In conclusion, the mission is marked by these major milestones, ensuring that each phase progresses efficiently and effectively. By sticking to this roadmap, the mission team can navigate challenges, adapt to dynamic requirements, and achieve its objectives, contributing valuable scientific and operational knowledge for future explorations.

1.8.4. Budget Overview

The Mare Tranquillitatis exploration mission has a budget cap of $425 million and an estimated final cost of $375 million, carefully distributed across personnel, subsystem development, testing, outreach, and contingency. This allocation prioritizes scientific excellence, operational reliability, and public engagement.

Budget Breakdown

Personnel Costs (30%; $113 million)

- Salaries and benefits for a multidisciplinary team, including scientists, engineers, technicians, and project managers.
- Includes recruitment and training phases tailored to the mission timeline, scaling resources efficiently during key milestones.
- A portion is allocated for operational staff during testing, integration, and mission support phases.

Subsystem Costs (43%; $160 million)

- Covers design, development, and integration of critical mission hardware:
- Mechanical Systems ($30 million): Structural elements for the lander and rover, including deployment mechanisms and mobility systems.
- Command and Data Handling (CDH) ($25 million): Ensures robust control, data processing, and communication links.
- Power Systems ($20 million): Solar panels, batteries, and power management units tailored for lunar conditions.
- Thermal Management ($35 million): Insulation and heating systems to counteract extreme lunar temperature fluctuations.
- Payload (Science Instruments) ($50 million): Advanced tools for surface and subsurface analysis, including GPR, spectrometers, and LiDAR.

Testing and Simulation Costs (13%; $50 million)

- Comprehensive testing of spacecraft subsystems, focusing on durability under extreme lunar conditions.
- Includes analog environment simulations to replicate Mare Tranquillitatis and validate hardware, software, and scientific capabilities.

Travel and Outreach (5%; $20 million)

- Travel: Team deployment for field tests, integration activities, and launch preparations.
- Outreach: Public engagement programs, such as educational workshops and mission updates, to inspire broader interest in space exploration.

Contingency Fund (16%; $60 million)

- Reserved for mitigating technical, operational, or logistical risks.
- Provides flexibility to address unforeseen challenges, ensuring mission integrity without compromising objectives.

Strategic Budget Prioritization

This allocation emphasizes critical mission objectives, with nearly half the budget

devoted to developing state-of-the-art subsystems. A robust contingency fund and investment in personnel and outreach ensure adaptability and long-term impact, aligning the mission with NASA's goals of sustainable lunar exploration and public involvement.

This financial strategy provides a strong foundation for the mission's success, supporting innovative science and robust operations while remaining within the prescribed budget cap.

2. Overall Vehicle and System Design

2.1. Spacecraft Overview

The spacecraft consists of two main components: the Main Lunar Lander and the Exploration Rover, both crucial for fulfilling the mission's goal of exploring lunar lava pits for potential human habitation and in-situ resource utilization. The lander serves as a stable base of operations and communications hub, while the rover enables detailed exploration of the lunar surface and subsurface environments, particularly targeting the selected lava pit. During transit, the rover remains securely attached to the lander, in a dormant state. Upon landing, the rover is deployed from the lander and begins its autonomous mission. The lander is equipped with various instruments for surface-level data collection and is tasked with transmitting mission data back to Earth via a relay satellite. Its large solar arrays also serve as power generators for both the lander and the rover, recharging the rover when it docks.

The Main Lunar Lander is designed to remain stationary and acts as the mission's base. It houses a suite of scientific instruments aimed at surface analysis, including a neutron spectrometer for water ice detection and seismic sensors for monitoring tectonic activity. Its stable platform allows for continuous environmental monitoring, ensuring consistent data collection from the lunar surface. The lander also supports the mission’s power needs, with solar panels capable of producing energy during lunar day periods, and battery packs providing energy storage for operations during the lunar night. The Exploration Rover, in contrast, is a mobile platform tasked with investigating the lunar surface and subsurface, with a particular focus on entering and exploring the lava pits. Equipped with instruments such as a ground-penetrating radar (GPR) and laser-induced breakdown spectroscopy (LIBS), the rover is capable of mapping underground features and analyzing soil composition. Its mobility system is designed to navigate the rough and uneven terrain of the lunar surface, ensuring access to hard-to-reach areas, including the steep and rocky inclines of the Mare Tranquillitatis pit.

Once deployed, the rover will traverse the lunar surface independently, mapping the pit's interior and surrounding areas. It is designed to enter the pit via its overhangs, exploring possible cave-like structures that may indicate the presence of lava tubes. The rover’s sensors will help identify any potential resources, such as water ice, hydrogen, or other volatiles, crucial for future human missions. These subsurface measurements will be key to assessing the structural integrity and habitability of the cave environments. Communication between the rover and the lander is maintained through a dedicated radio link, with the lander acting as a relay to transmit data back to Earth. This setup ensures that even in areas where direct communication with Earth might be compromised, such as within the pit itself, the mission’s data flow remains

uninterrupted. The rover’s ability to operate in a semi-autonomous mode allows it to carry out scientific objectives even when real-time communication is delayed.

Both the lander and rover are equipped with robust thermal control systems to handle the extreme temperature fluctuations of the lunar environment. During the lunar day, radiators dissipate heat to prevent overheating, while Multi-Layer Insulation (MLI) and onboard heaters ensure that components remain functional during the cold lunar night. The rover’s thermal system is especially important, as it must remain operational in the colder, permanently shadowed areas of the pit, where temperatures can drop significantly below those experienced on the surface. The rover will also conduct high-resolution imaging and mapping of the pit’s interior using LiDAR and visual cameras. This will help generate 3D models of the cave system, which are essential for assessing the potential of the site as a future habitat. These models will provide insights into the size, shape, and stability of the cave systems, as well as identify possible entrances and overhangs that could serve as shelter for future astronauts.

By combining the stationary capabilities of the lander with the mobility and advanced instrumentation of the rover, the mission is well-suited to gather comprehensive data on the lunar environment. The mission’s focus on exploring the Mare Tranquillitatis pit, a location known for its depth, overhangs, and stable temperatures, positions it to make significant contributions to the understanding of lunar geology and the potential for sustainable human habitation .

Main Lunar Lander

The Main Lunar Lander serves as the mission's central hub for both initial surface operations and as a critical platform for continuous scientific data collection. Its primary function is to ensure the safe delivery of the Exploration Rover and its own scientific payload to the lunar surface, specifically the Mare Tranquillitatis pit. Once landed, it remains stationary, providing a stable base for long-duration scientific experiments, environmental monitoring, and communications relay. Additionally, the lander houses key power generation systems that not only sustain its own instruments but also recharge the Exploration Rover during its operational cycles. As a result, the lander plays a pivotal role in mission longevity and the continuous transmission of data back to Earth.

The lander's strategic positioning near the Mare Tranquillitatis pit enables in-depth environmental analysis, such as real-time monitoring of seismic activity and radiation levels, while the Exploration Rover carries out more dynamic investigations in the pit's interior. The lander ensures mission resilience by maintaining communication with Earth via a relay satellite, even when the rover is exploring regions out of direct line-of-sight.

Mechanical Subsystem:

The mechanical subsystem provides the robust structural framework necessary for a safe and precise lunar landing. The lander must endure the forces of launch, transit, and lunar surface landing while maintaining the rover's secure attachment. Upon arrival, it facilitates the smooth deployment of the rover, ensuring the separation process is free of mechanical complications. The lander's chassis is built to handle the rugged and uneven terrain of the lunar surface, particularly around the Mare Tranquillitatis pit. This region is known for its irregularities, with overhangs and steep inclines that require the lander's feet to stabilize on different surface gradients. The lander also incorporates shock-absorbing materials and precise thruster controls to mitigate landing impacts.

Power Subsystem:

The power subsystem is critical for the lander's long-term operation and supports the rover during recharging cycles. The system includes high-efficiency solar panels optimized for the lunar day and a robust battery storage system to ensure continuous operation during the lunar night, when temperatures plummet and solar energy is unavailable. The solar panels are positioned and designed to capture optimal sunlight near the equatorial Mare Tranquillitatis pit, ensuring sufficient energy is stored even during periods of reduced solar exposure. Given the extreme temperature swings, the battery packs are built with materials capable of retaining charge in the cold lunar night, supporting not only the lander's instruments but also recharging the rover between its exploration cycles. The lander's power management system ensures that energy is distributed efficiently, prioritizing life-critical instruments such as communication devices and heating elements during low-power phases.

Command and Data Handling (CDH) Subsystem:

The CDH subsystem governs all command functions, data processing, and communication between the lander, the rover, and Earth. This includes collecting and processing data from scientific instruments and ensuring that this data is stored and transmitted reliably via an orbiting relay satellite. The CDH system integrates seamlessly with the lander's stationary instruments as well as the rover's dynamic exploration tools, managing the flow of data from both platforms. A high-gain antenna enables communication between the lander and Earth, while a dedicated radio link ensures that data from the rover is efficiently relayed back through the lander. The CDH is also responsible for managing the timing of operations, including scientific measurements, instrument calibration, and power consumption cycles, ensuring optimal resource allocation during the mission's duration.

Thermal Subsystem:

The thermal subsystem ensures that the lander's critical components remain within operational temperature ranges in the harsh lunar environment, where temperatures fluctuate dramatically between day and night. The subsystem employs a combination of passive and active thermal control methods, including Multi-Layer Insulation (MLI) and electric heaters. At the Mare Tranquillitatis pit, temperatures can range from over 100°C during the lunar day to -150°C at night. The lander's thermal design leverages MLI to minimize heat absorption during the day and prevent rapid heat loss at night. Electric heaters, powered by the lander's battery system, activate during the lunar night to protect sensitive electronics and scientific instruments. The thermal subsystem also safeguards the rover's recharging bay, ensuring the vehicle remains operational even after extended periods in shadowed areas of the pit.

Science Instruments:

The lander is equipped with a range of scientific instruments designed to collect long-term environmental and geological data. This includes monitoring seismic activity to detect moonquakes and potential underground lava tubes, as well as measuring radiation levels in the lunar environment. The scientific payload aboard the lander is tailored to study the structural integrity and habitability of lunar caves, particularly in the context of future human missions. Instruments include a Lunar Hydrogen Neutron Spectrometer (LHNS) for detecting water ice and hydrogen concentrations in the soil, thermal infrared spectrometers for mapping surface temperature fluctuations, and seismic sensors to assess tectonic activity beneath the lunar surface. These instruments will provide continuous data on environmental stability, offering crucial insights into the feasibility of using lunar pits as long-term habitats. Additionally, the lander is equipped with radiation dosimeters to measure exposure levels inside and outside the pit, vital for assessing the long-term habitability of the region for human explorers.

Exploration Rover:

The Exploration Rover is designed specifically for in-depth exploration of the Mare Tranquillitatis Pit, targeting both surface and subsurface regions that are critical to understanding the potential for human habitation and in-situ resource utilization (ISRU). This rover's mission is to investigate the geological formations within the pit, assess structural integrity, and identify resources such as water ice and volatiles. Equipped with a sophisticated suite of instruments, it will collect data from areas inaccessible to the stationary lander, particularly focusing on the potential for lava tubes or caves that may offer stable environments for future lunar bases. The rover's mission extends beyond surface-level exploration, as it is engineered to enter and investigate deep subsurface features within the lava pit, providing key insights into the Moon's geological history and its potential for sustaining human life. Its autonomy and mobility make it essential for covering ground that would otherwise be unreachable, offering a dynamic platform for

scientific discovery in areas where communication and visibility are limited.

Mechanical Subsystem:

The rover's mechanical subsystem is robust and specifically designed to withstand the extreme lunar environment. It features a mobility system capable of traversing the rough, uneven terrain of the Mare Tranquillitatis Pit. The chassis, built from lightweight but durable materials, allows the rover to handle significant gravitational stresses and high-vibration environments without compromising structural integrity. The system includes a rugged set of wheels with adaptive suspension, enabling the rover to navigate rocky outcrops, steep slopes, and loose regolith deposits that characterize the pit's surface and interior. A key feature is its compact, foldable design during transit, which expands to full operational capacity once deployed from the lander. This expansion includes the deployment of additional tools, sensors, and scientific instruments mounted on robotic arms.

Power Subsystem:

The rover's energy needs are met by a combination of high-efficiency solar panels and rechargeable battery packs. These solar panels, made from advanced photovoltaic cells, are designed to generate maximum energy during the lunar day, with an efficiency rate exceeding 25%. They fold during launch and expand upon deployment to cover a larger surface area, ensuring sufficient energy capture even in low-light conditions near the pit. The battery packs are engineered for long-term reliability and can operate in extremely cold conditions, a critical feature when the rover navigates permanently shadowed regions within the pit where temperatures can drop to -150°C. The batteries also feature integrated thermal protection and are capable of storing enough energy to allow for continued operation during lunar night cycles or when sunlight is obstructed.

Command and Data Handling (CDH) Subsystem:

The rover's CDH system is vital for managing its autonomous operations and maintaining communication with the lander and mission control. Given the challenges of operating deep within the Mare Tranquillitatis Pit, where direct line-of-sight communication may be compromised, the CDH subsystem employs a fault-tolerant communication protocol. It features dual-band radio systems for both short-range communication with the lander and long-range data relays back to Earth via orbiting satellites. The onboard computer is responsible for processing all scientific data collected by the rover's instruments, ensuring real-time decision-making capabilities, such as selecting optimal routes through the pit or adjusting its operation schedule based on energy availability. In addition, the CDH system includes autonomous navigation software capable of obstacle detection, allowing the rover to chart its path safely through challenging terrain without constant human oversight.

Thermal Subsystem:

The thermal subsystem is crucial for ensuring that the rover's components function optimally in the extreme temperature ranges of the lunar environment. The system uses a combination of passive and active thermal management strategies. Multi-Layer Insulation (MLI) provides the primary defense against extreme heat loss during lunar nights and shields sensitive equipment from solar radiation during the day. Additionally, heating elements are embedded near critical components like batteries, electronics, and scientific instruments, preventing them from freezing in shadowed regions. The rover's design also includes radiators that dissipate excess heat generated by its internal systems, ensuring operational stability during high-energy activities such as surface excavation or subsurface scanning.

Science Instruments:

The rover's scientific payload is highly specialized for both surface and subsurface exploration. It includes:

- *Ground-Penetrating Radar (GPR):*
    - The GPR system is a primary tool for subsurface mapping, capable of detecting geological structures such as buried lava tubes, fractures, and rock layers up to several meters below the lunar surface. Operating in the 30 MHz to 3 GHz frequency range, the GPR provides high-resolution images of subsurface features, offering clues to the geological history and stability of the pit.
- *Near-Infrared Spectrometer (NIRS):*
    - This instrument measures the reflectance spectra from the lunar surface and shallow subsurface, identifying water ice, hydrogen, and other volatiles by analyzing how light interacts with minerals. By working in the 1-3 μm wavelength range, the NIRS can detect water content at concentrations as low as 1% by weight, critical for assessing ISRU potential.
- *Laser-Induced Breakdown Spectroscopy (LIBS):*
    - LIBS allows the rover to analyze the elemental composition of the lunar soil and rocks by firing a laser at the surface and measuring the spectral lines emitted by vaporized material. This data will help identify essential elements such as silicon, iron, and titanium, which are vital for understanding the Moon's resource potential.
- *Thermal Infrared Spectrometer (TIR):*
    - This sensor measures the surface and subsurface temperatures within the pit, providing critical data on heat retention and the thermal behavior of rock formations, which are key for determining the feasibility of using the pit for human habitation.

- *LiDAR:*
    - The Light Detection and Ranging (LiDAR) system maps the interior of the pit, creating detailed 3D models of its topography. These models will be crucial for future mission planning, as they will identify areas of interest, such as overhangs or cave entrances that could provide shelter for astronauts.
- *Seismic Sensors:*
    - In collaboration with the stationary lander, the rover deploys seismic sensors to monitor tectonic activity and assess the structural integrity of the lunar surface and pit walls. This data is vital for evaluating the risks of moonquakes or other geological events that could impact long-term human presence

The Exploration Rover plays a pivotal role in the mission by extending its operational reach into the most scientifically rich and geologically significant areas of the Mare Tranquillitatis Pit. Its advanced mobility, autonomous capabilities, and comprehensive instrument suite make it an indispensable part of the mission's exploration and data collection efforts, offering insights that will guide future lunar exploration and potential settlement.

2.1.1. Mechanical Subsystem Overview

Rover Design

The rover's mechanical subsystem includes multiple subassemblies designed for surface and subsurface navigation. The rover's compact, expandable frame fits the specified launch dimensions, ensuring it can navigate and enter lunar caves while supporting scientific instrumentation.

1. **Chassis and Frame**
   - **Design**: The chassis is a lightweight, durable frame made of titanium alloy for high strength-to-weight ratio. The frame is foldable to fit the launch configuration and expand upon deployment, providing mounting points for scientific instruments.
   - **TRL**: 7 (The frame and chassis design are based on previous lunar rover designs, with slight modifications.)
   - **Mass**: 20 kg
   - **Dimensions**: 1.9 m x 1.2 m x 1.1 m
   - **Max Power Draw**: Negligible

2. **Drive System**
   - **Design**: Includes six ruggedized, motorized wheels with independent suspension, each fitted with a combination of lightweight, shock-absorbing tires capable of traversing lunar terrain. Each wheel's motor and suspension system allows adaptation to rugged surfaces and inclines up to 30 degrees.
   - **TRL**: 6 (Although similar designs have been deployed, the unique requirements for cave exploration introduce additional challenges.)
   - **Mass**: 30 kg
   - **Dimensions**: 0.5 m diameter per wheel
   - **Max Power Draw**: 200 W

3. **Deployment Mechanism**
   - **Design**: Includes foldable support arms and actuators that engage once the rover is deployed, allowing it to extend to full operational dimensions.
   - **TRL**: 5 (Newly integrated design to meet mission-specific compaction requirements.)
   - **Mass**: 10 kg
   - **Dimensions**: Fits within the folded rover configuration
   - **Max Power Draw**: 100 W

4. **Instrument Mounts and Payload Bay**

- **Design**: Specialized mounting brackets and compartments allow for secure placement of scientific instruments and sample collection tools.
- **TRL**: 7 (Instrument mounts are based on previous mission designs.)
- **Mass**: 15 kg
- **Dimensions**: Variable, adaptable to payload
- **Max Power Draw**: Negligible

Lander Design

The lander's mechanical subsystem provides a stable platform for the rover's operations and sample analysis.

1. **Lander Frame and Base**
    - **Design**: A fixed, stable titanium alloy frame, which includes vibration-damping materials to ensure stability during lander operations.
    - **TRL**: 7 (Based on previous stationary lander designs for lunar missions.)
    - **Mass**: 50 kg
    - **Dimensions**: 1.5 m x 1.5 m x 1.2 m
    - **Max Power Draw**: Negligible

2. **Lander Legs and Stability System**
    - **Design**: Four telescopic, shock-absorbing legs provide stability on uneven surfaces. The legs deploy upon landing, absorbing impact and providing a level base.
    - **TRL**: 6 (The design incorporates some new stability features due to lunar surface variability.)
    - **Mass**: 20 kg
    - **Dimensions**: Adjustable height up to 1.5 m
    - **Max Power Draw**: 50 W

3. **Sample Handling and Analysis Platform**
    - **Design**: A rotating, modular platform capable of handling various instruments for in situ analysis. The platform includes robotic arms for sample handling and manipulation.
    - **TRL**: 5 (New integration to support specific scientific objectives.)
    - **Mass**: 25 kg
    - **Dimensions**: 1 m x 1 m x 0.5 m
    - **Max Power Draw**: 150 W

2.1.1.1. Mechanical Subsystem Requirements

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Relevant Subsystem	Met/Not Met
SC-1	The mission shall utilize a combination of a mobile rover and a stationary lander to achieve science objectives.	The mission requires both surface and subsurface exploration	Mission Objective	Rover and Lander Design	Inspection	Mechanical Design	Met
SC-3	The mobile rover shall be capable of traversing the lunar surface and entering selected lunar caves.	The rover must navigate the lunar surface and explore caves to gather important scientific data.	Mission Configuration	Distance to Science Site	Demonstration	Mechanical and Drive System	Met
SC-5	The mobile rover shall have a compact design that expands after deployment, adhering to the dimension constraints of 2 m x 1.25 m x 1.25 m.	The rover must be compact during launch and expandable for further operations	Mission Configuration		Inspection	Mechanical System	Met
SC-6	The stationary lander shall provide a stable platform for scientific operations, including sample analysis.	The lander must provide a stable platform for conducting scientific operations	Mission Configuration		Test	Mechanical System	Met
SC-7	The total mass of the spacecraft, including both the rover and lander, shall not exceed 350 kg.	The mission's total mass must not exceed launch vehicle limitations.	Mission Configuration		Analysis	Mechanical System	Met

SC-11	All design and functional tests shall confirm that the spacecraft can operate within the specified performance constraints.	All systems must function as intended within performance constraints to ensure mission success	Mission Objective		Test	Systems	Met

Table 3: Mechanical Subsystem Requirements

2.1.1.2. **Mechanical Sub-Assembly Overview**

The mechanical subsystem for the Mare Tranquillitatis exploration mission comprises the Main Lunar Lander and the Exploration Rover, each designed to operate seamlessly in the Moon's harsh environment. These components are tailored for maximum functionality under extreme conditions, such as temperature swings from -150°C to 100°C, exposure to abrasive lunar dust, and operation on uneven, rocky terrain. The subsystem prioritizes modularity and integration, ensuring the lander and rover collectively meet the mission’s objectives of surface and subsurface exploration while maintaining structural and operational integrity.

The Main Lunar Lander serves as the mission's stable base for operations, communications, and scientific data collection. It features a robust titanium alloy chassis that supports its scientific payload and houses critical systems such as power generation, thermal management, and communication units. The lander's landing legs are telescopic and equipped with shock absorbers, enabling safe landings on slopes of up to 10° and uneven terrain around the Mare Tranquillitatis pit. Additionally, instrument mounts within the lander minimize vibrations and thermal interference, ensuring consistent data collection. The lander's structural design incorporates lightweight materials to maximize payload capacity while adhering to mass constraints, with a total system weight of approximately 120 kg. Its adaptability allows it to support extended operations during lunar nights using battery reserves and active thermal management systems.

The Exploration Rover, designed for mobility and precision, complements the lander's stationary capabilities by investigating the Mare Tranquillitatis pit’s interior and surrounding areas. Its chassis, crafted from titanium alloy, is compact and expandable, ensuring secure storage during transit and rapid deployment upon landing. The rover's six motorized wheels, paired with adaptive suspension, allow it to traverse steep inclines of up to 30° and navigate loose regolith, rocky outcrops, and shadowed regions within the pit. Equipped with state-of-the-art instruments, including ground-penetrating radar (GPR) and laser-induced breakdown spectroscopy (LIBS), the rover excels in subsurface mapping and resource analysis. To mitigate the challenges of lunar dust, its mobility system incorporates dust-tolerant seals and materials that maintain performance in abrasive conditions.

Both the lander and rover employ advanced thermal management solutions to address the Moon's extreme temperature fluctuations. Multi-Layer Insulation (MLI) and embedded heaters protect sensitive components during the frigid lunar night, while radiators dissipate excess heat generated during the day. The integration of these

thermal systems ensures the longevity and reliability of both platforms. The rover also features an actuator-based deployment mechanism that transitions it from a compact launch configuration to its operational form, supporting its full suite of scientific instruments and mobility functions. This deployment mechanism, along with the rover's semi-autonomous navigation capabilities, reflects the mission's emphasis on resilience and adaptability in unexplored terrains.

By combining the stationary data-gathering power of the lander with the dynamic exploratory potential of the rover, the mechanical subsystem offers a comprehensive solution for investigating Mare Tranquillitatis. The thoughtful design, reliance on proven technologies, and adaptations for mission-specific challenges ensure the subsystem is well-equipped to achieve the scientific and operational goals of this ambitious lunar mission.

Main Lunar Lander Mechanical Sub-Assemblies

- ***Chassis and Base***

  The lander chassis is a critical component of the mission, serving as the primary structural framework that supports all operational and scientific activities. Constructed from a titanium alloy, the chassis offers a superior strength-to-weight ratio, ensuring that the structure is lightweight enough to adhere to the mission’s strict mass constraints (120 kg for the lander) while maintaining the durability required for lunar conditions. The chassis also features integrated mounting points for securing scientific instruments, communication equipment, and power systems, ensuring their stability during both transit and operations.

  To mitigate the impact forces during landing, the base incorporates shock-absorbing materials that reduce stress on the lander and its components. This is essential for protecting sensitive instruments from vibrations caused by the descent and touchdown phases. The chassis is designed to handle micrometeoroid impacts, leveraging a combination of layered shielding and impact-resistant coatings to ensure long-term survivability. Additionally, the chassis has been optimized to minimize thermal expansion and contraction, which is critical given the Moon’s extreme temperature fluctuations from -150°C at night to over 100°C during the day. The design incorporates cutouts and modular panels that facilitate quick assembly and integration of subsystems during mission preparation.

  TRL: 7, based on heritage designs from prior lunar lander missions, with minor modifications for this mission.

- ***Landing Legs***

  The lander is equipped with four telescopic landing legs, designed to ensure stability and resilience during descent and touchdown on the Moon's uneven surface. Each leg features a multi-jointed structure with adjustable lengths, allowing the lander to accommodate surface irregularities such as slopes of up to 10° and minor obstacles like small boulders or depressions. The legs are constructed from a lightweight aluminum alloy with shock-absorbing joints that dissipate kinetic energy during landing. This design minimizes the risk of structural damage and prevents the instruments onboard from being affected by impact forces.

  To ensure long-term operational stability, the landing legs incorporate self-leveling mechanisms that adjust the platform's inclination after landing, providing a stable base for scientific operations. This feature is particularly crucial near the Mare Tranquillitatis pit, where the terrain is rugged and uneven. Thermal-resistant coatings on the legs protect against the extreme lunar environment, while dust-tolerant seals prevent abrasive lunar regolith from interfering with joint movements. The telescopic design allows the legs to retract partially during transit, saving space and ensuring the compactness of the lander.

  TRL: 6, as the self-leveling mechanism is a novel feature requiring additional testing in analog environments.

- ***Instrument Mounts and Payload Bay***

  The lander features custom-designed mounts and a payload bay tailored to securely house its suite of scientific instruments, including neutron spectrometers, seismometers, and radiation dosimeters. These mounts are engineered to minimize vibration and thermal interference, ensuring the accuracy and reliability of data collected by sensitive instruments. The mounting system includes modular brackets that can be adjusted to accommodate varying instrument dimensions and configurations. This adaptability allows for upgrades or changes in the payload without requiring major modifications to the lander structure.

  The payload bay is equipped with thermal insulation to shield instruments from extreme temperature fluctuations and direct solar radiation. Active thermal control elements, such as heaters and radiators, are integrated into the mounts to maintain optimal operating temperatures for critical systems. Additionally, dust shields and protective enclosures safeguard the instruments from lunar regolith, which could otherwise impair their functionality. The payload bay also features

quick-access panels, enabling astronauts or automated systems to easily interact with the instruments for maintenance or upgrades during future missions.

TRL: 7, leveraging proven designs from previous lunar exploration missions while integrating mission-specific customizations.

- ***Overall Lander TRL***

  The overall Technology Readiness Level (TRL) of the lander is assessed at 6, driven by the landing leg subsystem, which includes novel features such as the self-leveling mechanism. While other subsystems, such as the chassis and instrument mounts, have reached TRL 7 due to their reliance on proven designs, the innovative elements of the landing legs require additional testing in simulated lunar conditions to validate performance. Future validation tests in analog environments will be conducted to increase the overall subsystem maturity before launch.

  This comprehensive design ensures that the Main Lunar Lander can withstand the challenges of the lunar environment, provide a stable platform for scientific exploration, and support the mission's ambitious objectives with a robust and reliable mechanical subsystem.

Exploration Rover Mechanical Sub-Assemblies

The Exploration Rover for the Mare Tranquillitatis mission is a cornerstone of the mechanical subsystem, engineered to traverse the Moon's harsh terrain while performing critical scientific investigations. Its design prioritizes durability, mobility, and adaptability, enabling efficient exploration of the pit's surface and subsurface features. Below is a comprehensive breakdown of the rover's mechanical sub-assemblies.

Chassis and Frame

- Description: The chassis is crafted from a titanium alloy, chosen for its excellent strength-to-weight ratio, corrosion resistance, and ability to withstand micrometeoroid impacts. This frame serves as the backbone of the rover, housing internal components such as the avionics, power systems, and scientific instruments. The chassis is compact in its stowed configuration for launch but expands upon deployment to provide a stable and functional structure for surface operations.
- Features:
  - Modular design with integrated mounting points for scientific payloads such as LiDAR, ground-penetrating radar (GPR), and cameras.
  - Reinforced structural elements to protect sensitive electronics and maintain

alignment under stress from uneven terrain.
    - Lightweight but durable enough to endure high-impact forces during lunar pit entry maneuvers.
- Environmental Factors: The chassis is designed for operations in the Moon's low-gravity environment (1/6th of Earth's gravity), where it must remain stable on uneven surfaces while supporting scientific instrumentation. Additionally, it is optimized for resistance to radiation and lunar dust infiltration.
- Technology Readiness Level (TRL): 7. This design is based on successful heritage from prior rover missions with minor modifications for mission-specific requirements.

Mobility System

- Description: The mobility system comprises six independently motorized wheels equipped with adaptive suspension, designed to overcome the challenges of traversing the rugged terrain of Mare Tranquillitatis. The wheels are constructed from lightweight, high-durability materials, with treads designed to grip both loose regolith and solid rock surfaces effectively. The adaptive suspension provides superior stability and allows the rover to navigate inclines of up to 30°.
- Features:
    - Each wheel has a dedicated electric motor, enabling precise control and redundancy in case of individual wheel failure.
    - Suspension system allows for differential elevation adjustment, ensuring that all wheels maintain contact with the ground on uneven terrain.
    - Dust-tolerant seals and bearings to maintain mobility in abrasive lunar conditions.
- Environmental Factors: The system is built to handle the Moon's steep slopes and loose soil, where traction and balance are critical. High resistance to mechanical wear from prolonged exposure to sharp lunar regolith ensures long-term reliability.
- Technology Readiness Level (TRL): 6. Although the system incorporates proven components, extended validation is required to ensure its effectiveness in deep lunar pits with challenging terrains.

Deployment Mechanism

- Description: The rover features an actuator-based deployment mechanism that facilitates its transition from a compact launch configuration to its operational state on the lunar surface. This system is crucial for minimizing the rover's footprint during transit and ensuring rapid readiness upon landing.
- Features:
    - Actuated hinges and locking systems that expand the rover's chassis and deploy its wheels, antennas, and instrument mounts.
    - Integrated fail-safe mechanisms to prevent misalignment or partial

deployment.
    - Autonomous calibration system that verifies full deployment and adjusts for minor deviations caused by transit stresses.
- Environmental Factors: The deployment mechanism is engineered for precise operation in extreme lunar temperatures and under variable gravitational loads. High-resilience materials and redundant actuators ensure reliability during critical transitions from dormant to active states.
- Technology Readiness Level (TRL): 5. The system includes new integrations specifically designed for this mission, necessitating further testing to validate its performance in operational environments.

Instrument Mounts and Robotic Arms

- Description: The rover is equipped with articulated instrument mounts and robotic arms designed to support its suite of scientific tools. These mounts and arms ensure that instruments maintain optimal alignment and positioning for data collection, even under dynamic conditions.
- Features:
    - Vibration-isolated mounts for sensors such as the laser-induced breakdown spectroscopy (LIBS) system and cameras.
    - Robotic arms with multi-degree-of-freedom joints, capable of precise positioning for sampling tasks or instrument calibration.
    - Thermal shielding to protect instruments from temperature extremes during operations in shadowed regions or under direct sunlight.
    - Tool exchange capability for swapping between sampling tools, such as drills or scoops, based on mission needs.
- Environmental Factors: These components are designed to function in shadowed regions within the Mare Tranquillitatis pit, where temperatures can plummet and dust accumulation poses a risk. Thermal insulation and low-power heating elements protect moving parts and sensitive instruments.
- Technology Readiness Level (TRL): 7. The mounts and arms are based on established designs with proven functionality in space missions, adapted for the specific demands of lunar exploration.

Overall Rover Technology Readiness Level (TRL)

The Exploration Rover's mechanical subsystem has an overall TRL of 5, driven by the deployment mechanism’s current development status. Upcoming testing phases, including simulations in analog environments and field trials, are planned to advance the subsystem’s maturity to a flight-ready state. This systematic approach ensures that all components meet the mission’s stringent performance and reliability requirements.

Subsystem-Level Considerations

Extreme Temperature Management

The lunar environment presents one of the most extreme thermal challenges faced by any mission, with temperatures ranging from -150°C during the lunar night to 100°C during the day. To ensure continuous functionality, the mechanical subsystem integrates a combination of passive and active thermal management solutions.

- Multi-Layer Insulation (MLI) serves as the primary defense against temperature extremes. MLI, composed of alternating layers of reflective films and spacers, minimizes heat transfer through radiation and conduction, significantly reducing thermal losses during the frigid lunar night and preventing overheating during prolonged sun exposure. The MLI is tailored to encapsulate both the Main Lunar Lander and the Exploration Rover, protecting critical subsystems like electronics and power storage units.
- Active thermal control supplements MLI, particularly in regions exposed to direct solar radiation or permanently shadowed areas of the Mare Tranquillitatis pit. These systems include resistive heaters placed near sensitive components, such as scientific instruments and batteries, ensuring optimal operating temperatures are maintained. Advanced thermal sensors provide real-time feedback, enabling dynamic adjustment of heating elements to conserve power and maximize efficiency.
- Radiators are used to dissipate heat generated internally by the lander's instruments and rover's mobility systems, preventing overheating during high-energy activities. These radiators are designed with low-emissivity coatings to optimize heat rejection in a vacuum environment.

This layered approach ensures mission-critical systems remain operational across diverse temperature profiles, supporting extended mission durations and nighttime operations.

Dust Mitigation

Lunar regolith, composed of fine, sharp-edged particles, poses a significant risk to mechanical systems. To combat the abrasive and adhesive nature of this dust, the mission employs several mitigation strategies:

- Sealing mechanisms protect moving parts and critical subsystems, such as the rover’s mobility system and the lander’s deployment components. Dust-tolerant seals and filters prevent the ingress of regolith into joints, bearings, and actuators, ensuring smooth operations.
- Dust-resistant materials have been selected for exposed surfaces. Titanium alloys and specially coated polymers are used for high-wear components, minimizing

erosion and reducing the risk of contamination.

- Electrostatic control reduces dust adhesion on surfaces prone to regolith accumulation, such as solar panels and instrument mounts. Static charge dissipation systems prevent dust from clinging to critical areas, maintaining efficiency and accuracy.
- Maintenance protocols, including self-cleaning mechanisms and redundant systems, provide additional layers of resilience. For example, the rover's instruments feature vibration modes to dislodge accumulated dust, while the lander includes redundant hinges and actuators to ensure reliability.

These measures ensure the mechanical subsystem's durability, even in the harsh, dust-laden environment of the Moon.

Structural Integrity and Compactness

The structural design of the lander and rover balances robustness with compactness, adhering to strict mass constraints of ≤350 kg for the entire spacecraft. Key considerations include:

- Lightweight Materials: High-strength titanium alloys are extensively used for chassis and frames due to their excellent strength-to-weight ratio. These materials provide durability while allowing additional payload capacity for scientific instruments and power systems.
- Shock-absorbing elements are integrated into the lander's landing legs and the rover's suspension system, protecting subsystems from the high impact forces of lunar landing and traversal over rocky terrain.
- Compact Design: Both the lander and rover are designed to fit within the constraints of the launch vehicle's payload fairing. The rover's chassis is foldable, expanding to its full operational size upon deployment. The lander's modular design allows efficient packing while ensuring ease of assembly and operation on the lunar surface.
- Load Testing: Components are rigorously tested under simulated conditions to ensure structural integrity during launch vibrations, lunar landings, and operations.

By prioritizing these features, the mechanical subsystem achieves operational readiness while adhering to payload and mission constraints.

Technology Readiness Level (TRL)

The overall TRL of the mechanical subsystem is currently at 5, with the rover's

deployment mechanism driving this level. The deployment system is a novel design requiring further validation under operational conditions to ensure reliability. Key steps planned to advance the subsystem maturity include:

- Analog testing in simulated lunar environments, such as volcanic fields or thermal-vacuum chambers, to validate subsystem functionality under representative conditions.
- Integration testing to evaluate the interaction between mechanical, thermal, and electrical systems, ensuring seamless performance.
- Iterative design improvements, informed by testing results, to address identified weaknesses and optimize system performance.

The mechanical subsystem's comprehensive design and testing plan ensure it can meet the mission’s objectives while overcoming the challenges posed by the lunar environment. By advancing the TRL through planned activities, the subsystem will be fully mission-ready, enabling groundbreaking exploration of Mare Tranquillitatis.

2.1.1.3. Mechanical Subsystem Recovery and Redundancy Plans

1. Mechanical Subsystem

Redundancy

The mechanical subsystem, which provides both the structural backbone and the means of movement, generally does not rely on full duplication of parts. Instead, key structural elements—such as the chassis, load-bearing beams, joints, and arm assemblies—are designed with high safety margins. In other words, they are built stronger than they need to be for normal conditions, so that even if unexpected stresses occur, these components are unlikely to fail. This approach avoids adding heavy, expensive duplicates of large structural parts.

For mobility, a partial form of redundancy is woven into the design. By using multiple wheels, for example, the system can still move if one or two fail. The control software can compensate for a damaged or stuck wheel by shifting the load and steering through the remaining functioning wheels. This setup strikes a good balance, allowing the machine to keep working without carrying unnecessary backup components that would add weight and complexity.

Recovery

When it comes to handling mechanical problems that do arise, the emphasis is on flexible responses rather than hardwired backups. If a wheel begins to falter or a joint starts to stiffen, the system's control software steps in. By monitoring the machine's performance—keeping an eye on vibrations, the smoothness of joint motion, and how well each wheel is turning—this software can adjust motor outputs, redistribute weight, and find alternate paths forward. If the front-left wheel is dragging, for instance, the system might rely more on the other wheels, or alter its path to reduce strain on that component.

In addition, ongoing measurement of shocks and structural vibrations can signal the start of a problem before it becomes a critical failure. If the data show that a particular joint is experiencing unusual stress, the system can try slower movements or gentler turns to preserve its integrity. Instead of having an entire spare arm ready to bolt on, the machine adapts in real time, using intelligent controls and carefully monitored performance data. Through this approach, minor mechanical setbacks rarely spell disaster, because the system can continually tweak its own operation to maintain functionality without carrying around a warehouse of backup parts.

2.1.1.4. Mechanical Subsystem Manufacturing and Procurement Plans

1. Manufacturing and Procurement Plans

Essential to the success of the Mare Tranquillitatis pit mission are well-established NASA industry contractors and suppliers. These include firms with proven flight experience, strong contract performance records, and NASA itself. By using NASA facilities for both component manufacturing and testing, as well as for verification and validation of individual parts and subsystems, the program ensures that every element meets the demanding requirements of lunar exploration. Detailed information—such as primary suppliers, testing centers, lead times, and other specifics—can be found in Table 3.

For the Mechanical Subsystem and the Command and Data Handling (CDH) system, Northrop Grumman has been selected as the principal supplier. With a long history of delivering reliable mechanical structures for space missions, Northrop Grumman is well-known for its lightweight yet high-strength aerospace components. These qualities are critical to minimizing total spacecraft mass while ensuring that all hardware can withstand the intense conditions of a mission to the Mare Tranquillitatis pit. Northrop Grumman has a strong record of supporting missions to the International Space Station (ISS) and has contributed to Mars rovers and lunar landers, further demonstrating its capability in engineering spacecraft components that remain durable and efficient under extreme environments.

SpaceX has been chosen as a secondary supplier of mechanical systems. Their broad experience includes payload fairings, structural assemblies, and the development of reusable spacecraft. Their work with NASA missions—such as resupplying the ISS and deploying satellites—has proven their ability to deliver robust mechanical systems that meet strict aerospace standards. With their Falcon 9 rocket and uncrewed Dragon spacecraft operating from Kennedy Space Center’s Launch Complex 39A, SpaceX has repeatedly demonstrated reliability, efficiency, and engineering excellence. This proven track record underlines their suitability as a supplier for a challenging lunar mission.

Subsystem	Subassembly	Primary Component Supplier	Place of Subsystem Assembly and V&V	Testing Locations/ Centers	Estimated Lead Time (months)	Details
Mechanical	Chassis and frame	Northrop Grumman Corporation	Northrop Grumman Aerospace Systems, CA: They specialize in lightweight, durable spacecraft structures and has experience in lunar surface systems	Glenn Research Center	14	Lead time includes design, fabrication, and integration of chassis and wheel systems while considering the surface hazards of the pit and including material selection and testing phases.

	Drive System	Lockheed Martin Space Systems	Lockheed Martin Space Systems, CO: They have Expertise in robust mobility systems for challenging terrain, leveraging rover technologies.	Glenn Research Center	11	Includes the design and integration of the drive system, ensuring it can handle the lunar terrain.
	Deployment Mechanism	Maxar Technologies Inc	Maxar technologies CO: Maxar has a strong record in spacecraft mechanisms, particularly for deployment systems like solar panels and robotic arms.	Glenn Research Center	14	Includes customization and transfer to the research center

	Lander Frame and Base	Northrop Grumman Corporation	Northrop also has a proven track record in lander structure development for extreme environments .	Glenn Research Center	14	Includes customization and transfer to the research center
	Lander Legs & Stability	Space Exploration Technologies	SpaceX Headquarter s, TX: They are well known for robust lander designs, ensuring stability and durability on the lunar surface. Moreover, they have demonstrate d landing systems expertise through its Starship and Falcon 9 programs.	Glenn Research Center	12	Includes customization and transfer to the research center

	Sample Handling Platform	Honey Bee Robotics	Honeybee Robotics: Specializes in sample handling and robotic systems for space exploration.	Glenn Research Center	14	Includes customization and transfer to the research center
	Instrument Mounts and Payload Bay	Jet Propulsion Laboratory	JPL specializes in designing, integrating, and testing scientific payloads for planetary and lunar missions, such as those for Mars rovers and lunar landers. Standard off-the-shelf components integrated in-house to ensure compatibility with various	Glenn Research Center	14	Includes customization and transfer to the research center

			scientific instruments.			

Table 4: Mechanical Manufacturing and Procurement Plans

By using these suppliers, the mission can achieve a balance of proven expertise, reliability, and adherence to project timelines. Opting for backup suppliers may extend the project schedule slightly but ensures continuity and adaptability in procurement. Not having back up suppliers for some subsystems could halt progress on critical components, but also the quality and safety of the overall spacecraft.

2.1.1.5. Mechanical Subsystem Verification Plans

Essential to the success of the Mare Tranquillitatis pit mission are well-established NASA industry contractors and suppliers. These include firms with proven flight experience, strong contract performance records, and NASA itself. By using NASA facilities for both component manufacturing and testing, as well as for verification and validation of individual parts and subsystems, the program ensures that every element meets the demanding requirements of lunar exploration. Detailed information—such as primary suppliers, testing centers, lead times, and other specifics—can be found in Table 3.

For the Mechanical Subsystem and the Command and Data Handling (CDH) system, Northrop Grumman has been selected as the principal supplier. With a long history of delivering reliable mechanical structures for space missions, Northrop Grumman is well-known for its lightweight yet high-strength aerospace components. These qualities are critical to minimizing total spacecraft mass while ensuring that all hardware can withstand the intense conditions of a mission to the Mare Tranquillitatis pit. Northrop Grumman has a strong record of supporting missions to the International Space Station (ISS) and has contributed to Mars rovers and lunar landers, further demonstrating its capability in engineering spacecraft components that remain durable and efficient under extreme environments.

SpaceX has been chosen as a secondary supplier of mechanical systems. Their broad experience includes payload fairings, structural assemblies, and the development of reusable spacecraft. Their work with NASA missions—such as resupplying the ISS and deploying satellites—has proven their ability to deliver robust mechanical systems that meet strict aerospace standards. With their Falcon 9 rocket and uncrewed Dragon spacecraft operating from Kennedy Space Center’s Launch Complex 39A, SpaceX has repeatedly demonstrated reliability, efficiency, and engineering excellence. This proven track record underlines their suitability as a supplier for a challenging lunar mission.

For the Power Subsystem, Maxar Technologies Inc. is the primary provider. They have extensive experience in designing high-efficiency solar arrays and power systems, and their contributions to NASA’s OSIRIS-REx mission highlight their technical expertise. Among their accomplishments is the development of the OSIRIS-REx Laser Altimeter (OLA)—a high-resolution scanning LIDAR instrument used to map the asteroid Bennu. This sensor played

a critical role in identifying a suitable sample collection site, showcasing Maxar's capacity for creating specialized, mission-critical hardware. Their skills in crafting dependable power generation solutions are perfectly aligned with the demands of sustained operations in a lunar environment.

As a backup supplier for the power subsystem, Raytheon Technologies Corporation offers significant experience in power generation and energy management, especially in demanding military and space applications. Raytheon's contributions include power systems and electrical solutions for NASA's spacecraft, such as the Mars Science Laboratory's Curiosity rover and the Parker Solar Probe. Notably, they developed Curiosity's descent stage, a critical element that ensured the rover's successful landing on Mars. Raytheon's history of delivering rugged, reliable technologies positions them as a solid alternate source for backup power systems.

Lockheed Martin serves as a backup supplier for the CDH system. They bring decades of experience in avionics and spacecraft integration for NASA programs. One of their landmark efforts is the Orion spacecraft, which includes advanced avionics and CDH systems. Orion's CDH capabilities are central to its navigation, communication, and control during deep space missions to the Moon and beyond. Lockheed Martin's involvement with Orion and other complex projects highlights their proficiency in developing sophisticated and dependable electronic systems, making them well-suited as a contingency supplier for the CDH subsystem.

2.1.2. Power Subsystem Overview

The power subsystem is critical to the functionality of the team’s rover spacecraft (rover and lander), ensuring reliable energy delivery to support operational phase cycles. The power subsystem integrates JinkoSolar Tiger Pro solar panels, Tadiran LiSOCL2 batteries, the EXA COLOSSUS Power Control and Distribution Unit (PCDU), and TL7700-SEP Radiation Hardened Supply Voltage Supervisor to ensure efficient power generation, storage, distribution, and protection throughout the mission. Each component is selected for its high performance, reliability, and suitability for the mission's challenging conditions.

Each phase of the mission presents a unique set of power requirements, necessitating careful consideration and allocation of energy resources to ensure that all subsystems operate efficiently and effectively during their designated tasks.

1. **Pit Analysis**

During the Pit Analysis phase, the spacecraft performs initial assessment and diagnostic tasks to gather data from the surrounding environment, assess system functionality, and monitor health parameters. This phase requires moderate power allocation to analytical instruments, sensors, and onboard diagnostic systems that process environmental data, temperature, and structural integrity. The voltage supervisors play a key role by protecting the spacecraft's electrical systems from voltage spikes or surges, which could occur due to sudden system startup or electrical fluctuations. Additionally, battery safety is critical during this phase to prevent overcharging or deep discharge. To ensure consistent performance and minimize fluctuations, power is primarily drawn from the batteries, which provide stable energy even during periods of reduced sunlight.

2. **Rover Deployment**:

The Rover Deployment phase is one of the most power-intensive stages of the mission since here, the spacecraft must deploy the rover and activate its associated systems. These include actuators that unfold and position the rover, navigation systems that establish the rover's path, and thermal control systems that manage temperature fluctuations in the rover's command and data handling, and instruments. Each of these subsystems requires a significant amount of energy, particularly the actuators and navigation systems, which involve complex mechanical movements and communication with the spacecraft. Given the high demand for power during this phase, both solar panels and battery packs work in tandem to meet the peak energy needs. The solar panels provide a steady energy supply from sunlight, while the batteries supplement this power during moments of higher demand or when solar energy is insufficient, such as in shadowed areas or during rover movements that draw substantial current. This dual-source power management ensures that the rover is deployed successfully and all its systems remain operational without power constraints.

3. **Entry, Descent, and Landing (EDL)**:

This is the most critical and power-demanding segment of the mission. It requires precise power management to ensure that critical subsystems, such as thrusters, guidance systems, and sensors, function optimally during the spacecraft's descent to the surface. The thrusters are essential for controlling the descent speed and ensuring a safe landing, while guidance and navigation systems ensure that the spacecraft stays on the correct trajectory. This phase's power consumption is intense because of the continuous operation of high-power systems that require uninterrupted energy. Battery power takes precedence during EDL, as the spacecraft may pass through phases where solar generation is minimal or absent, such as during atmospheric entry or in the shadow of the moon's surface. The batteries are engineered to provide the necessary high peak power without risking system failure. Therefore, ensuring that the batteries are charged and functioning efficiently before entering this phase of the mission.

4. **Moon Transit**:

During the Moon Transit phase, the Mare Tranquillitatis spacecraft travels from one location to another on the moon's surface or across lunar orbit. During this phase, the energy demand is moderate, as most subsystems require continuous, steady power for navigation, communication, and monitoring. The spacecraft relies heavily on the solar panels for energy generation during transit. These panels provide a continuous power supply as long as the spacecraft is in sunlight, making them the ideal source of power for this extended phase. However, the spacecraft may enter periods of shadow, particularly when transitioning between lunar locations. During these times, the spacecraft must rely on its batteries to ensure that no systems are compromised. Batteries are used more sparingly in this phase, but they must remain available to support critical maneuvers, such as attitude control or course correction, when solar power cannot be relied upon. The spacecraft's energy management system dynamically adjusts the use of power to balance between using the solar panels and battery reserves, ensuring the spacecraft can maintain its operations smoothly throughout the transit.

5. **Data Transmission**:

The Data Transmission phase is **vital** for the success of the mission, as it involves sending the collected data back to Earth for analysis. This process requires a substantial amount of energy, particularly for the communication systems that transmit high-volume scientific data and system status updates. The energy consumption during this phase peaks when data transmission rates are high, as the communication system requires significant power to operate the onboard transmitters and antennas. While the solar panels provide the bulk of the energy required during this phase, battery reserves are used to handle power surges or to sustain the communication systems during periods when the spacecraft might be in partial or full shadow, or when transmission needs exceed the available solar generation. The spacecraft's power management system will prioritize data transmission and ensure that adequate power is available, diverting energy from non-essential systems to maintain the integrity of the communication subsystem. The batteries act as a buffer, ensuring that the spacecraft maintains its power balance and does not experience power shortages that could disrupt the transmission or cause data loss.

2.1.2.1. Power Subsystem Requirements

The Mare Tranquillitatis pit, known for its high depth and irregular structure, presents various challenges for power storage and generation due to its high temperatures (very hot during the day to extremely cold at night). The surface hazards with partially buried impact craters and basaltic infill are also conditions to generate and distribute power in the spacecraft.

The power subsystem is a critical component of any spacecraft, responsible for generating, storing, and distributing electrical power to all other subsystems to enable the spacecraft to operate and perform the required tasks. This document labeled the power requirements and constraints needed to ensure the science objectives are met. The power subsystem must provide sufficient power to all instruments during operational phases.

The Power Generation is responsible for converting external sources of energy to electrical power to ensure the system is always working. The Power Storage stores the energy in the battery packs in case the power generation is not sufficient. The Power Distribution allocates the electrical power to all the subsystems. The Safety subsystem will protect the spacecraft from electrical hazards to prevent the overheating of the components. Given the high temperatures on Mare Tranquillitatis Pit, the preferred power source is lithium battery packs, and the power system must be designed to perform reliably in low-temperature environments. Since the distance from the Mare Tranquillitatis Pit to the sun is advantageous for solar power, solar panels are good for their renewable source.

Requirement ID	**Subsystem**	**Requirement**	**Parent Requirement**	**Child requirement**	**Subassembly**	**Verification Method**	**Met/ Not met**
EE-1	Power Generation	Due to the depth of the pit, Solar Panels must	Science Data Collection	Instrument Functionalit	Solar Panels	Test	Met

		be strategically placed to effectively capture the sunlight and restore it to the instruments, by adhering to the mass constraint.		y and Power Supply			
EE-2	Power Generation	The solar panels must generate a minimum of 500 Watts of power under standard test conditions and should maintain at least 90% of their power output after 4 years of operation in space.	Science Data Collection	Instrument Functionality and Power Supply	Solar Panels	Analysis	Met
EE-3	Power Storage	The batteries must provide a minimum of 730 Wh/kg of energy density and have a cycle life of at least 2000 cycles.	Science Data Collection	Instrument Functionality and Power Supply	Battery Pack	Analysis	Met
EE-4	Power Storage	The batteries must operate in temperatures ranging from -80°C to 125°C.	Science Data Collection	Instrument Functionality and	Battery Pack	Test	Met

				Power Supply			
EE-5	Power Distribution	PCDU must distribute power to all subsystems to ensure the stability of the power supply	Science Data Collection	Instrument Functionality and Power Supply	Power Control and Distribution Unit (PCDU)	Analysis	Met
EE-6	Power Distribution	The PCDU must handle up to 1.6 kW of total power	Science Data Collection	Instrument Functionality and Power Supply	Power Monitoring	Analysis	Met
EE-7	Safety	The voltage supervisor must ensure that voltage does not drop below safe operating thresholds	Mission Objective	Instrument Functionality and Power Supply	Circuit Protection	Inspection	Met
EE-8	Safety	The voltage supervisor must operate in temperatures ranging from -55°C to +180°C without failure	Mission Objective	Instrument Functionality and Power Supply	Circuit Protection	Test	Met

Table 5 : Power Subsystem Requirements

These requirements and constraints serve as the foundation for designing and choosing the power system components and technologies for the Mare Tranquillitatis mission. The following sections of this report will explore the detailed design, components, and strategies needed to meet these requirements and ensure the spacecraft is effectively powered throughout its mission to Mare Tranquillitatis.

2.1.2.2. Power Sub-Assembly Overview

The power subsystem is critical to the successful execution of the mission, providing continuous, reliable energy throughout the spacecraft's operational phases. This subsystem is composed of Power Generation, Power Storage, Power Distribution and Control, and Power Safety components. Each component is carefully selected based on its reliability, efficiency, and compatibility with the spacecraft's operational requirements and environmental conditions. Below is a thorough description of each sub-assembly, with a focus on how it contributes to the overall power system.

- **Power Generation**

The JinkoSolar Tiger Pro 72HC-TV is chosen as the primary power generation source for the spacecraft. These panels are highly efficient, capable of generating up to 545 Watts of power, measure 2274mm × 1134mm × 35mm and weigh 28.9 kg , making them suitable for large-scale installations, and reliable for the spacecraft requirement.[1] . They have a low annual power degradation rate of 0.45%, ensuring a longer lifetime power yield. With a temperature coefficient of -0.35%/°C, a reliable performance in high-temperature environments and is efficient at up to 25.45%, which is among the highest in the industry. The decision on which solar cell or panel to use will consider efficiency, size, weight, and operational temperature range. It must also be compatible with the spacecraft's systems and align with our mission objectives. ensuring that the spacecraft can generate substantial power even in challenging environments like the lunar surface or during transit.[2] Solar panels are a well-established technology in space missions, providing a reliable source of power for spacecraft. The Mare Tranquillitatis Pit presents unique environmental challenges, such as irregular light conditions, varying angles of sunlight, and extended periods of shadow during lunar nights. While the JinkoSolar Tiger Pro panels have not yet been used for lunar operations, their high efficiency (25.45%) and robust performance in space environments make them a strong candidate for this mission. Based on these factors, the TRL for Power Generation would be assessed as TRL 6, indicating that the technology has been demonstrated in a relevant environment but requires further validation under actual lunar conditions. Nonetheless, the technological advancements and trust in JinkoSolar's products make the panels a highly promising choice for the mission.

- **Power Storage**

Battery technology, particularly for space missions, demands exceptional reliability and longevity. For power storage, team 3 will utilize Tadiran LiSOCL2 batteries, which offer high energy density and exceptional performance in extreme environments. Tadiran LiSOCL2 batteries, which offer high energy density and

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[1] JinkoSolar, *Tiger Pro 72HC-TV 525-545 Watt*, accessed October 24th 2024, https://www.jinkosolar.com/uploads/JKM525-545M-72HL4-TV-F1-EN.pdf.

[2] JinkoSolar, *Tiger Pro 72HC-TV 525-545 Watt*, accessed October 24th 2024.

exceptional performance in extreme environments. These batteries provide an energy density of up to 730 Wh/kg, significantly surpassing traditional lithium-ion batteries. They are designed to operate effectively in temperatures ranging from -80°C to 125°C, which is particularly beneficial for lunar and deep-space missions where temperature variations can be extreme.[3] However, a key limitation of these batteries is that they are **non-rechargeable**, thus needing to be frequently replaced. Replacing them can be challenging and costly, and they lack detailed mass data.

The TRL of the Tadiran LiSOCL2 batteries is 7, as they are a proven technology used in numerous NASA and space missions.[4] Their reliability and robustness in harsh environmental conditions make them a reliable choice for power storage. The spacecraft will be designed with redundancy in mind, so even if some batteries degrade, others will continue to provide power without significant disruption. To mitigate the challenge of using non-rechargeable batteries, mission planners along with electrical engineers will carefully manage battery replacement schedules and ensure that battery management systems are in place to optimize power usage, monitor battery health, and prevent complete depletion.

- **Power Distribution and Monitoring**

The Power Control and Distribution Unit (PCDU) is responsible for managing the distribution of power across the spacecraft's subsystems, ensuring that each system receives the appropriate voltage and current levels. PCUs are commonly used in space-rated missions, providing stability and reliability even in the harsh conditions of space. For the Mare Tranquillitatis Pit mission, team 3 will use the EXA COLOSSUS PCDU, with its high power-to-weight ratio and efficiency in managing up to 1.6 kW. It integrates EPS, TITAN2 Batteries, and Solar Management including release and deploy control with a powerful MCU and 32GB onboard storage in a single unit from a minimum of 400W/hr to 8000 W/hr, from 8 to 56 power rails with power outputs as high as 150W per rail. The EXA COLOSSUS PCDU also ensures that the spacecraft's subsystems are powered in a prioritized and efficient manner. For instance, during Rover Deployment, the unit will prioritize energy to actuators and thermal systems, while ensuring that the spacecraft's communication systems and scientific instruments receive power during data collection and data transmission phases. Moreover, it supports nominal to 4C charge/discharge rates, allowing for high-power operations during peak energy demand phases, such as data transmission or mobility tasks, features Redundant RLCL (Relay-Latched Current Limiters) in each cell to enhance fault tolerance and maintain system stability, and it tolerates space radiation for up to 2 years in LEO and 4 years with shielding, while resisting Single-Event Upsets (SEU) up to 70 MeV×cm²/mg, addressing the high-radiation risks posed by Mare Tranquillitatis.[5] This tolerance ensures uninterrupted functionality in the high-radiation environment of Mare Tranquillitatis and

[3] Tadiran Batteries, "LiSOCL2 LTC Batteries," accessed December 7, 2024, https://tadiranbatteries.de/lisocl2-ltc-batteries/.

[4] Barbara McKissock, Patricia Loyselle, and Elisa Vogel, *Guidelines on Lithium-ion Battery Use in Space Applications* (Glenn Research Center, Cleveland, 2009).

[5] *COLOSSUS PCDU Family: 400 to 8000 W/hr*, Satsearch, accessed November, 16, 2024, https://satsearch.co/products/exa-colossus-pcdu-family-400-to-8000-w-hr.

that cells are heated to optimal levels, allowing for reliable operation during the frigid lunar nights and the intense heat of lunar days. The EXA COLOSSUS is built with robust Aluminum T6061 enclosures for structural integrity, ensuring resilience against dust and micrometeoroid impacts common in lunar terrain, and protecting against dust infiltration and micrometeoroid impacts, safeguarding the system from physical hazards.

The TRL of the EXA COLOSSUS PCDU is 8, as it has been successfully used in several operational missions. It is a reliable solution for spacecraft power distribution and is known for its robustness and efficiency in space conditions.

With its low power dissipation and fault tolerance, the EXA PCDU can efficiently manage energy storage and distribution, critical for periods of low solar energy during extended shadows. The EXA COLOSSUS PCDU's advanced features make it an ideal solution for powering missions in the demanding and unpredictable environment of the Mare Tranquillitatis Pit. Its combination of efficiency, safety, and resilience ensures reliable energy management for sustained exploration.

- **Safety (Circuit Protection)**

The TL7700-SEP Radiation Hardened Supply Voltage Supervisor is a highly reliable component designed to monitor and regulate the voltage levels of spacecraft power systems, ensuring that all systems operate within safe voltage limits. This device is critical for detecting undervoltage conditions, providing reset functions, and preventing system failures that could otherwise disrupt the spacecraft's power integrity. The TL7700-SEP's design is tailored for space applications, offering enhanced resilience to space-specific environmental factors such as radiation and extreme temperature conditions. The TL7700-SEP is manufactured using Space Enhanced Plastic (SEP), making it resistant to the damaging effects of radiation, which is a key requirement for any component operating in space. It provides protection against voltage spikes, ensuring that the spacecraft's power systems remain within safe operating conditions despite fluctuations in voltage. This characteristic is especially crucial for missions operating in challenging environments, such as the Mare Tranquillitatis Pit, where the spacecraft will experience highly variable lighting conditions, temperature extremes, and elevated radiation exposure. The bipolar integrated circuit features an operating temperature range from -55°C to +125°C, making it ideal for the harsh lunar environment.[6] This is particularly important for the mission to Mare Tranquillitatis, where the spacecraft will experience both extreme heat during the lunar day and extreme cold during the night. The TL7700-SEP helps maintain power stability by providing monitoring functions for the spacecraft's power subsystems during these challenging conditions.

The Technology Readiness Level (TRL) of the TL7700-SEP is TRL 7. This level indicates that the device has been demonstrated in a relevant environment, but it still requires further testing under mission-specific conditions, such as those encountered at

[6] *TL7700-SEP Radiation Hardened Supply-Voltage Supervisor in Space Enhanced Plastic*, Satsearch, accessed December 6, 2024, https://satsearch.co/products/texas-instruments-radiation-hardened-supply-voltage-supervisor-in-space-enhanced-plastic.

Mare Tranquillitatis. The TL7700-SEP has already been proven in various space applications, but its full integration into the power subsystem for a lunar mission requires validation through operational testing in the lunar environment. Given its radiation-hardened design and ability to withstand extreme temperatures, the TL7700-SEP is well-suited for the lunar mission. It ensures the spacecraft’s power subsystem remains protected from voltage irregularities and failures, supporting the longevity and success of the mission. Additionally, the device will be integrated into the spacecraft's broader safety architecture, which includes thermal control and fault-tolerant systems to further enhance the spacecraft’s overall safety and reliability.

In summary, the TL7700-SEP provides vital protection against undervoltage and power anomalies in space missions, making it an essential component for ensuring the spacecraft's power integrity. The combination of its radiation tolerance, wide temperature range, and proven performance in space applications make it a reliable choice for the mission to the Mare Tranquillitatis Pit.

The table below summarizes the key subassemblies within the spacecraft's power subsystem, highlighting their critical features and Technology Readiness Levels (TRLs). Each subassembly is selected to meet the mission's specific requirements, ensuring reliable performance in the harsh environment of the Mare Tranquillitatis Pit.

Subsystem	**Subassembly**	**Key Features**	**TRL**
Power Generation	JinkoSolar Tiger Pro Solar Panels	- Efficiency: 25.45% - Power Output: Up to 545 W - Size: 2274mm × 1134mm × 35mm - Weight: 28.9 kg - Annual degradation: 0.45% - Temperature coefficient: -0.35%/°C - Proven robust performance for harsh environments	6
Power Storage	Tadiran LiSOCL2 Batteries	- Energy Density: Up to 730 Wh/kg - Temperature range: -80°C to 125°C - High reliability in extreme conditions - Non-rechargeable with robust longevity	7
Power Distribution	EXA COLOSSUS PCDU	- Power management up to 1.6 kW - Supports 8-56 power rails with outputs up to 150W per rail - Integrated energy storage and solar management - Fault-tolerant with RLCL protection - Radiation tolerance up to 70 MeV×cm²/mg	8

		- Resilient Aluminum T6061 enclosure	
Safety (Circuit Protection)	TL7700-SEP Voltage Supervisor	- Radiation-hardened for space use - Undervoltage protection and reset functions - Operating range: -55°C to +125°C - Space Enhanced Plastic (SEP) construction - Voltage monitoring and spike prevention	7
Total Subsystem TRL		The Power subsystem TRL reflects the demonstrated reliability of the subassemblies in relevant environments such as the Mare Tranquillitatis pit.	**7**

Table 6: Power Sub-Assembly

These components work in synergy to deliver a stable and efficient power supply, a critical foundation for the success of the Mare Tranquillitatis exploration.

2.1.2.3. Power Subsystem Recovery and Redundancy Plans

- **JinkoSolar Tiger Pro Solar Panels**

Redundancy Plan

The JinkoSolar Tiger Pro solar panels are vital for power generation, but due to their size, cost, and space limitations, creating redundant arrays is not feasible. These panels are considered a Single Point Failure (SPF) in the power subsystem. In the event of a failure or degradation of one or more panels, the spacecraft will continue to operate using the remaining panels. However, during periods of low power generation, such as lunar nights or eclipse phases, the spacecraft will rely on Tadiran LiSOCL2 batteries for power. While solar panel failure is unlikely due to their robustness and radiation resistance, it remains a risk that must be mitigated through careful monitoring and power management strategies.

Recovery Plan

In the event of panel degradation or partial failure, the spacecraft will autonomously manage power consumption by prioritizing essential systems and shutting down non-critical components. The EXA COLOSSUS PCDU will reallocate power as necessary, using battery storage to ensure continued operation. Diagnostic software will continuously monitor panel performance and alert the onboard system to any anomalies. If necessary, the spacecraft can adjust panel orientation to optimize power generation during available sunlight periods, while the thermal control system will also protect the panels from excessive temperature fluctuations.

- **Tadiran LiSOCL2 Batteries**

Redundancy Plan

The Tadiran LiSOCL2 batteries are non-rechargeable and provide essential power storage. Multiple battery units will be included in the spacecraft design, ensuring that a failure in one unit does not result in the total loss of power. These batteries are not physically redundant, but their operational strategy relies on battery pack redundancy, meaning the spacecraft has sufficient capacity even if one unit fails. While non-rechargeable batteries are a potential risk, their high energy density and robustness make them a reliable choice for long-term energy storage during mission phases where solar power may not be sufficient.

Recovery Plan

If a battery degrades or fails, the spacecraft will switch to its backup battery pack or rely on solar power as a backup system. The PCDU will monitor the state of charge and

health of the batteries and will automatically reallocate power from the backup units when needed. In the event of a critical battery failure, the system will initiate an autonomous power conservation mode, ensuring that only essential systems are powered while non-essential systems are turned off. Battery degradation will be detected by telemetry systems, and the spacecraft will respond by adjusting its power management strategy accordingly.

- **EXA COLOSSUS Power Control and Distribution Unit (PCDU)**

Redundancy Plan
The EXA COLOSSUS PCDU is responsible for distributing power across the spacecraft. This unit includes internal redundancy, with dual power distribution paths that ensure continued functionality if one path fails. The system is designed to handle power surges, over-voltage, and thermal conditions that could potentially damage critical components. Redundancy in this context ensures that if a power distribution path is compromised, the secondary path will automatically take over to prevent system failure.

Recovery Plan
In case of a failure in the PCDU, the system will autonomously reroute power to critical systems through the redundant power distribution path. If both paths are compromised, the spacecraft will rely on its backup battery and solar power systems to maintain power to essential components. The thermal control system will also monitor the PCDU's temperature to prevent overheating, and if a thermal issue arises, the spacecraft will activate cooling mechanisms, such as increasing thermal dissipation or switching to a lower-power mode. Diagnostic software will continuously monitor the status of the PCDU and initiate recovery procedures if necessary, including soft reboots or isolation of damaged components.

- **TL7700-SEP Radiation Hardened Supply Voltage Supervisor**

Redundancy Plan
The TL7700-SEP radiation-hardened supply voltage supervisor monitors the spacecraft's power supply and ensures that voltage levels remain within the safe operating range. While there is no direct hardware redundancy for this component, its function is supported by voltage regulation systems in the PCDU, which can step in to manage voltage if an issue arises. Additionally, the power distribution system is designed with over-voltage protection and thermal shutdown features to protect sensitive components, ensuring that the system continues to function even if the voltage supervisor detects a failure.

Recovery Plan
If the TL7700-SEP detects an under-voltage or other fault condition, it will trigger

protective actions, including cutting power to affected components or switching to a backup power source. The PCDU will take over to manage the distribution of power in response to voltage abnormalities, while autonomous software algorithms will adjust the spacecraft's power management strategy. In the event of a supervisor failure, the voltage regulation systems and redundant power pathways will continue to monitor and maintain voltage levels to prevent damage to the spacecraft's systems. If a failure is detected, the system will isolate the faulty component and initiate autonomous recovery protocols, such as rerouting power and activating thermal protection systems to maintain system integrity.

2.1.2.4. Power Subsystem Manufacturing and Procurement Plans

Necessary to the manufacturing process and subsequent mission success of the Mare Tranquillitatis Pit mission are industry-leading suppliers and contractors with a proven record in space technologies. For the Power Subsystem, primary suppliers with specialized expertise in space-grade components have been selected for their reliability and performance in extreme environments. Backup suppliers, including NASA contractors, have been identified to ensure continuity in case of delays or unavailability from the primary suppliers. The power generation, storage, distribution, and protection components will be sourced from reputable companies, ensuring that the system is both reliable and resilient.

Subassembly	Primary Supplier	Justification	Primary Lead Time	Backup Supplier	Justification	Backup Lead Time
JinkoSolar Tiger Pro 72HC-TV	JinkoSolar	JinkoSolar was selected due to its expertise in high-efficiency solar panel manufacturing and proven track record in developing durable and reliable solar technologies. The Tiger Pro series offers industry-leading efficiency and robust design for challenging environments.	7 months (includes customization for space-grade panels)	Maxar Technologies Inc	Maxar Technologies CO is a leader in high-efficiency solar panel technology for space applications, having designed and manufactured systems for interplanetary missions and lunar projects.	12 months (Includes customization)

Tadiran LiSOCL2 Battery	Tadiran Batteries	Tadiran specializes in high-performance batteries tailored for extreme environments, such as space missions. Their LiSOCL2 batteries are reliable and widely used in aerospace applications.	8 months (based on battery size and quantity)	Raytheon Technologies Corporation	Raytheon has provided power systems and electrical solutions for NASA's spacecraft, such as on the Mars Science Laboratory (Curiosity Rover) and the Parker Solar Probe. They developed the rover's descent stage, which was crucial for the successful landing of Curiosity on Mars. Their work with electrical systems, along with their specialization in rugged, reliable technologies, makes them a strong contender for backup power systems.	9 months (due to customization and delivery requirements)
EXA COLOSSUS PCDU	EXA Corporation	EXA is selected for its cutting-edge COLOSSUS PCDU, which is highly efficient, lightweight, and radiation-tolerant. EXA specializes in power management	10 months (including testing and validation)	Airbus Defence and Space	Airbus provides robust PCUs designed for demanding space environments, capable of similar functionality but with longer lead times for customization.	Up to 14 months (due to additional design and validation requirements)

		solutions for space applications.				
TL7700-SEP Radiation Hardened Supply Voltage Supervisor	Texas Instruments	Texas Instruments is chosen for its proven expertise in designing radiation-hardened components. The TL7700-SEP is tailored for space applications and provides reliable circuit protection for spacecraft power systems.	8 months (includes testing and space adaptation)	Microchip Technology	Microchip offers radiation-hardened voltage supervisors (e.g., their RAD-HARD series) with similar performance characteristics.	8 months months (additional validation processes and lead time)

Table 7: Power Manufacturing and Procurement Plans

2.1.2.5. Power Subsystem Verification Plans

The following table outlines the verification methods and preliminary verification plans for the Power Subsystem of the Mare Tranquillitatis mission. These methods include analysis, inspection, and testing approaches to ensure the subsystem's components meet the requirements outlined in the mission's design specifications. Each verification method is selected based on the nature of the requirement and the most appropriate technique to demonstrate that the mission objectives will be met.

Req #	Requirement Summary	Verification Method	Rationale for Method	Preliminary Verification Plan
EE-1	The solar panels must generate a minimum of 500 Watts/hour of power under standard test conditions (STC).	Test	A test is required to ensure that the panels perform as expected under controlled conditions, replicating the environment in which the spacecraft will operate.	Perform performance testing in a thermal vacuum chamber to simulate the space environment and verify power generation capacity.
EE-2	The solar panels should maintain at least 90% of their power output after 4 years of operation in space.	Analysis	This is a longevity prediction and can be modeled based on manufacturer data and historical degradation rates.	The solar panels will be verified through performance analysis based on manufacturer data and the results of similar mission operations.Then power output's prediction loss over the course of the mission.
EE-3	The batteries must provide a minimum of 730 Wh/kg of energy density.	Analysis	Energy density is a critical factor that needs to be verified under actual use conditions to ensure the batteries provide adequate power storage capacity.	Energy density and cycle life will be verified through analysis of Tadiran's LiSOCL2 batteries and historical data from similar space missions.
EE-4	The batteries must operate in temperatures ranging	Test	This is a critical environmental condition that must be validated through direct testing to ensure the batteries can withstand	Conduct thermal cycling tests on the batteries in a vacuum chamber to simulate temperature variations during the mission.

	from -80°C to 125°C.		temperature extremes.	
EE-5	The Power Control and Distribution Unit (PCDU) must manage power distribution across all spacecraft subsystems effectively.	Analysis	A functional test of the PCDU is required to ensure that power is being distributed correctly and to verify its operational readiness.	The PCDU's power distribution capability will be validated by analyzing its ability to distribute power to all subsystems while ensuring stability and reliability under space conditions.
EE-6	The PCDU must handle up to 1.6 kW of total power and operate under radiation exposure up to 70 MeV×cm²/mg.	Analysis	The PCDU will be subjected to an analysis of its design and performance to ensure it can handle the full 1.6 kW of power load required by the mission.	The analysis will also include simulations of operational scenarios to assess any risks of overheating or power instability under extreme space conditions. It involves reviewing its design documents, performing simulations of the power distribution system under expected space conditions.
EE-7	The voltage supervisor (TL7700-SEP) must ensure that voltage does not drop below safe operating thresholds.	Inspection	Voltage regulation must be verified through functional testing to ensure that the voltage supervisor operates within the specified thresholds. It also ensures that the voltage supervisor is physically inspected for defects and functionality, especially under failure conditions such as power surges or under-voltage scenarios.	The supervisor will be visually and functionally tested for proper response to under-voltage and over-voltage conditions, ensuring it performs as intended under operational conditions.

EE-8	The voltage supervisor must operate in temperatures ranging from -55°C to +125°C without failure.	Test	The TL7700-SEP voltage supervisor will undergo inspection and thermal cycling tests to confirm it operates correctly across the required temperature range (-55°C to +180°C).	The voltage supervisor will be subjected to temperature extremes and then checked for functionality, ensuring it protects the spacecraft in both hot and cold environments in a Thermal vacuum chamber. After exposure, the supervisor's functionality will be inspected to ensure it can withstand these temperature extremes without failure, to ensure its reliability.

Table 8: Power Verification plans

The results of these verification plans will ensure that all components meet the necessary operational standards and contribute to the overall success of the Mare Tranquillitatis mission.

2.1.3. CDH Subsystem Overview

The Command and Data Handling subsystem manages data across spacecraft subsystems, ensuring reliable communication between spacecraft and ground control, with multiple subassemblies for specific mission functions.

Subassemblies, Components, and Technology Readiness Level (TRL)

1. Communications
    - Radio Frequency Communication Interface

This component establishes and maintains a communication link with an intermediary orbiting spacecraft, relaying mission data back to Earth. The requirement id is SC-2 and enables data observation by scientists on Earth, critical for mission analysis. It has the parent requirement of mission objective, verification test. The TRL for this would be 7 because it is a system prototype in an operational environment.

    - Communication interface Board

Provides a communication rate (bps) between subsystems and instruments, supporting the transfer of large data volumes during communication windows. The requirement id is SC-5 and supports data transmission across subsystems and instruments. It has the parent requirement of mission objective and verification test. The TRL would be 6 because it is a representative model tested in a relevant environment.

2. Data Handling
    - Data Processor

    Manages data from scientific instruments, ensuring it’s processed and sent to ground control. The requirement id is SC-2 and it ensures real-time processing of scientific and telemetry data. The parent verification is a mission objective and the verification is a demonstration. The TRL would be 6 because it is a representative model tested in a relevant environment but not in an operational environment.

3. Data Storage
    - Solid-State Recorder

      Stores collected data for transmission and inter-subsystem sharing. The recorder holds large volumes of scientific and telemetry data until it can be transmitted. The requirement id is SC- 5 and it is critical for data retention between transmissions.The parent requirement is mission objective and the verification is analysis. The TRL is 7 because it represents a major step in actual system prototype in an operational environment.

4. Software Architecture
    - Flight Software

      The software oversees spacecraft operations, manages data, and enables autonomous function and fault management to reduce dependency on ground commands. The requirement id is SC-2 and manages all spacecraft activities and allows autonomous operation. The parent requirement is a mission objective and the verification is a test. The TRL is 6 because it is a system model in a relevant environment.

The overall TRL of the CDH subsystem is 6 due to the lowest readiness level from the Software Architecture subassembly, Data Handling and Communications. To increase the TRL additional testing is required and has to verify the levels in the operational environment and different tests and evaluations.

2.1.3.1. CDH Subsystem Requirements

Requirement ID	Relevant Subsystems	Subassembly	Requirement	Rationale	Parent Req	Child Req	Verification Method	Requirement Met/ Not Met?
CDH-01	Communications	Radio Frequency Communication Interface	The spacecraft shall establish a communication link with an orbiting spacecraft to relay data back to Earth.	Allow for data to be observed by scientists on Earth for analysis	Science Data Collection	Instrument Functionality and Power Supply	Test	Met
CDH-02	Communications	Communication interface Board	The system shall transmit at 300,000 bps.	Allows for spacecraft to send all stored data per communication window between various subsystems and instruments	Science Data Collection	Instrument Functionality and Power Supply	Test	Met

CDH-03	Data Handling	Data Processor	The mission shall successfully collect and relay scientific data to Earth, verified by successful communication with the orbiting spacecraft.	Ensures the spacecraft processes and stores data from scientific instruments and subsystems promptly.	Science Data Collection	Instrument Functionality and Power Supply	Demonstration	Met
CDH-04	Data Storage	Solid-State Recorder	The system shall store 10 TB of data.	Allows the spacecraft to store large volumes of scientific data and telemetry between components	Science Data Collection	Instrument Functionality and Power Supply	Analysis	Met
CDH-05	Software Architecture	Flight Software	The system shall manage all spacecraft operations and data handling and support autonomous	Allows the spacecraft o control and coordinate all subsystem activities and data flows and enables the	Mission Objective	Instrument Functionality and Power Supply	Test	Met

			operations and fault management	spacecraft to operate independently throughout the mission				

Table 9: CDH System Requirements

2.1.3.2. CDH Sub-Assembly Overview

The Command and Data Handling (CDH) subsystem is a key part of ensuring that all aspects of mission data collection, processing, and communication are efficiently coordinated. This subsystem integrates various hardware and software components to manage data flow across the spacecraft's systems. The CDH subsystem not only handles operational commands from Earth but also processes data from scientific instruments and subsystems, making it a vital enabler for mission success. The CDH subsystem has different key subassemblies and their roles are unique, and they work together to handle the subsystem.

Subassembly Roles and Interactions

The CDH subsystem includes several key subsystems: communications, data handling, data storage, and software architecture. Each of these subassemblies plays a critical role in ensuring the spacecraft's ability to communicate, store and process data, and operate it. The integration of subassemblies is essential for the spacecraft to function effectively throughout its mission.

Communications Subsystems

The communications Subsystems is responsible for the spacecraft's ability to send and receive data, both between different subsystems on board and with external entities, such as Earth-based mission control and orbiting spacecraft. The key components of the communications subassembly include:

SUBASSEMBLIES

- **Radio Frequency Communication Interface (RF Interface):** This component establishes the communication link between the spacecraft and Earth. It ensures that data can be transmitted during designated communication windows, which is crucial for both real-time commands and scientific data downlinks.

- **Communication Interface Board:** This board manages the communication between the spacecraft and external systems. It is responsible for transmitting data at the specified rate (in bits per second, bps), which is determined by the spacecraft's data transmission needs and available bandwidth. The communication interface board acts as a translator between the spacecraft's internal systems and external communication channels.

These components must function together to allow uninterrupted communication between the spacecraft and Earth. The communication system needs to handle a variety of data types, including real-time telemetry, scientific measurements, and mission updates. This data is often used during specific communication windows when the spacecraft is within range of Earth-based ground stations or an orbiting relay satellite.

Data Handling Subsystems

The data handling subassembly is responsible for processing and managing data within the spacecraft. This sub assembly ensures that all scientific data collected from the spacecraft's instruments, as well as telemetry data from other subsystems, is properly processed and formatted for transmission to Earth.

SUBASSEMBLIES

- **Data Processor:** The data processor is a central component of the data handling subassembly. It processes raw data from scientific instruments, performs calculations or transformations as necessary, and formats the data for storage or transmission. The processor also ensures that data integrity is maintained throughout the process. Different scientific instruments aboard the spacecraft have unique data needs. Instruments such as spectrometers, cameras, or sensors generate large amounts of raw data that must be quickly processed for both on-board decision-making and eventual downlink. The data processor ensures that this data is organized, compressed, and formatted into packets for transmission.

The data handling system is also responsible for managing the spacecraft's onboard memory and ensuring that data is written to storage during periods when communication is not available. This storage capability is crucial for ensuring that no data is lost during gaps in communication windows.

Data Storage Subsystems

The data storage subassembly ensures that all data collected by the spacecraft is safely stored for later transmission. Given the large amounts of data generated by scientific instruments and subsystem telemetry, having a robust and reliable storage system is critical for mission success.

SUBASSEMBLIES

- **Solid-State Recorder :** The Solid-State Recorder stores both scientific data from instruments and telemetry data from the spacecraft's systems. The data is stored in non-volatile memory, ensuring that it is preserved even if the spacecraft experiences power loss. The Solid-State Recorder's capacity must be large

enough to store data over extended periods between communication windows. Data is written to the storage device in a way that ensures it can be retrieved quickly when communication becomes available. The system must also prioritize data, storing mission-critical data while potentially discarding less important information in case of storage limitations.

The data storage system must be able to handle high-speed data writes during data collection periods and be capable of holding large volumes of data until the spacecraft is in range to transmit it. This capability ensures that the spacecraft can continue its scientific operations without interruptions, even if communication is not immediately available.

Software Architecture Subsystems

The flight software is the brain of the spacecraft's operations. This software manages the entire spacecraft's activities, from system health monitoring and data handling to communication management and fault detection. The flight software is responsible for ensuring that all subsystems work together to meet mission objectives.

SUBASSEMBLIES

- **Flight Software:** The flight software allows the spacecraft to operate autonomously, performing tasks such as scheduling observations, handling data transfers, and adjusting spacecraft orientation based on mission parameters. This autonomy is especially important in deep space missions, where real-time communication with Earth may not be possible. The software is also designed to detect faults within subsystems and take corrective actions when necessary. This could involve rebooting systems, switching to backup hardware, or altering the spacecraft's operations to avoid damage. The flight software controls the flow of data between subsystems, ensuring that data is processed, stored, and transmitted as needed. It plays a crucial role in ensuring that scientific data is preserved and transmitted efficiently.

The flight software ensures the spacecraft can operate independently throughout the mission, with minimal human intervention. It also allows for on-the-fly adjustments to the spacecraft's operations based on changing mission needs or unforeseen issues.

Integration of Subassemblies

The integration of the communications, data handling, data storage, and software architecture subsystems and its subassemblies are vital for the spacecraft's overall success. These subsystems must work in harmony, and the exchange of data between them is crucial for maintaining spacecraft operations in a flow.

Data Flow and Coordination
Data flow between subsystems is continuous and must be carefully managed. The data processor coordinates with the communication interface to ensure that data is processed, formatted, and ready for transmission when communication windows are available. At the same time, the flight software manages the timing and sequencing of all operations, ensuring that data flows smoothly and that the spacecraft's objectives are met.
For example, when a scientific instrument captures data, the data is sent to the data processor for initial analysis. The processor then formats the data and stores it in the solid-state recorder. Once the spacecraft enters a communication window, the flight software coordinates the transmission of this data to Earth.

Data Needs of Instruments
Each scientific instrument on the spacecraft has its own data needs. Instruments may require different levels of processing or data formats before the information can be transmitted. For example, a spectrometer might generate continuous raw data that must be processed into usable scientific information, while a camera might generate large image files that must be compressed for transmission. The data handling system must manage these differing requirements while ensuring that all data is stored appropriately in the system.

Collaboration with Other Subsystems
In addition to the interactions between CDH subassemblies, the CDH subsystem must collaborate with other spacecraft subsystems, such as power management and thermal control. For instance, the flight software architecture must ensure that data collection and communication tasks are scheduled in such a way that they do not conflict with the spacecraft's power needs. Similarly, thermal control systems must manage the heat generated by the CDH subsystems, ensuring that components such as the solid-state recorder do not overheat during extended periods of data collection.

Challenges and Considerations
While the CDH subsystem is designed to function autonomously, there are several challenges that must be considered during the mission:

- **Data Volume and Transmission Rates:** Scientific instruments can generate large amounts of data, which can strain the spacecraft's storage and communication capacity. The spacecraft's data handling and communication

systems must be designed to efficiently manage and compress this data for transmission.

- **Autonomy and Fault Management:** In the event of a system failure, the flight software must be able to detect and manage faults autonomously. This requires sophisticated algorithms and onboard decision-making capabilities to ensure that the spacecraft can continue its operations without human intervention.

- **Communication Windows:** The spacecraft's communication windows are often limited, particularly in deep space missions. This requires efficient management of data storage and transmission, ensuring that critical data is downlinked during the available windows.

- **Energy Consumption:** The CDH subsystems must operate within the spacecraft's power constraints. Data processing and communication can be power-intensive, so the spacecraft's energy usage must be carefully managed to ensure it does not exceed the available power.

The Command and Data Handling subsystem is a complex and integrated set of components responsible for managing the flow of data and ensuring the spacecraft operates autonomously. By coordinating the activities of the communications, data handling, data storage, and flight software subassemblies, the CDH subsystem enables the spacecraft to meet its mission objectives. It ensures that scientific data is collected, processed, stored, and transmitted, all while managing power, thermal, and operational constraints. Successful integration of these subsystems is essential for the mission's success, making the CDH subsystem one of the most critical components of the spacecraft mission.

SUBASSEMBLY	MASS	Dimensions	Max Power Draw
Communications - Radio Frequency Communication Interface	2.0 kg	250 x 250 x 100	15 W
- Communication Interface Board	0.5 kg	150 x 150 x 50	10 w
Data Handling - Data Processor	3.0 kg	100 x 100 x 50	20 W
Data Storage - Solid-State Recorder	2.5 kg	120 x 80 x 40	5 W
Software Architecture - Flight Software	TBD	TBD	TBD

Table 10 : CDH Subsystem and Subassemblies

2.1.3.3. CDH Subsystem Recovery and Redundancy Plans

A Command and Data Handling Subsystem Recovery and Redundancy Plan can help to ensure spacecraft reliability by preparing for potential failures and enabling system recovery.

Fault Detection, Isolation, and Recovery

Automatic systems within the CDH detect irregularity, isolate the fault, and initiate actions that can help to restore functionality.

To Implement this there should be real-time health monitoring of data buses, processors, and memory. Use of timers to reset stalled processors .Event-driven or periodic diagnostic checks to detect and recover.

Redundant Architectures

Processor Redundancy incorporates dual or more redundant processors. If one fails, the backup can take over without interrupting operations.

Data Storage to use as a dual solid-state recorder to store mission-critical data, ensuring redundancy in case of a failure.

Communication Interfaces implement redundant communication channels to avoid single points of failure in data transmission to Earth or other spacecraft.

Cross-Checking and Synchronization

Use multiple processors to cross-check operations and detect mismatches. Synchronize data between primary and backup storage to prevent data loss during recovery.

Reconfigurable Systems

Employ software-defined architectures where systems can dynamically reconfigure themselves to bypass damaged components.

Utilize modular design in CDH hardware to allow re-routing of power or signals around faulty modules.

Error Correction and Mitigation

Use Error Correction Code memory to mitigate transient radiation-induced errors.

Implement redundant data paths and cyclic redundancy checks (CRC) for error detection during data transmission.

Recovery Procedures

Manual Commands can help to ground control and initiate specific recovery sequences when automatic systems fail.

Safe Mode Activation with the spacecraft enters a pre-programmed safe mode to conserve resources and await ground instructions.

Testing and Validation

Conduct fault-injection testing during development to simulate hardware and software failures.

Perform integrated system tests to validate redundancy mechanisms under operational conditions.

Radiation Hardening

Use radiation-hardened components to minimize susceptibility to high-energy particles.

Employ shielding and redundancy to safeguard critical CDH functions.

These can help to prepare for any potential failures or problems through thorough planning and by having these steps as part of the recovery plan.

2.1.3.4. CDH Subsystem Manufacturing and Procurement Plans

For the Communication and Data Handling (CDH) subsystem of the Mare Tranquillitatis mission, it is essential to select reliable, high-performance suppliers that can meet the stringent requirements of deep space exploration. Below is a detailed justification for the primary and backup suppliers for each of the key components in the this system:

1. COMSEC Converter AT-CC-160-2 (RF Interface)

- **Primary Supplier: Amergint Technologies**
  Rationale:
  Amergint Technologies is a leading supplier of secure RF communication systems for space missions, with extensive experience in developing communication interfaces for aerospace applications. Their COMSEC Converter AT-CC-160-2[7] is widely used for secure communication in space missions due to its high reliability, robust performance, and ability to operate in harsh space environments. Amergint is known for producing space-grade, radiation-hardened communication systems that are capable of providing high data throughput while ensuring secure transmission using advanced encryption methods. Amergint Technologies has a proven history of supporting space programs that require high-performance RF communication systems. This includes communication systems for lunar missions and other deep space programs, where uninterrupted communication is critical. Their systems have been used in NASA and commercial space applications, making them a trustworthy partner for lunar missions like yours.
- **Lead Time:** 6 months
- **Backup Supplier: L3 Harris Technologies**
  Rationale:
  L3Harris Technologies is a global aerospace and defense contractor with an extensive portfolio of communication systems for space missions. Their communication products, such as the ATN-151, are designed for deep space missions, providing reliable data transmission under extreme environmental conditions, such as the lunar surface and deep space. They specialize in space communication systems with a focus on data security, which is essential for lunar missions that require encrypted data transmissions. L3Harris has a history of providing communication solutions for NASA, including Lunar Gateway and Artemis programs, making them a suitable backup supplier for the RF Interface in a lunar mission.[8] Their expertise with high-bandwidth, secure communication systems ensures that they can meet the needs of lunar exploration.
- **Lead Time:** 9 months

2. Communication Interface Board

[7] Satsearch, *COMSEC Converter AT-CC-160-2*, accessed December 8, 2024, https://www.satsearch.co/products/comsec-converter-at-cc-160-2.

[8] L3 Harris Technologies, *Artemis Missions*, accessed December 8, 2024, https://www.l3harris.com/artemis-missions.

- **Primary Supplier: Astrosat Communications**
  Rationale :
  Astrosat Communications is a well-regarded supplier of communication interface boards specifically designed for space applications. Their space-grade communication boards are optimized for high-data-rate applications, which is crucial for spacecraft that need to transmit large amounts of data back to Earth, especially during scientific observations and communication windows. Astrosat's boards are proven in space, with extensive use in low Earth orbit (LEO) missions and interplanetary space communication. Their experience in developing reliable communication systems for space exploration makes them a strong candidate for lunar missions, where stable and high-throughput communication is essential.
- **Lead Time:** 8 months
- **Backup Supplier: Teledyne Technologies**
  Rationale:
  Teledyne Technologies is a leader in providing space-qualified electronics, including communication interface systems. They specialize in high-reliability electronics and communication systems, having delivered solutions for space exploration programs, including NASA's Mars Science Laboratory and Juno missions. Their boards are known for excellent data handling capabilities, making them ideal for lunar communication systems. Teledyne has provided communication systems for NASA's robotic missions, and their products are capable of handling the demanding data rates required for deep space and lunar missions.
- **Lead Time:** 8 months

3. EDGE-1100 Payload Processor (Data Processor)

- **Primary Supplier: Ibeos**
  Rationale:

  Ibeos is recognized for their highly reliable payload processors, including their flagship processor used in spacecrafts' systems that demand significant data processing capabilities. Their systems are designed to efficiently handle large amounts of data from scientific instruments, such as cameras, spectrometers, and sensors, which are essential for lunar exploration. Ibeos' processors are built to support autonomous spacecraft operations, a critical feature for lunar missions where remote operations and limited communication windows are common.

- **Lead Time:** 6 months

- **Backup Supplier: Innovative Solutions in Space (ISIS)**
  Justification:
  ISIS is known for their EDGE-1100 Payload Processor, which is a versatile and efficient processor designed for space missions requiring significant data processing capabilities. It is specifically built to handle the demands of scientific instruments aboard spacecraft, including data from cameras, spectrometers, and other sensors, which are common on lunar missions. The EDGE-1100 is built for

autonomous spacecraft operations, which is critical for lunar missions that involve remote or autonomous operations, particularly when direct communication with Earth is not possible. ISIS's EDGE-1100 has been used in various small satellite and interplanetary missions, which makes it a great fit for lunar missions that require advanced payload data processing. Their systems are highly reliable in space environments and are customizable to meet mission-specific needs.

- **Lead Time:** 9 months

4. LRO Solid-State Recorder (Data Storage)

- **Primary Supplier: NASA's Goddard Space Flight Center**
  Rationale:
  NASA's Goddard Space Flight Center is the manufacturer of the Lunar Reconnaissance Orbiter's (LRO) Solid-State Recorder, which is used to store both scientific data and spacecraft telemetry. This system has been tested and proven in lunar missions, making it an excellent choice for your mission, as it meets the necessary data storage capacities and endurance for long-duration missions. The LRO Solid-State Recorder has been designed specifically for lunar exploration, capable of storing large volumes of scientific data over extended periods between communication windows. Its reliability and large storage capacity are crucial for lunar missions, ensuring data retention during long communication gaps.
- **Lead Time:** 7 months
- **Backup Supplier: SpaceX**
  Rationale:
  SpaceX's commercial-grade solid-state recorders have been developed for a variety of space missions and have been qualified for use in low-Earth orbit (LEO) and lunar missions. Their products are cost-effective and reliable, with the capability to handle significant data volumes while ensuring high throughput and integrity. SpaceX has been involved in several high-profile space programs, including their Starlink satellite constellation and NASA's Artemis program, ensuring compatibility with lunar mission data storage needs.
- **Lead Time:** 8 months

5. NASA's Core Flight Software (cFS) (Flight Software)

- **Primary Supplier: NASA's Jet Propulsion Laboratory (JPL)**
  Rationale:
  NASA's Core Flight Software (cFS) is the industry standard for space missions. It is used extensively by NASA and other space agencies for a variety of missions, including lunar, planetary exploration, and deep space operations. cFS is highly modular and customizable, making it an ideal solution for autonomous operation of spacecraft. cFS has been used in a variety of lunar missions including Lunar Reconnaissance Orbiter and Artemis missions, proving its ability to autonomously control spacecraft operations. This is essential for lunar missions,

where real-time communication with Earth may be limited or unavailable.

- **Lead Time:** 8 months
- **Backup Supplier: Blue Origin**
  Rationale:
  Blue Origin has developed autonomous flight software for their New Shepard and New Glenn rockets. Their software has been tested for deep space and lunar applications, making it a suitable backup for missions that require autonomy, fault management, and flexible operations. Blue Origin has supported NASA's Artemis program and has experience with spacecraft systems that operate autonomously in deep space.
- **Lead Time:** 10 months

2.1.3.5. CDH Subsystem Verification Plans

The Verification Plan for the Command and Data Handling (CDH) subsystem is required to ensure that all functional, performance, and interface requirements are met for the mission. Verification involves testing, analysis, inspection, and demonstration, often carried out in multiple stages.

The verification plan includes confirming compliance with all specified requirements. Ensuring that the subsystem functions as intended under expected and extreme conditions. Validating other subsystems, such as communications, power, and storage.

Verification Methods

Analysis

- Conduct simulations to verify data processing and communication functionalities.
- Perform thermal, radiation, and power analysis to ensure robustness in space conditions.

Inspection

- Physically inspect hardware for manufacturing defects or design flaws.
- Confirm compliance with design specifications (e.g., connector types, board layouts).

Testing

- Unit Tests to validate individual components, such as processors or memory modules, for basic functionality.
- Integration Tests to confirm proper data transfer and control signals between CDH and other subsystems.
- End-to-End Tests to verify that the CDH subsystem can handle simulated mission operations, from data acquisition to storage and communication.
- Environmental Test to include thermal vacuum, vibration, and compatibility tests to simulate launch and space environments.

Demonstration

- Demonstrate critical functions such as sequences, data relay, and fault recovery in a mission.

Requirements

Each verification method is mapped to a specific requirement to ensure complete coverage. Such as Data Handling Requirements are verified through functional tests and data simulations. Redundancy Features are tested through fault injection and failover tests. Communication Interfaces are assessed with different tests simulating data transfer with ground stations and orbiting spacecraft.

Documentation and Reporting

Create a Verification Document to track results. Record observations, pass/fail outcomes, and deviations in test logs. Submit detailed test reports for review by system engineers and project stakeholders to further verify.

Processor Validation should run benchmarking and stress tests to evaluate performance under maximum data loads.

Data Integrity to Inject deliberate errors into the data stream to verify error detection and correction algorithms.

Communication Testing uses ground-based simulations to validate the timing and reliability of data transfers.

Verification Tools and Resources

Software Tools such as MATLAB for simulations, JMARS for data analysis, and Siemens NX for hardware modeling.

Hardware Facilities such as Thermal-vacuum chambers, vibration test tables, and cleanrooms for component assembly and inspection.

Flight Models use engineering or qualification models for testing before transitioning to the flight model.

These steps and methods are the plans for the Command and Data Handling (CDH) subsystem verification.

2.1.4. Thermal Control Subsystem Overview

In order to mitigate the severe temperatures anticipated in lunar conditions, the thermal management subsystem is made to ensure ideal operating temperatures for instruments and components. To maintain system resilience and stability, this subsystem combines passive and active temperature management components.

Subassemblies and Components

Thermal Insulation Layers:

- Design:
  - The insulation is made up of several layers of Multi-Layer Insulation (MLI), which was chosen because of its remarkable ability to reduce heat transfer. In order to minimize radiation absorption from the lunar surface and prevent heat loss to the vacuum of space, the MLI employs alternating layers of reflecting and spacer materials.
- Sub-components:
  - To provide a high barrier against thermal conduction and radiation, each MLI layer is made up of reflective materials such as Mylar and Kapton combined with low-conductive spacer layers. Long-term defense against the Moon's harsh surface conditions is ensured by the insulating layers' unique construction, which can withstand the accumulation of lunar dust without degrading.
- Purpose:
  - To lessen heat transfer between external and interior components, particularly during lunar daylight (up to 121°C) and nighttime (down to -133°C).

Radiators and Heat Sinks:

- Design:
  - To disperse excess heat produced by the rover's onboard electronics, radiators are positioned strategically. High-conductivity materials, such as aluminum, are included into heat sinks to give high-power components more passive cooling.
- Sub-components:
  - To help heat move to radiating surfaces, which subsequently release it into space, each radiator has conductive channels. High-thermal-conductivity metals used to make the heat sinks guarantee efficient heat transmission away from delicate electronics.
- Purpose:
  - Keeping the internal operating range below 50°C during high-energy operations in order to control and dissipate heat generated within.

Heating Elements:

- Design:
  - To combat the intense lunar nighttime temperatures, electrical heating components are included into the frames of the lander and rover. These heaters are set up to turn on automatically in response to temperature sensor data, guaranteeing that vital subsystems continue to function during the lunar night.
- Sub-components:
  - Resistive elements, which transform electrical energy into heat, make up each heater. The heaters provide necessary warmth without using excessive amounts of power since they are placed strategically close to temperature-sensitive equipment.
- Purpose:
  - To shield sensors from freezing temperatures on moon nights so they can operate reliably for long stretches of time.

Temperature Sensors:

- Design:
  - To offer accurate, real-time temperature fluctuation monitoring, sensors are positioned throughout the subsystem. With an accuracy of ±2°C, each sensor enables rapid thermal control.
- Sub-components:
  - Thermocouples and resistance temperature detectors (RTDs), which are made to function well in vacuum and dusty conditions, are part of the sensor suite.
- Purpose:
  - By providing the command and data handling (CDH) system with temperature data from these sensors, the thermal control strategy can be modified in real time as necessary.

Technology Readiness Level (TRL)

With previous field testing conducted under lunar-like circumstances, each subassembly is at TRL 5. The insulating layers' thermal interface material has the lowest TRL, contributing to the subsystem's overall TRL of 5.

Component	Mass (Kg)	Dimensions (cm)	Power Draw (W)
Insulation Layers	2.5	150x150	0
Radiators	3.0	100x50	0
Heating Elements	1.0	20x20	10
Temperature Sensors	0.2	5x5	1

Justification of Design Choices

The subsystem can balance passive and active temperature regulation thanks to the radiators, heating components, and MLI. MLI was chosen because of its low bulk and great thermal efficiency, which are crucial for the Moon's particular environment. An efficient passive cooling solution is offered by radiators, which are especially important during lunar day operations when temperatures rise quickly. In contrast, the heating components provide a dependable remedy for the frigid lunar night circumstances, guaranteeing that subsystems can tolerate the significant temperature changes that are a natural part of the lunar surface.

The design of this thermal management subsystem offers a reliable way to deal with the thermal difficulties of lunar exploration, extending the mission's lifespan and guaranteeing that the scientific equipment runs within the ideal temperature range the entire time.

2.1.4.1. Thermal Control Subsystem Requirements

Requirement ID	Requirement Description	Rationale	Verification Method	Relevant Subsystem/Components
TMS-1	The internal instrument temperatures must be kept between -20°C and +50°C via the thermal management subsystem.	Guarantees that equipment runs within acceptable temperature ranges, preventing damage or malfunction from extremes.	Thermal testing and simulation	Thermal insulation, heating elements
TMS-2	All delicate components exposed to lunar surface temperatures between -208°F (-133°C) and +250°F (121°C) must be insulated by the thermal management subsystem.	Shields vital systems from drastic changes in the moon's temperature.	Material performance testing	Insulation layers, exterior coating
TMS-3	Active thermal monitoring will be part of the thermal subsystem to detect temperature changes with an accuracy of ±2°C.	Make sure that subsystems' operating temperatures are maintained in real time.	In-situ thermal sensor testing	Temperature sensors, data handling unit
TMS-4	To keep the interior temperature below 50°C, the thermal management system will use passive cooling to disperse heat produced by internal electronics..	Keeps internal electronic components from overheating.	Passive thermal testing	Radiators, heat sinks

TMS-5	For heat-sensitive components, the subsystem must have thermal interface materials with a minimum thermal conductivity of 2.5 W/m·K.	Increases the effectiveness of heat transport to avoid localized overheating.	Thermal conductivity testing	Thermal interface materials, heat-conductive surfaces

2.1.4.2. Thermal Control Sub-Assembly Overview

To optimize the thermal management subsystem for lunar conditions, comprehensive trade studies were conducted for each major subassembly: Thermal Insulation Layers, Radiators, and Heating Elements. The primary criteria used for these studies were Thermal Performance, Durability, Mass Efficiency, and Cost. These factors were chosen based on mission priorities to maintain temperature stability, ensure long-term functionality, stay within the spacecraft's mass constraints, and adhere to budget limitations. Below is a breakdown of each trade study and the decisions that guided the final selections.

Criteria and Weighting

- Thermal Performance (40%): Thermal performance was given the highest weight because it is the thermal subsystem's principal function to manage temperature extremes. This criterion evaluated how well each material insulated or dissipated heat.
- Durability (25%): Over the course of the mission, the materials must endure impacts from micrometeorites, radiation, and lunar dust. In order to guarantee that the subsystem could function dependably without suffering appreciable degradation, this criteria was essential.
- Mass Efficiency (20%): Lighter materials were chosen to optimize payload capacity due to stringent mass limits.
- Cost (15%): Both the initial outlay and possible long-term savings from improved performance and durability were taken into account.

With a total score of 8.1, MLI had the highest score. Although aerogel was lightweight and provided good thermal insulation, its lower durability score made it less suitable for lunar settings where dust and micrometeorite strikes could be dangerous. Although they were heavier and marginally less effective at insulating than MLI, thermal blankets offered a strong durability profile. Because it provided a compromise between good thermal performance, adequate durability, and lightweight design—all within a reasonable price range—the MLI was selected.

Because of their ability to balance mass efficiency, thermal performance, and durability, carbon fiber radiators received the highest score of 7.8. Despite being less expensive, aluminum radiators did not have the same thermal performance as carbon fiber. Although copper radiators had good thermal performance, their mass efficiency score was affected by their weight. Carbon fiber was chosen because it is strong and lightweight, which will improve performance and mission dependability over the long run.

With a score of 7.95, resistant wire heaters were the best choice. They provided

outstanding endurance and thermal performance, which were critical under lunar night settings. Although PTC heaters offered reliable temperature control and good durability, their thermal performance was marginally worse. The lightest choice was a film warmer, but it wasn't sturdy enough for long flights. Because of their consistent heat output, positioning flexibility, and capacity to endure extended use in cold temperatures, resistive wire heaters were chosen.

Because of their exceptional thermal performance and mass efficiency—two factors that are crucial in lunar applications where accurate, dependable measurements are needed—thermocouples received the highest overall score of 8.3. Thermocouples are ideal for lunar circumstances because of their exceptional ability to endure abrupt temperature swings.

2.1.4.3. Thermal Control Subsystem Recovery and Redundancy Plans

In lunar missions, the thermal system is critical for maintaining the spacecraft's temperature within operational limits, especially during the extreme temperature variations between lunar day and night. The components selected for this thermal system—MLI (Multi-Layer Insulation), carbon fiber radiators, resistive wire heaters, and thermocouples—have been chosen based on their balance of performance, durability, and mass efficiency. However, redundancy and recovery mechanisms are essential in case any of these components fail during the mission. The following plan outlines strategies for ensuring continued thermal management throughout the mission.

Redundancy Strategies

1. **MLI (Multi-Layer Insulation)**:
    - **Redundancy Plan**: Since MLI provides crucial insulation to prevent heat loss or gain, additional layers of MLI may be used on critical parts of the spacecraft to ensure that even if one layer is damaged (e.g., from micrometeorite strikes or dust accumulation), the others can still provide adequate insulation.
    - **Recovery Strategy**: If the insulation material is compromised, onboard repair kits could be used to patch up exposed sections of the spacecraft's thermal system. Additionally, reorientation or deployment of alternative passive thermal shielding techniques may be considered.
2. **Carbon Fiber Radiators**:
    - **Redundancy Plan**: A secondary carbon fiber radiator or an alternative cooling method could be installed. These could be switched on if the primary radiator is compromised or underperforming.
    - **Recovery Strategy**: If the radiator fails or efficiency drops due to micrometeorite impact or damage, the system could reroute heat dissipation to secondary radiators. Onboard monitoring systems should provide early detection to allow preemptive corrective actions.
3. **Resistive Wire Heaters**:
    - **Redundancy Plan**: Multiple resistive wire heaters should be distributed across the spacecraft in different critical areas. If one heater fails, others can compensate by increasing their heat output to maintain required temperatures.
    - **Recovery Strategy**: The mission system should be programmed to detect heater failure and reroute power to operational heaters. In case of extensive failure, a secondary heater system using thermoelectric devices could be used for smaller localized heating needs.
4. **Thermocouples**:
    - **Redundancy Plan**: Thermocouples should be installed in key areas to measure temperature at multiple points across the spacecraft. If one fails,

other thermocouples can provide continuous readings.
    - **Recovery Strategy**: If a thermocouple fails or provides inaccurate readings, onboard systems should automatically switch to alternative temperature sensing devices (e.g., resistance temperature detectors). The mission's software should also trigger diagnostic checks for temperature sensors to ensure accurate measurements.

System-Level Redundancy:

- **Thermal Management Software**: A software redundancy approach will be implemented to monitor and adjust the operation of the thermal components. This will ensure that if one component fails, the others will compensate, and the spacecraft's temperature remains within operational limits.
- **Temperature Monitoring and Alerts**: The system will be capable of detecting anomalies in temperature readings and can adjust the output of the heaters or shift load from the radiators to mitigate temperature extremes.
- **Autonomous Fault Detection and Recovery**: In the event of a failure, the thermal system should be capable of autonomously diagnosing the fault and reconfiguring the system to maintain operational stability. This would include rerouting power to secondary systems or activating backup thermal components.

Impact of Failure and Recovery Actions:

1. **Failure of MLI**: A failure in MLI would likely result in heat loss or gain, impacting spacecraft performance. A secondary layer of MLI or reorientation of the spacecraft could mitigate this risk.
2. **Failure of Carbon Fiber Radiator**: If the radiator's heat dissipation is compromised, backup radiators or redistribution of heat through alternative cooling methods should ensure the spacecraft stays within operational temperature limits.
3. **Failure of Resistive Wire Heaters**: The spacecraft will need to switch to secondary heaters or use other forms of passive heating, like insulating blankets, to prevent freezing of sensitive components during lunar nights.
4. **Failure of Thermocouples**: If a thermocouple fails, backup sensors will take over. Software will use algorithms to detect and compensate for missing temperature data, ensuring the spacecraft does not overheat or freeze.

Recovery Testing:

- **Pre-Launch**: Test redundancy systems in simulation environments where multiple failures (e.g., heater or thermocouple failure) can be tested in parallel.
- **In-Mission**: Implement continuous health checks of all thermal system components, including the insulation, radiators, heaters, and thermocouples. Use telemetry data to monitor performance and adjust parameters as necessary.

2.1.4.4. Thermal Control Subsystem Manufacturing and Procurement Plans

The thermal control subsystem ensures the spacecraft's operational temperatures are maintained during all mission phases. Key components include radiators, thermal coatings, multi-layer insulation (MLI), flexible heaters, and thermal straps. These elements are critical for managing the extreme lunar thermal environment, where temperatures range from 250°F (121°C) during the day to -208°F (-133°C) at night.

Manufacturing Process

- ***Radiators and Thermal Coatings:***
  - Radiators are manufactured using aluminum alloys due to their high thermal conductivity and low weight. The manufacturing process involves precision CNC machining followed by anodization to enhance durability.
- ***Multi-Layer Insulation (MLI):***
  - MLI blankets are fabricated from layers of polyimide and aluminized Mylar. These are precision cut and layered in cleanroom environments to minimize contamination.
- ***Heaters and Sensors:***
  - Flexible heaters and Resistance Temperature Detectors (RTDs) are procured from aerospace-certified vendors. Assembly involves bonding heaters to critical components using high-temperature adhesives.

Procurement Plans

- ***Vendor Selection:***
  - Vendors are selected based on their experience in producing space-qualified thermal components. Notable vendors include suppliers for aerospace MLI and high-performance adhesives.
- ***Quality Assurance:***
  - Each component undergoes incoming inspection to ensure adherence to specifications. MLI blankets and heaters are tested for vacuum compatibility and durability against micrometeorite impacts.
- ***Timeline:***
  - Procurement begins during Phase B and is timed to align with the integration phase. Lead times for thermal coatings and MLI are accounted for in the mission schedule.

2.1.4.5. Thermal Control Subsystem Verification Plans

The thermal control subsystem will be verified to ensure that all spacecraft systems remain operational under expected lunar conditions. This includes maintaining temperature stability during lunar day-night cycles and protecting critical instruments.

Verification Methods

Thermal Vacuum Testing:

Subsystem components are placed in a thermal vacuum chamber to simulate the lunar environment. This verifies insulation performance, radiator efficiency, and heater functionality.
Thermal Balance Testing:

Conducted to validate thermal models by comparing predicted and measured heat dissipation and retention under various operational scenarios.

Component-Level Testing:
RTDs and heaters are individually tested for accuracy, response time, and durability under vacuum and thermal cycling conditions.

Environmental Testing

High-temperature gradient tests simulate lunar day-night transitions, ensuring MLI and radiators operate without degradation.

Vibration testing confirms that thermal components withstand launch and deployment stresses without compromising functionality.

Pass/Fail Criteria

Components must maintain operational temperatures as defined in the thermal subsystem requirements.

Heaters and sensors must demonstrate response accuracy within ±1°C under simulated conditions.

2.1.5. Payload Subsystem Overview

Overview of Science Instruments

The payload subsystem consists of three primary science instruments integrated to achieve the mission's objectives of mapping and evaluating lunar lava pits:

Ground Penetrating Radar (GPR):

Purpose: Detect and map subsurface structures, including voids, lava tubes, and potential hazards.

Operation: Emits electromagnetic waves that reflect off subsurface features, providing data for structural analysis.

Laser-Induced Breakdown Spectroscopy (LIBS):

Purpose: Analyze regolith and rock composition to identify volatiles (e.g., water ice, hydrogen) for in-situ resource utilization (ISRU).

Operation: Uses high-energy laser pulses to vaporize material, analyzing emitted light for elemental composition.

Thermal Infrared (TIR) Sensors:

Purpose: Measure temperature variations in cave interiors to assess thermal stability for habitation.

Operation: Captures thermal radiation to create temperature distribution maps.
Support from Subsystems

Thermal Control Subsystem: Maintains instrument functionality during lunar night by preventing freezing of sensitive components.

Command and Data Handling (CDH): Manages data storage and transmission to the lander and Earth.

Power Subsystem: Supplies sufficient power for high-energy operations such as LIBS firing cycles.

Challenges and Mitigations

Lunar Dust Mitigation: Instruments are equipped with dust-resistant coatings and cleaning mechanisms.
Thermal Fluctuations: MLI and heaters ensure instruments remain within operational temperatures.
Linkage to Previous Reviews

Findings from the SRR and MCR confirmed the feasibility of selected instruments and their integration.
Environmental factors, including temperature and radiation, were evaluated during MCR, leading to enhanced design of the payload subsystem.

2.1.5.1. Science Instrumentation Requirements

The science instruments for the Mare Tranquillitatis pit mission must perform reliably in the harsh lunar environment, which includes extreme temperature fluctuations, high radiation levels, and significant mechanical vibrations during launch and landing. Each instrument is tailored to meet specific scientific objectives while ensuring durability and precision. For example, the spectrometer is required to detect fluorescence peaks between 200-500 nm to accurately analyze the mineral composition of the lunar surface. Similarly, the imaging camera must achieve a spatial resolution of 0.5 cm at a distance of 10 meters to capture detailed surface features and geological structures. Proper placement of these instruments is critical; the spectrometer, for instance, must have an unobstructed 120° field of view to maximize data collection and ensure comprehensive coverage of the exploration area.

Supporting subsystems play a vital role in meeting these instrumentation requirements. The thermal management system is essential for maintaining stable operating temperatures, preventing sensor drift, and ensuring consistent performance of all scientific instruments. The power subsystem must deliver a reliable and adequate energy supply, providing 5 V DC ±0.5 V and up to 15 W continuously to sustain the instruments throughout the mission. Additionally, the Command and Data Handling (CDH) system must support high data streaming rates of at least 5 Mbps to facilitate real-time data acquisition, processing, and transmission. To protect sensitive instruments, vibration-damped mounting platforms are used to minimize motion-induced noise, and dust mitigation measures are implemented to prevent dust accumulation on optical surfaces. These measures ensure that the scientific instruments can operate effectively, maintaining data quality and integrity even under challenging conditions.

2.1.5.2. Payload Subsystem Recovery and Redundancy Plans

Redundancy

The payload subsystem, encompassing the suite of scientific instruments and sensors essential for achieving the mission's objectives, generally does not carry exact duplicates of each device. Instead, redundancy is achieved through strategic overlap in instrument capabilities. For example, where possible, multiple instruments may be able to measure similar parameters or operate in complementary modes. This way, if one sensor underperforms or fails, at least a portion of its function can be taken over by another, ensuring that critical scientific data is not lost.

Additionally, components within certain payload instruments may have built-in backup features—such as secondary detector elements or internal electronics that can be switched in if the primary set encounters issues. All payload hardware is designed using high-quality, space-rated materials and protective housings to minimize the likelihood of mechanical or environmental damage. In short, instead of carrying full-scale spare instruments, the payload subsystem relies on robust design, overlapping capabilities, and internal backups that preserve mission value without adding excessive weight and complexity.

Recovery

When problems do arise with an instrument—such as unexpected data dropouts, calibration drift, or reduced sensitivity—the payload subsystem employs a flexible, software-driven approach to recovery. The control software continuously monitors performance metrics, comparing sensor outputs against expected baselines and cross-validating measurements with related instruments. If a particular device's readings begin to degrade, operators can adjust its operational parameters—from exposure times and scanning patterns to calibration settings—helping restore acceptable performance levels.

If an instrument's primary function cannot be fully recovered, ground-based teams can reconfigure the payload to emphasize other instruments that provide similar or supportive data. In some cases, the payload may be repositioned or operated in a mode that mitigates environmental factors like dust or glare, reducing their impact on sensitive optics. Through careful monitoring, adaptive algorithms, and the smart use of complementary capabilities, the payload subsystem can continue to deliver valuable scientific results, maintaining mission objectives despite encountering unforeseen technical challenges.

2.1.5.3. Payload Subsystem Manufacturing and Procurement Plans

This section details the direct costs associated with our mission, totaling approximately $375 million. Direct costs include expenses for the mission vehicle and all related systems. This analysis covers subsystem estimates, personnel expenses, testing and simulation costs, and the contingency fund.

Personnel Costs

The total personnel costs are estimated at $113.4 million, reflecting the salaries for our diverse team over the project's lifespan. Here is the breakdown:

Role	Annual Salary
Scientist	$80,000/year
Engineers	$80,000/year
Technicians	$80,000/year
Administration	$60,000/year
Managers	$120,000/year

Table 11: Personnel Costs

Scientist: Salaries for scientists who will develop the mission's scientific objectives and data analysis frameworks.

Engineers: Compensation for engineers who will design, develop, and test the spacecraft and its subsystems.

Technicians: Wages for technicians who will assist in assembling and testing the spacecraft components.

Administration: Pay for administrative staff who will support the project through clerical work, procurement, and coordination.

Managers: Salaries for project managers who will oversee the mission operations, ensuring project milestones are met within budget and on schedule.

These figures are derived from industry-standard rates and are adjusted for the project's duration and specific skill requirements.

Subsystem Costs

The subsystem costs are calculated to be $160 million, distributed among the following key areas:

Subsystem	Estimated Cost
Mechanical	$30 million
Command and Data Handling	$25 million
Power	$20 million
Thermal Management	$35 million
Payload (Science Instruments)	$50 million

Table 12: Subsystems estimated costs

The project's overall budget is divided among several key subsystems, each vital to the spacecraft's success. The **Mechanical System** covers the development and construction of the structural components, and its current allocation reflects a $5 million savings compared to the original System Requirements Review (SRR) estimate. This reduction was achieved by refining the design specifications and eliminating unnecessary complexity—an improvement that did not compromise the system's strength or reliability.

The **Command and Data Handling (CDH)** subsystem encompasses all the hardware and software needed to control the spacecraft, manage onboard operations, and handle the flow of data. The Power System budget supports the spacecraft's energy generation and storage needs, including solar panels and batteries, ensuring a stable power supply throughout the mission.

Maintaining the correct operating temperatures is critical for all onboard technologies, which makes the **Thermal Management** subsystem essential. Its budget also benefited from a $5 million reduction after the SRR phase, thanks to more efficient thermal designs that met requirements while lowering costs.

The **Payload**, consisting primarily of scientific instruments, represents another significant area of expense. By implementing cost-saving measures in both design and manufacturing, the team achieved a substantial $25 million reduction from the initial SRR budget, without impacting the performance of the instruments themselves.

To ensure these estimates are both accurate and reliable, the team employed the NASA Instrument Cost Model (NICM) and the Mission Concept Cost Estimating Tool (MCCET). These tools leverage standard cost-estimating relationships (CERs) to produce credible, data-driven cost projections for each subsystem.

2.1.5.4. Payload Subsystem Verification Plans

Ensuring the payload subsystem meets all mission requirements is crucial for the success of the Mare Tranquillitatis pit mission. Our verification plan is comprehensive, focusing on confirming that each scientific instrument and its supporting components perform reliably under expected mission conditions. This process involves multiple stages, including analytical assessments, laboratory testing, environmental simulations, and integrated system evaluations.

Initially, we conduct detailed analytical reviews using computer models to predict how each instrument will behave in space. These models help us verify optical alignments, thermal responses, and structural integrity before any hardware is built. By identifying potential issues early through simulations, we can refine designs and address problems before moving to the physical testing phase.

Once the designs are validated through modeling, we move to component-level testing in the laboratory. Each instrument, such as spectrometers, cameras, and thermal sensors, undergoes rigorous functional tests to ensure they meet performance specifications. For example, spectrometers are tested for their ability to detect specific wavelengths, while cameras are evaluated for image resolution and stability. Additionally, we subject these components to thermal cycling and vibration tests to mimic the harsh conditions of launch and space travel, ensuring they can withstand extreme temperatures and mechanical stresses without degradation.

After individual components pass their tests, we assemble the payload subsystem and perform integrated testing. This stage verifies that all instruments work together seamlessly and communicate effectively with the spacecraft’s command and data handling systems. We conduct system-level functional tests to ensure data flows correctly between instruments and the onboard computer, and that power distribution meets each instrument’s needs. Calibration tests are also performed using known reference targets to confirm that sensor outputs align with expected values, ensuring accurate scientific data collection.

Environmental simulations play a significant role in our verification process. We place the integrated payload in a thermal vacuum chamber to replicate the vacuum of space and extreme temperature variations. This testing ensures that all instruments operate correctly without the interference of atmospheric pressure or temperature fluctuations. Additionally, we perform

radiation exposure tests on sensitive electronics to verify their durability against cosmic rays and solar radiation, which are prevalent in the lunar environment.

Once the payload subsystem has passed all ground-based tests, we prepare for in-flight verification. After the spacecraft is launched and reaches its operational orbit, we conduct a series of checkouts to confirm that each instrument functions as expected in the actual space environment. This includes initial calibrations and performance assessments to ensure that the ground test results translate accurately to space conditions. Throughout the mission, we perform periodic calibrations and health checks to maintain data integrity and address any issues that arise during operations.

Documentation and traceability are integral to our verification plans. Every test and its results are meticulously recorded, allowing us to trace each verification step back to the original mission requirements. This thorough documentation ensures transparency and accountability, providing a clear record that all instruments meet the necessary standards before and during the mission.

In summary, our Payload Subsystem Verification Plans are designed to methodically validate each component and the integrated system through a combination of modeling, testing, and real-time assessments. This structured approach ensures that the payload subsystem is robust, reliable, and fully capable of delivering high-quality scientific data throughout the Mare Tranquillitatis pit mission.

2.2. Interface Control

The integration and seamless interaction between subsystems are pivotal for the success of the Mare Tranquillitatis mission. This section details the interfaces among the primary subsystems, emphasizing the roles of the Central Command & Data Handling (C&DH), Mechanical, Power, Thermal, and Payload subsystems. Each subsystem is interconnected to ensure the spacecraft operates cohesively in the extreme lunar environment.

Central Command & Data Handling (C&DH)

The C&DH subsystem functions as the mission's brain, coordinating operations, data management, and communication between all other subsystems. It ensures synchronized functionality through real-time monitoring and autonomous decision-making.

Interface with the Mechanical Subsystem:

- Receives deployment commands for the lander's legs and rover's deployment mechanism.
- Continuously monitors feedback, such as positional data from actuators and stress sensors, ensuring successful deployment and operation.
- Executes mobility commands for the rover, including trajectory planning and obstacle avoidance, based on input from sensors and instruments.

Interface with the Power Subsystem:

- Allocates energy resources efficiently, prioritizing critical functions during power constraints (e.g., lunar nights).
- Monitors power usage across subsystems and initiates corrective actions to prevent overdraw or inefficiencies.

Interface with the Thermal Subsystem:

- Controls heaters and radiators to maintain optimal operating temperatures for critical components, especially during lunar nights.
- Monitors thermal sensors and adjusts heating or cooling requirements dynamically.

Interface with the Payload Subsystem:

- Manages scientific data acquisition, processing, and storage.
- Directs instruments based on mission objectives, such as selecting specific areas for ground-penetrating radar scans or laser-induced spectroscopy.

Mechanical Subsystem

The mechanical subsystem includes the structural, mobility, and deployment systems for the lander and rover. It is tightly integrated with other subsystems to ensure functionality under dynamic conditions.

Interface with C&DH:

- Executes deployment sequences and provides feedback on system status, such as leg deployment completion or rover positioning.
- Sends stress and load data to C&DH to evaluate structural integrity during operations.

Interface with the Power Subsystem:

- Receives power for actuators and motors used in deployment and mobility systems.
- Monitors energy consumption during operations, sending data back to optimize power allocation.

Interface with the Thermal Subsystem:

- Includes thermal pathways to maintain mechanical components within operational temperature ranges.
- Relies on thermal sensors to detect hotspots or extreme cold that could impair mobility systems, triggering corrective actions.

Power Subsystem

The power subsystem is responsible for generating, storing, and distributing energy to support all spacecraft operations. It prioritizes subsystem needs based on mission phases and environmental conditions.

Interface with the Mechanical Subsystem:

- Provides energy to actuators and motors during deployment and rover mobility.
- Supplies power for the rover's robotic arms and payload handling systems.

Interface with the Thermal Subsystem:

- Powers heaters during lunar nights to protect subsystems and instruments.
- Supplies energy to radiators and temperature monitoring systems to ensure thermal stability.

Interface with the Payload Subsystem:

- Powers scientific instruments, ensuring continuous operation during data collection phases.
- Provides surge protection for high-energy instruments, such as lasers used in

spectroscopy.

Thermal Subsystem

The thermal subsystem ensures all components remain within operational temperature ranges, despite the Moon's extreme thermal conditions.

Interface with the Mechanical Subsystem:

- Heats or cools mechanical components, such as actuators, hinges, and joints, to prevent thermal deformation or freezing.
- Maintains structural stability by preventing thermal expansion or contraction in the chassis and legs.

Interface with C&DH:

- Communicates thermal data, enabling dynamic control of heaters and radiators.
- Implements commands from C&DH to preheat sensitive areas before deployment or operations.

Interface with the Payload Subsystem:

- Protects scientific instruments by regulating temperature and preventing condensation or overheating.
- Works with thermal shields and insulation to safeguard delicate optics and sensors.

Payload Subsystem

The payload subsystem includes scientific instruments designed for surface and subsurface exploration, operating under precise environmental conditions.

Interface with C&DH:

- Receives commands for instrument deployment, calibration, and operation.
- Sends data to C&DH for processing, storage, and transmission to Earth.

Interface with the Power Subsystem:

- Draws power for high-demand instruments like LiDAR and ground-penetrating radar.
- Relies on uninterrupted power supply to maintain the accuracy of data collection during critical phases.

Interface with the Thermal Subsystem:

- Depends on active thermal management to maintain functionality in the harsh lunar environment.
- Includes localized heaters to prevent freezing of optical lenses or other sensitive components.

These interconnected subsystems create a robust framework capable of handling the mission's demands while ensuring seamless operations. By leveraging real-time communication, adaptive resource allocation, and advanced environmental controls, the spacecraft is equipped to meet its objectives of exploring the Mare Tranquillitatis pit and advancing lunar exploration. This tightly integrated interface design ensures not only operational success but also adaptability to unforeseen challenges in the lunar environment.

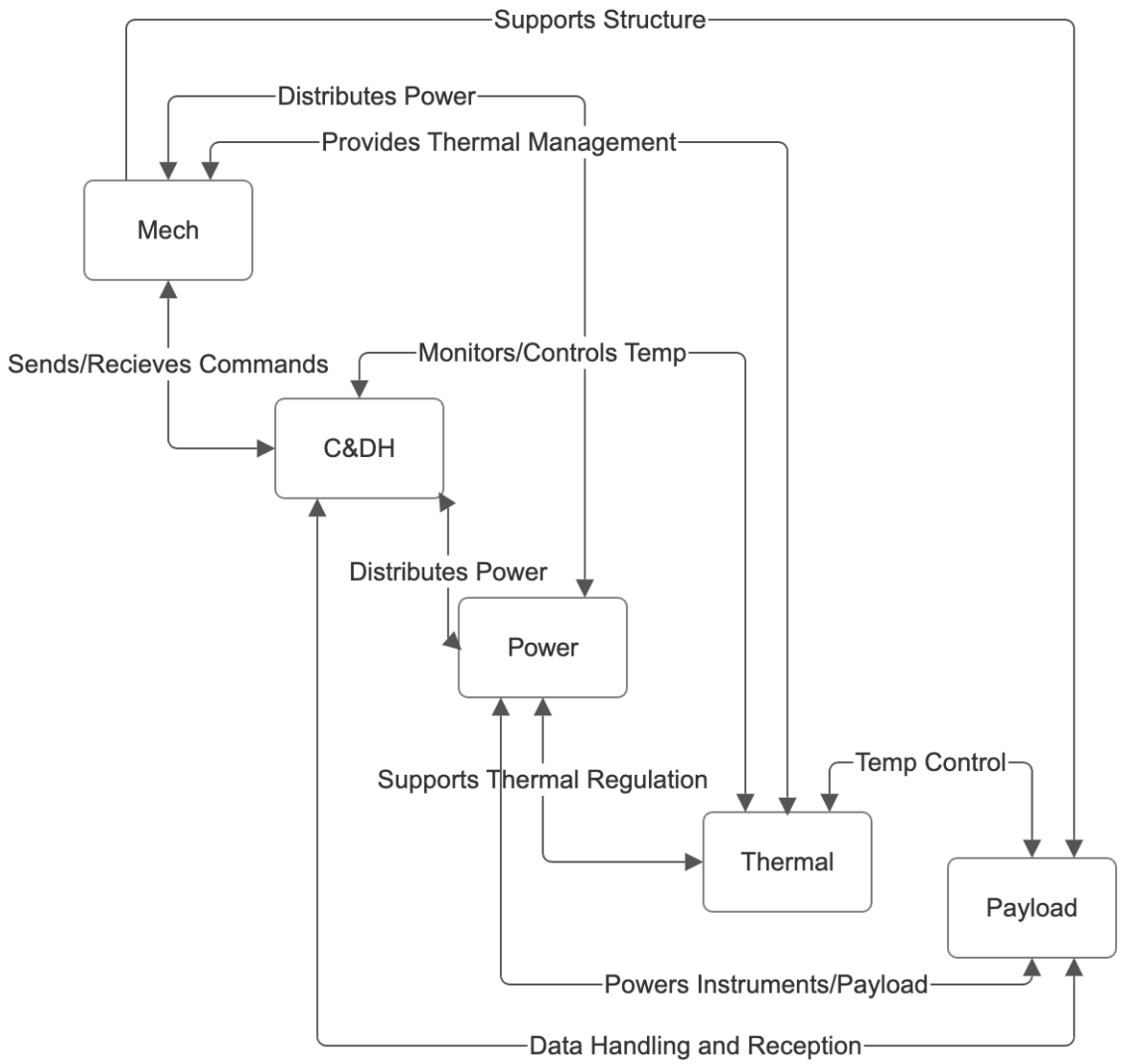


Figure 5 : N^2 Diagram for the Sub-systems

3. Science Mission Plan

3.1. Science Objectives

Our mission is structured around two overarching goals: the Science Mission Directorate (SMD) goal of advancing lunar scientific knowledge and the Exploration Systems Development Mission Directorate (ESDMD) goal of enabling safe and enduring human habitation on the Moon. These goals guide our scientific objectives, which are essential for addressing both immediate exploration needs and broader planetary science inquiries.

For the SMD goal, the mission aims to characterize the distribution and composition of water ice on the lunar surface and subsurface up to 1 meter. Water ice is a key resource for in-situ resource utilization (ISRU), with potential applications in generating oxygen, drinking water, and rocket fuel. Additionally, the mission seeks to investigate volcanic and tectonic activities to understand the Moon's geologic history and identify potential hazards to future exploration. Lunar soil composition will also be analyzed to assess the availability of critical resources such as metals, oxygen-bearing minerals, and helium-3, all of which are crucial for establishing a sustainable lunar presence. Finally, the properties and dynamics of lunar dust will be examined, focusing on its behavior during lander operations and its potential impact on equipment and human health. These objectives are vital for enabling resource utilization and ensuring the safety of lunar infrastructure.

For the ESDMD goal, the mission will evaluate the structural integrity of lunar pit caves to determine their suitability as potential habitats. These pits may offer natural protection from radiation, temperature extremes, and micrometeorite impacts, making them ideal candidates for human habitation. Additionally, in-situ resources within and around the pits will be inspected to assess their utility for sustainable living, including mapping regolith thickness and measuring isotopic content. Lastly, the environmental risks associated with potential habitation, such as radiation exposure, thermal fluctuations, and geological hazards, will be analyzed to ensure long-term safety and viability for human explorers.

To achieve these objectives, the mission incorporates many scientific instruments to achieve each objective. For example, the Lunar Prospector Neutron Spectrometer (LPNS) and Near-Infrared Spectrometer (NIRS) will map water ice distribution, while CP ZM500 vertical seismometer will monitor tectonic activities. Laser-Induced Breakdown Spectroscopy (LIBS) and X-ray fluorescence (PIXL) will analyze soil composition. Ground-penetrating radar (GPR) and LiDAR systems will map subsurface structures and assess pit cave stability. Together, these instruments provide the data needed to address both the SMD and ESDMD goals, ensuring that the mission contributes to advancing lunar science and supporting future human exploration.

3.2. Experimental Logic, Approach, and Method of Investigation

- **Experimental Logic**

Our mission focuses on using advanced instruments, careful mission planning, and efficient data collection methods to meet the science goals set by the Science Mission Directorate (SMD) and the Exploration Systems Development Mission Directorate (ESDMD). The payload is designed to ensure all instruments work together effectively, taking advantage of their unique capabilities to collect detailed and meaningful data. Every subsystem is carefully planned to operate efficiently, stay reliable, and produce high-quality scientific results in the challenging environment of the Moon. This approach will help us pave the way for future missions and contribute valuable insights to the scientific community.

- **Approach to Data Collection and Method of Investigation for each instrument:**

1. **Water Ice and Volatile Characterization (SMD):**

    a. **Instrument(s):** Lunar Prospector Neutron Spectrometer (LPNS) and Near-Infrared Spectrometer (NIRS).
    b. **Integration:** The LPNS measures neutron emissions to map hydrogen abundance across the lunar surface and subsurface, while the NIRS captures near-infrared reflectance spectra (1–3 μm) to identify water ice deposits.
    c. **Method of Investigation:** These instruments will be mounted on the rover and will traverse suspected water-rich areas identified through remote sensing as it will make its way to the pit and once lowered into the pit, it will start measuring the subsurface of the pit. The combination of surface and subsurface measurements will ensure a holistic understanding of water distribution. It may focus on permanently shadowed regions (PSRs) for detailed volatile mapping.

2. **Volcanic and Tectonic Activity Investigation (SMD):**

    a. **Instrument(s):** CP ZM500 vertical seismometer (High-Sensitivity Seismometer).
    b. **Integration:** The seismometer will record ground vibrations over a wide frequency range (0.5–100 Hz), detecting moonquakes and tectonic shifts.
    c. **Method of Investigation:** Seismometers are strategically placed in regions of suspected tectonic activity based on orbital imagery and

thermal data. Continuous monitoring will be performed over several lunar cycles to identify patterns and anomalies.

3. **Lunar Soil and Resource Analysis (SMD):**

    a. **Instruments:** ChemCam (Laser-Induced Breakdown Spectroscopy (LIBS)) and/or Planetary Instrument for X-ray Lithochemistry (PIXL).
    b. **Integration:** ChemCam and PIXL work together to analyze soil samples. ChemCam uses laser pulses to create plasma from the soil, while PIXL provides detailed elemental composition using X-ray fluorescence.
    c. **Method of Investigation:** Soil samples will be collected by a robotic arm and be analyzed for ISRU. Results will be used to map the spatial distribution of key resources like silicon, titanium, and helium-3. It will conduct ChemCam and PIXL analyses for each sample, logging data in real-time.

4. **Dust Property and Dynamic Studies (SMD):**

    a. **Instruments:** High-Speed Particle Tracking Sensors and/or Electrostatic Sensors.
    b. **Integration:** Particle tracking sensors monitor Lunar dust motion during rover operations, while electrostatic sensors measure dust particle charges.
    c. **Method of Investigation:** These sensors will be active during lander and rover maneuvers to observe dust ejecta patterns. Data helps assess dust adhesion and electrostatic lofting behavior.

5. **Lunar Pit Structural and Environmental Analysis (ESDMD):**

    a. **Instruments**: Ground-penetrating radar (GPR) and LiDAR.
    b. **Integration**: GPR provides subsurface layer mapping, while LiDAR creates 3D models of the pit's geometry.
    c. **Method of Investigation**: The instruments are deployed on a rover that explores the pit. GPR scans layers to assess structural stability, and LiDAR generates high-resolution maps of the pit walls for structural analysis.

6. **ISRU Inspection (ESDMD):**

    a. **Instruments**: PIXL and/or Quadrupole Mass Spectrometer (QMS).

b. **Integration**: PIXL detects mineral signatures (1–20 keV energy range), while QMS analyzes isotopic content and volatile concentrations (1–100 amu mass range).
c. **Method of Investigation**: The rover will examine regolith within and around the pit cave over a 100 m^2 area. Samples will then be analyzed with integration times of 0.5–5 seconds to map regolith thickness and mineral distribution at 1–5 cm^2 resolution.

7. **Habitat Feasibility and Environmental Risks (ESDMD):**

a. **Instruments**: Thermal Sensors, Radiation Dosimeters, and/or Spectroradiometers.
b. **Integration**: Thermal sensors monitor diurnal temperature fluctuations, radiation dosimeters measure cosmic ray exposure, and spectroradiometers analyze the energy reflected by the pit walls.
c. **Method of Investigation**: Thermal sensors, radiation dosimeters, and spectroradiometers will be deployed to evaluate temperature fluctuations and radiation risks within the pit cave once the rover is 100-200 meters in the pit cave. Measurements will be taken over a lunar day-night cycle (28 Earth days) to assess habitability risks and inform habitat designs.

- **Data Transmission and Processing:**

Once all the data is collected, it will transmit all the data to Earth periodically using the rover's high-gain antenna. Furthermore, it will store backup data locally to ensure no loss during transmission.

- **Development of the Experimental Plan:**

The experimental plan was developed using JMARS imagery to identify optimal landing sites (Figure 02) and Mare Tranquilis Pit (Figure 4). Images of permanently shadowed regions, suspected tectonic zones, and lunar pits provided visual context and informed instrument placement strategies. Furthermore, as illustrated in the image on the right of Figure 4, the pit shows noticeable overhangs, which strongly suggest that it was formed as a result of a cave-in, likely caused by tectonic or volcanic activity. This formation process indicates a high probability of the presence of an underlying lava tube cave. Such lava tubes are significant geological features that could provide natural shielding from harsh lunar environmental conditions, such as extreme temperature fluctuations, micrometeorite impacts, and radiation. Their potential structural stability and unique formation processes also make them key targets for scientific investigation and exploration.

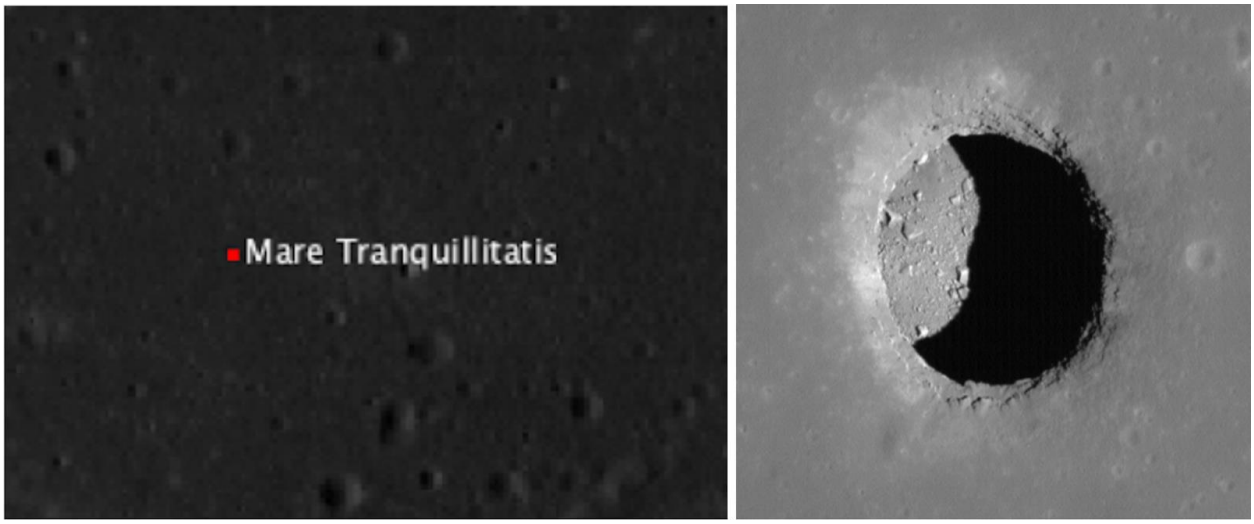


Figure 6: JMARS (Left) and Overhead View (Right) of Mare Tranquillitatis Pit

3.3. Payload Success Criteria

1. **Ground-Penetrating Radar (GPR)**

Success Criteria:

- The GPR must successfully generate high-resolution subsurface maps of lunar lava pits and caves, with a depth resolution of 10-30 cm.
- It should identify structural weaknesses and layer compositions within the subsurface.
- Instrument operations must be stable under the extreme thermal fluctuations on the Moon.

Failure Modes:

- **Data Quality Issues:** Degraded performance in areas with high regolith density, leading to incomplete or inaccurate subsurface mapping.
- **Mechanical Failure:** Antenna misalignment due to thermal expansion, affecting signal strength and accuracy.
- **Environmental Exposure:** Dust intrusion causing operational inefficiencies.

By revealing the subsurface structure and stability of lava pits, the GPR advances our understanding of the Moon's geological processes and evaluates potential sites for human habitation.

2. **Laser-Induced Breakdown Spectroscopy (LIBS)**

Success Criteria:

- The LIBS must accurately detect and quantify elemental compositions of lunar regolith and rocks with sensitivity down to 10^(-4) w/w.
- It should operate efficiently across a wide range of wavelengths (200-900 nm) and deliver real-time compositional data.

Failure Modes:

- **Calibration Drift:** Thermal stress leading to misalignment of laser optics and reduced precision in data collection.
- **Dust Contamination:** Obscuring optical paths, resulting in lower signal-to-noise ratios.
- **Power Issues:** Interruptions in the power supply affecting laser operation.

LIBS contributes to identifying key resources such as oxygen-rich minerals and

volatiles, enabling ISRU and supporting the development of sustainable lunar exploration.

3. Lunar Prospector Neutron Spectrometer (LPNS)

Success Criteria:

- The LPNS must detect hydrogen abundance in surface and subsurface materials down to 10^(-3) by weight, identifying water ice and volatiles critical for ISRU.
- It should achieve spatial resolutions of ~20 km and depth resolution of 1 meter.

Failure Modes:

- **Sensor Degradation:** Damage from radiation exposure leading to reduced sensitivity.
- **Thermal Instability:** Fluctuating temperatures impacting neutron detection accuracy.
- **Communication Lag:** Data transmission delays causing incomplete datasets.

By mapping water ice distribution, the LPNS enhances our understanding of resource availability and supports future lunar infrastructure development.

4. Light Detection and Ranging (LiDAR)

Success Criteria:

- LiDAR must generate high-resolution 3D models of cave interiors, achieving a surface feature accuracy of 1 cm and point spacing of ~1 cm.
- The instrument should remain functional under varying lighting conditions and dust exposure.

Failure Modes:

- **Signal Attenuation:** Dust scattering affecting laser returns and reducing model accuracy.
- **Mechanical Faults:** Misalignment of rotating sensors leading to gaps in data coverage.
- **Thermal Fluctuations:** Interference in sensor calibration due to extreme lunar temperatures.

LiDAR’s detailed mapping capabilities enable the evaluation of structural stability and habitability potential, providing critical data for site selection of lunar bases.

5. Thermal Infrared Spectrometer (TIR)

Success Criteria:

- The TIR must analyze thermal properties of regolith and cave walls with temperature sensitivity of ±0.1°C and spatial resolution of 5 meters.
- It must operate continuously over a lunar day-night cycle to track thermal dynamics.

Failure Modes:

- **Temperature Drift:** Prolonged exposure to extreme conditions causing sensor inaccuracies.
- **Dust Accumulation:** Obstructing the infrared lens and degrading measurement fidelity.
- **Power Deficits:** Insufficient energy availability during critical data collection periods.

TIR data elucidates thermal behavior, contributing to our understanding of lunar heat retention and assisting in assessing environmental risks for human operations.
By meeting these success criteria, the science instrumentation collectively provides a comprehensive dataset to evaluate the potential of lunar lava pits for human habitation and resource utilization, advancing both scientific knowledge and exploration strategies for the Moon and beyond. Let me know if you need further refinement!

3.4. Testing and Calibration Measurements

- **Lunar Prospector Neutron Spectrometer (LPNS):**
  - LPNS will measure background neutron emissions in a region with minimal hydrogen or water ice content, such as the landing site area. This serves as a baseline measurement for calibration. By comparing these readings to pre-launch simulations and known lunar neutron flux data, we can verify the spectrometer's accuracy. Any deviations will initiate adjustments to the data processing algorithm to account for local environmental factors. Testing will include varying the instrument's orientation to confirm consistent results regardless of angular alignment.

- **Near-Infrared Spectrometer (NIRS):**
  - The NIRS will first conduct a scan of a reference area with no expected water ice, such as a sunlit region's regolith. This will help calibrate reflectance spectra against pre-mission baseline data. Additionally, we will use onboard calibration targets with known spectral properties to ensure the instrument's sensitivity in the 1–3 µm range. These tests will verify the spectrometer's ability to distinguish subtle differences in reflectance and ensure accurate identification of water ice deposits.

- **CP ZM500 vertical seismometer:**
  - The seismometer will undergo calibration by measuring lunar ground vibrations caused by known environmental stimuli, such as lander movements or controlled rover operations. This allows us to verify its sensitivity to displacements as small as 1 nm/s^2 across the frequency range of 0.5–100 Hz. A control test will involve deploying a small hammering device near the seismometer to generate vibrations with known amplitudes and frequencies, confirming the instrument's accuracy and response range.

- **ChemCam (LIBS) and Planetary Instrument for X-ray Lithochemistry (PIXL):**
  - To calibrate LIBS and PIXL, we will analyze a calibration sample containing known elemental compositions stored onboard the rover. LIBS will fire its laser on the calibration target, and PIXL will conduct X-ray fluorescence spectroscopy, comparing the results to pre-mission standards. This ensures the accuracy of both instruments in detecting and quantifying key elements such as silicon, titanium, and helium-3. Field tests will include analyzing regolith in a homogenous area to ensure consistency across multiple measurements.

- **High-Speed Particle Tracking Sensors:**
  - The particle tracking sensors will be tested by observing dust dynamics during controlled rover maneuvers, such as turning in

place or driving short distances. The initial observations will focus on particle velocities and trajectories to confirm the sensors can accurately detect particles as small as 50 μm. Calibration will involve comparing the measured data to pre-launch simulations of expected dust motion under similar conditions.

- **Electrostatic Sensors:**
  - Calibration of the electrostatic sensors will involve measuring baseline electrostatic charges in a dust-free environment, such as during the lunar night when particle activity is minimal. This will establish a zero-charge baseline. The sensors will then be exposed to lunar dust stirred up during controlled rover movements to ensure they can accurately measure charges down to ±1 nC.

- **Ground-Penetrating Radar (GPR):**
  - GPR testing will begin with scans of a flat, homogenous regolith area to ensure accurate depth readings and subsurface layer differentiation. Preloaded radar reflectors with known properties will be placed in the rover's path for initial scans, allowing calibration of the instrument's depth sensitivity and spatial resolution. These tests will confirm the system's ability to detect subsurface anomalies, such as potential lava tubes.

- **LiDAR:**
  - The LiDAR system will perform initial scans of a calibration target with known geometry and reflectance placed near the lander. The resulting 3D model will be compared to preloaded design specifications to confirm accuracy. Subsequent tests will involve scanning nearby terrain to ensure consistent performance in mapping surface features with up to 1 cm accuracy.

- **PIXL and Quadrupole Mass Spectrometer (QMS):**
  - The PIXL and QMS instruments will be calibrated using onboard samples with known mineral and isotopic compositions. PIXL will analyze these samples to verify their elemental detection range (1–20 keV), while QMS will measure isotopic concentrations and volatiles to ensure accuracy down to ppm levels. These tests will confirm the instruments' ability to perform reliable in-situ resource analysis.

- **Thermal Sensors, Radiation Dosimeters, and Spectroradiometers:**
  - Thermal sensors will undergo testing by monitoring the lander's shadow temperature, which has predictable thermal behavior. Radiation dosimeters will be calibrated against known cosmic radiation levels recorded by prior missions, such as the Lunar Reconnaissance Orbiter (LRO). Spectroradiometers will use calibration targets with known reflectance properties to validate their sensitivity and spectral range (0.1–2.5 μm).

- **Control Variables:**
    - The calibration process will also consider control variables such as temperature fluctuations, lunar day-night cycles, and lander vibrations. Testing will also ensure that instruments perform consistently across different orientations and locations. Regular re-calibration will occur throughout the mission to account for instrument wear or environmental changes, ensuring accurate and reliable data collection.

3.5. Precision and Accuracy of Instrumentation

- **Lunar Prospector Neutron Spectrometer (LPNS):**
  - The Lunar Prospector Neutron Spectrometer (LPNS) detects hydrogen with a sensitivity of 10 ppm for polar regions and 50 ppm globally, enabling it to identify water ice at levels below 0.01%. This high sensitivity ensures accurate mapping of hydrogen abundance, but minor uncertainties may arise from local variations in cosmic-ray flux or spacecraft spin. The dual helium-3 detector system, with precise calibration at the 765 keV energy interval, mitigates these issues and ensures reliable data for identifying lunar water ice and hydrogen resources. (Feldman 2022)

- **Near-Infrared Spectrometer (NIRS)**
  - The Near-Infrared Spectrometer (NIRS) operates with a spectral accuracy of ±0.01 μm in the 1–3 μm range, allowing it to reliably differentiate water ice from other surface materials. This precision ensures accurate mapping of volatile deposits, but factors like surface reflectance variability and environmental dust scattering could introduce minor uncertainties. Regular recalibration using onboard spectral targets mitigates these effects, ensuring the validity of collected data for resource identification. (Fouchet, Johnson, Forni, Reess 2021)

- **CP ZM500 Vertical Seismometer:**
  - The CP ZM500 Seismometer is a highly sensitive vertical velocity sensor with a nominal velocity sensitivity of 9000 $V/(m \cdot s^{-1})$ and a frequency range of 0.5 Hz to 100 Hz. This precision allows for the accurate detection of seismic activity, such as moonquakes and subsurface movements, critical for understanding lunar tectonics. Minor uncertainties may arise due to environmental vibrations or sensor drift, but these are mitigated through careful calibration using controlled vibrations and comparison with theoretical response data. (Larsonnier 2019)

- **ChemCam:**
  - The ChemCam LIBS (Laser-Induced Breakdown Spectroscopy) operates with high precision to detect elements down to trace levels, with an accuracy of approximately ±10 ppm for major elements and ±30 ppm for trace elements. This accuracy ensures reliable identification of surface compositions and is validated through calibration targets that simulate expected lunar materials. Environmental factors, such as dust or variable atmospheric conditions, are mitigated through frequent recalibration and careful control of laser firing parameters. (Maurice and Drouet 2019)

- **PIXL:**
  - The PIXL instrument is capable of detecting elemental concentrations down to a few ppm, with a spatial resolution of 120 μm. This precision allows for detailed chemical mapping of lunar regolith and rocks, enabling insights into their composition and formation. Regular calibration using inbuilt standards ensures data integrity, and sub-millimeter alignment of

the instrument minimizes positional errors, making the data highly reliable for geological analysis. (Allwood 2020)

- **Electrostatic Sensors**
  - Calibration of the electrostatic sensors will involve measuring baseline electrostatic charges in a dust-free environment, such as during the lunar night when particle activity is minimal. This will establish a zero-charge baseline. The sensors will then be exposed to lunar dust stirred up during controlled rover movements to ensure they can accurately measure charges down to ±1 nC.

- **GPR:**
  - The Ground Penetrating Radar (GPR) achieves a vertical resolution of a few centimeters with high-frequency antennas (500–1,000 MHz) at shallow depths and approximately 1 meter with low-frequency antennas (50–100 MHz) at greater depths. This accuracy ensures reliable imaging of subsurface features like regolith thickness and potential lava tubes.

- **LiDAR:**
  - LiDAR generates 3D models with a surface feature accuracy of ±1 cm. This ensures precise mapping of pit cave geometries and surrounding terrain. Misalignments or shadowed regions may slightly reduce accuracy, addressed by scanning from multiple angles.

- **Quadrupole Mass Spectrometer (QMS):**
  - The QMS measures isotopic concentrations with an accuracy down to ±1 ppm. This precision supports the analysis of volatile content and regolith composition. Instrument drift or contamination may introduce errors, mitigated by cross-referencing measurements with onboard standards.

- **Thermal Sensors, Radiation Dosimeters, and Spectroradiometers:**
  - Thermal sensors measure temperature fluctuations with an accuracy of ±0.1°C, radiation dosimeters detect cosmic radiation with a precision of ±10^-6 W/m²/sr, and spectroradiometers resolve reflectance data with a sensitivity of ±10^-6 W/m²/sr. These precise measurements ensure reliable environmental assessments. Potential errors from sensor drift or extreme temperature variations are minimized through regular calibration.

3.6. Expected Data & Analysis

- **Lunar Prospector Neutron Spectrometer (LPNS):**
  - The LPNS will provide hydrogen abundance maps with a sensitivity of 10 ppm for polar regions and 50 ppm globally. Example data from prior missions like Lunar Prospector show neutron count rate maps, where darker regions correlate with higher hydrogen concentrations, indicating possible water ice deposits. Similar data from this mission would allow for mapping hydrogen-rich areas in permanently shadowed craters. Data analysis would involve comparing neutron count rates between cadmium- and tin-wrapped detectors to isolate thermal neutrons and calculate hydrogen concentrations. Error margins due to detector sensitivity (±10 ppm) will be accounted for using statistical models to ensure data reliability. The results will inform water ice distribution and guide future ISRU efforts.

- **Near-Infrared Spectrometer (NIRS):**
  - The NIRS will provide reflectance spectra in the 1–3 μm range, revealing the presence of water ice and volatiles. Example data typically include reflectance versus wavelength plots, where water ice exhibits strong absorption features around 1.5 μm and 2.0 μm. Expected data will feature similar patterns, but variations could arise due to lunar dust contamination or uneven illumination. The analysis will involve baseline corrections and spectral deconvolution to isolate water ice signals. Instrument precision (±0.01 μm) will be factored into error propagation models to ensure a good interpretation. The figure below shows the data obtained in the System Thermal Test (STT) at the Rover level before shipping to Cape Canaveral for the launch of the Mars rover (Fouchet, Johnson, Forni, Reess 2021). After calibration, it should provide us with similar data for water-ice.

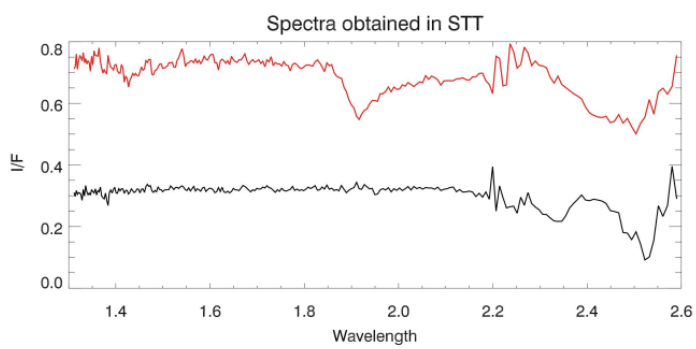


Figure 7: Spectra of Calcite (black line) and Gypsum (red line) obtained in STT

- **CP ZM500 Vertical Seismometer:**
    - The CP ZM500 seismometer will record vertical ground velocity data, capturing seismic events with a sensitivity of 9000 V/(m·s⁻¹) across the 0.5–100 Hz frequency range. The figure below is an example of data including time-series plots of ground velocity changes during seismic events, which provide insights into tectonic activity and subsurface structures (Larsonnier 2019). Similar data from this mission would allow identification of moonquakes and their origins. Analysis involves Fourier transforms to extract dominant frequencies and amplitudes, correlated with expected sensitivity variations (±10%). The results will map tectonic activity zones and assess structural integrity for potential habitats.

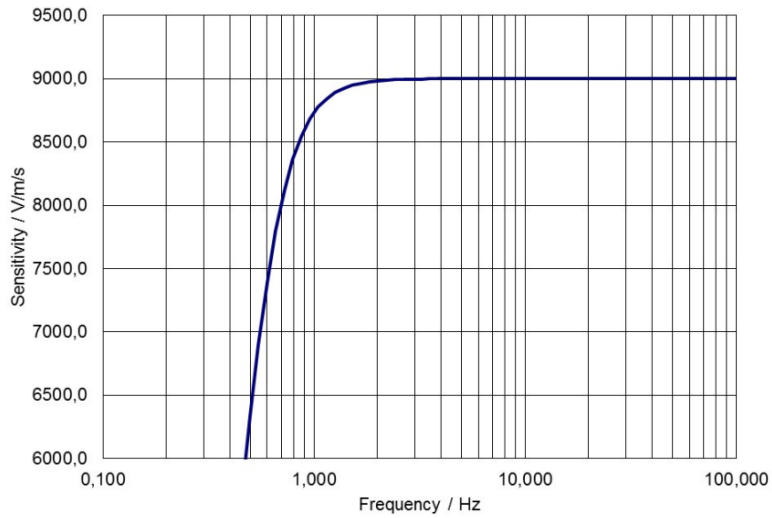


Figure 8: CP ZM500 theoretical amplitude velocity as func. of freq.

- **ChemCam:**
    - The ChemCam instrument will generate spectral data from laser-induced plasma, revealing elemental compositions with a precision of ±10 ppm for major elements. The example graph below includes emission spectra with peaks corresponding to specific elements like Iron and Manganese across 3 different samples taken by the Mars Curiosity rover (Maurice and Drouet 2019). Expected mission data will display similar spectral features but may include variability due to dust interference but will have peaks corresponding to the elements we want to look for after calibration. The analysis will involve spectral line fitting and intensity normalization to quantify elemental abundances. Error calculations based on instrument precision will ensure the validity of the compositional maps, aiding in resource identification.

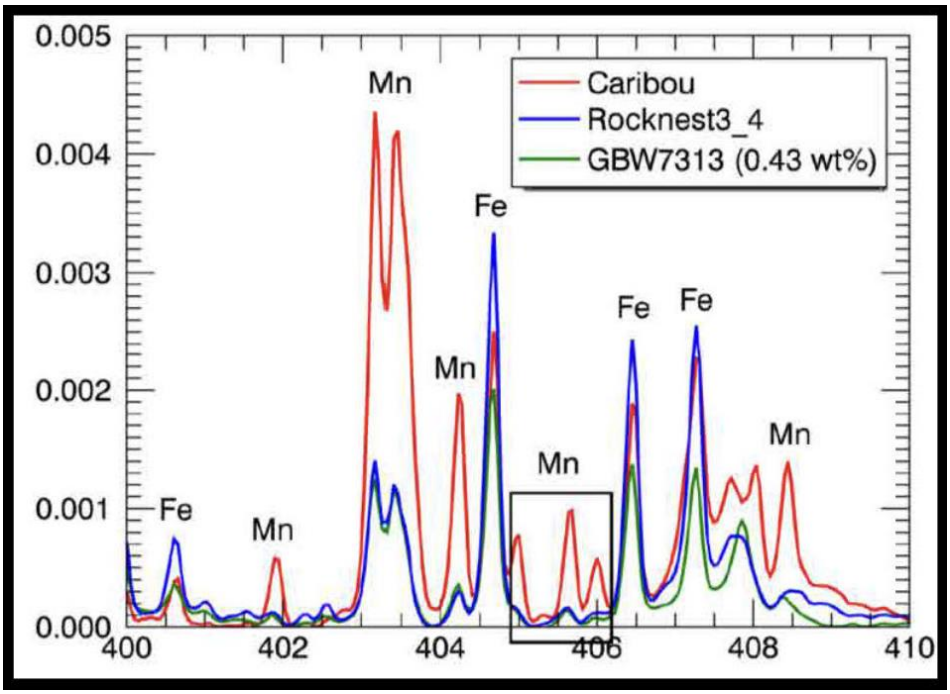


Figure 9: Emission Spectra Graph

- **PIXL:**
  - PIXL will produce high-resolution chemical maps, showing elemental distributions at spatial resolutions of 250 µm. Example data below includes heatmaps with intensity scales for elements like iron, calcium, and magnesium (Allwood 2020). Mission data will resemble these maps but may vary due to regolith heterogeneity. Data analysis involves correlating X-ray fluorescence intensities with calibration standards to determine concentrations, with precision down to a few ppm. These maps will provide detailed insights into lunar geology and resource potential.

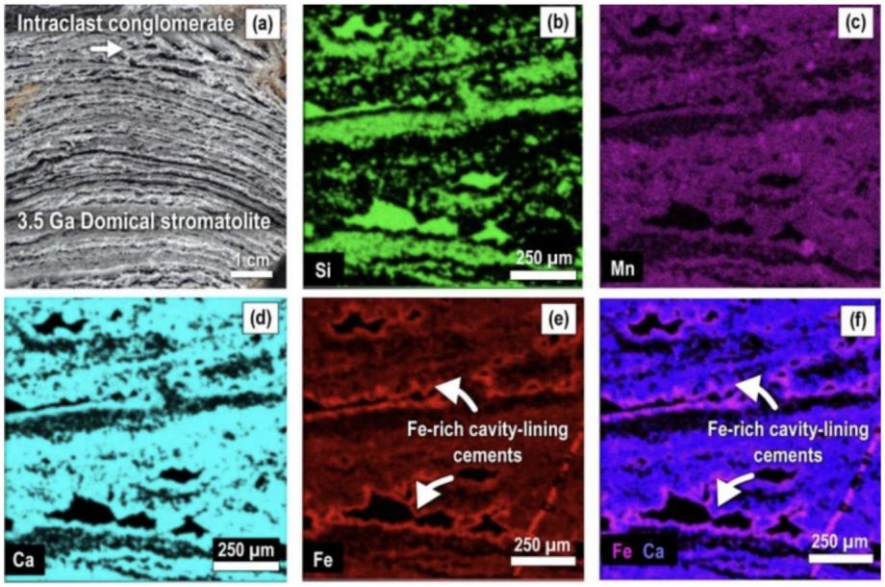


Figure 10: Heatmaps with Intensity Scales

- **GPR:**
  - The Ground Penetrating Radar (GPR) will create radargrams, or time-depth plots, showing subsurface layers, voids, and material changes with centimeter-scale resolution. The illustration below shows an example of WISDOM data both in the frequency and time domains and illustrates the generation of a radargram (Gao and Nie 2019). Similar to the illustrations below, we expect to see data revealing regolith thickness, lava tubes, and areas with water ice, or salts. Using the travel time of radar waves, we can calculate depth and identify materials based on how they reflect the signal. The system can penetrate over 10 meters with an accuracy of about ±10 cm, and calibration will help reduce errors caused by material differences or interference. This data will be crucial for understanding the lunar surface and planning for habitats and resources.

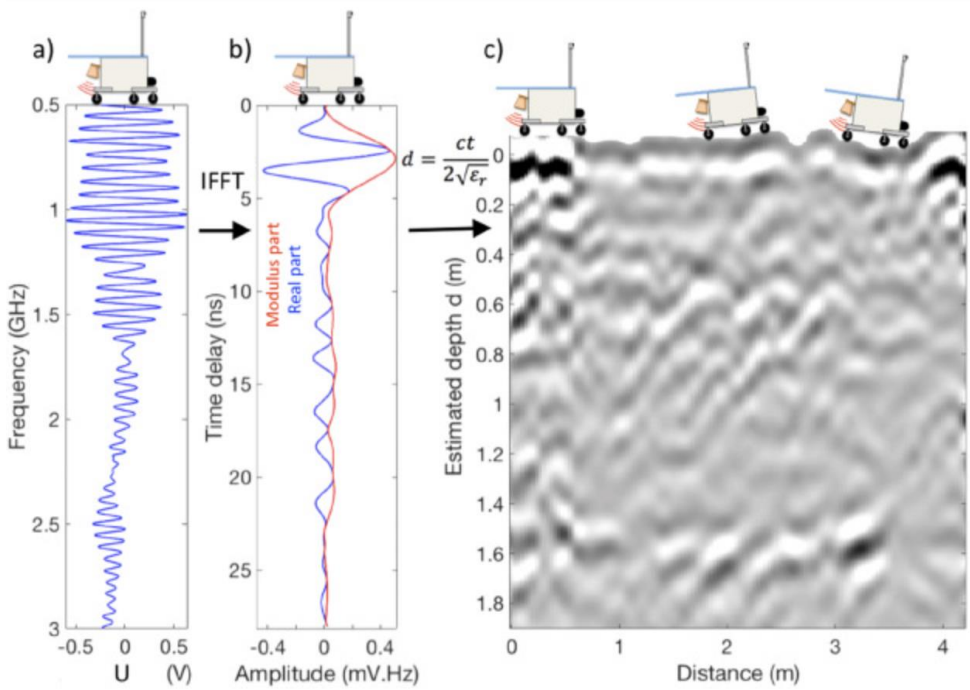


Figure 11: WISDOM Data

- **LiDAR:**
  - The LiDAR system will generate high-resolution 3D maps of the lunar surface, capturing detailed terrain features, regolith layers, and potential cave entrances. Data will be collected as discrete point clouds in .las or compressed .laz formats, which are widely used in geospatial analysis. These point clouds can be processed into raster files like GeoTIFFs for visualization and further analysis. Using Python libraries such as laspy and PDAL, the .las files can be analyzed to extract elevation profiles, calculate slope and aspect, and identify structural features. With centimeter-level accuracy in horizontal and vertical dimensions, the data will help locate safe landing sites, assess pit stability, and support habitat planning. Calibration will ensure reliable measurements by correcting for alignment and environmental variations. (Wasser 2024).

- **Quadrupole Mass Spectrometer (QMS):**
    - The Quadrupole Mass Spectrometer (QMS) will provide data similar to the example shown, where temperature-programmed desorption (TPD) profiles are recorded for specific ionized species such as H_2O^+ (m/z = 18) and CO_2^+ (m/z = 44). The plots below represent ion counts against temperature, highlighting changes in volatile release during different experimental conditions (Zhu et al. 2020). Expected data from the mission would similarly display peaks correlating to desorption events from the lunar regolith under varying thermal and environmental exposures. The analysis will involve comparing ion intensities and desorption profiles under different controlled scenarios to identify and quantify volatiles like water and carbon dioxide. Error propagation will account for sensitivity limits of the QMS, typically down to ppm levels, ensuring reliable quantification. This data will enhance our understanding of volatile abundance and distribution within the lunar regolith, informing in-situ resource utilization strategies.

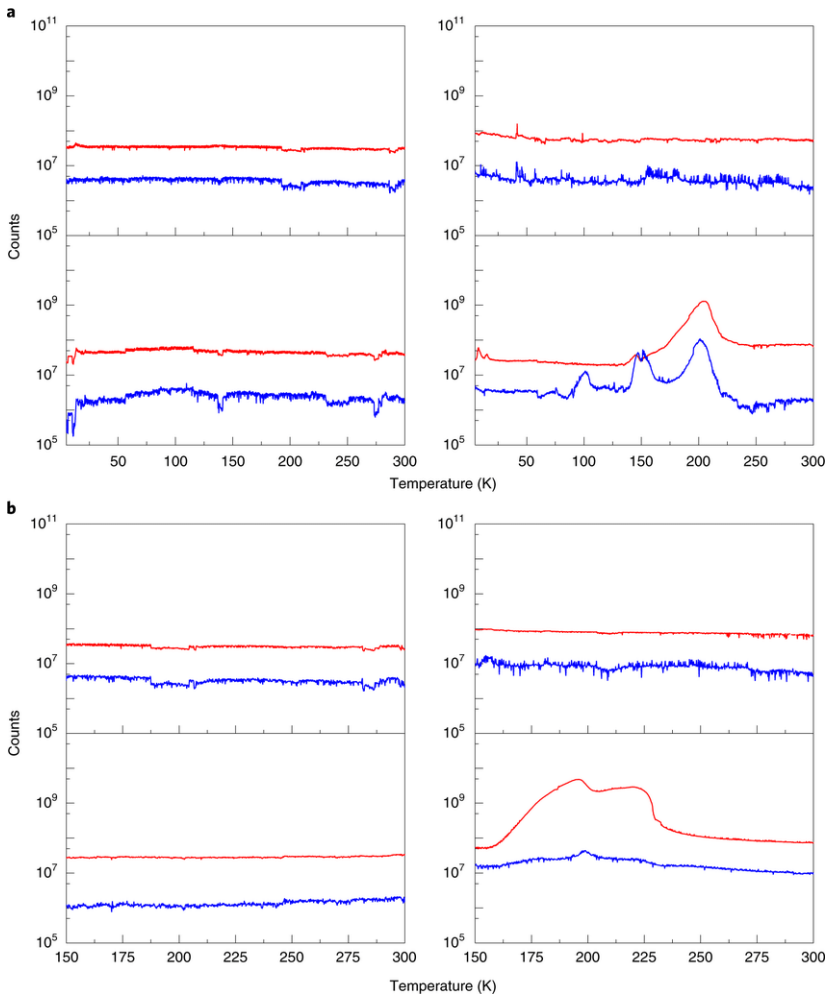


Figure 12: Ion counts against temperature plots

4. Mission Risk Management

4.1. Safety and Hazard Overview

Our team has carefully analyzed potential risks to ensure the mission remains safe and efficient throughout its lifecycle. Using methods like Failure Mode and Effect Analysis (FMEA) and subsystem-level reviews, critical risks have been identified and mitigation strategies were developed which will be listed in this section. While lower-cost missions often involve higher risks due to experimental technologies, focusing on ensuring that critical systems are managed at acceptable levels supports this mission's success.

Thermal risks, such as extreme temperature fluctuations on the lunar surface, are one of the major challenges that could damage insulation and internal systems. To address this, we’re using tested insulation materials and active thermal management systems. Similarly, thermal expansion, which could misalign instruments, is being tackled by using low-expansion materials and calibration routines. Mechanical risks include high radiation in lunar pits and uneven terrain affecting the rover’s mobility. We’ve mitigated these by adding radiation shielding and equipping the rover with advanced stabilization and traversal systems to handle difficult terrain.

Communication and data handling (CDH) risks, like delays from being on the lunar far side or antenna alignment failures, could disrupt operations. To prevent this, we’re using high-gain antennas and relay systems to maintain a strong connection to Earth. Power risks, such as battery inefficiency in extreme cold and potential solar panel malfunctions, are also being addressed. Batteries with integrated heating elements and solar panels designed for flexible deployment ensure reliable power generation. Instrumentation risks, like issues with payload deployment or exposure to lunar dust, are managed by using protective coatings and testing mechanisms to operate reliably in harsh conditions. Finally, programmatic risks, including budget overruns or delays, will be closely monitored with contingency plans in place.

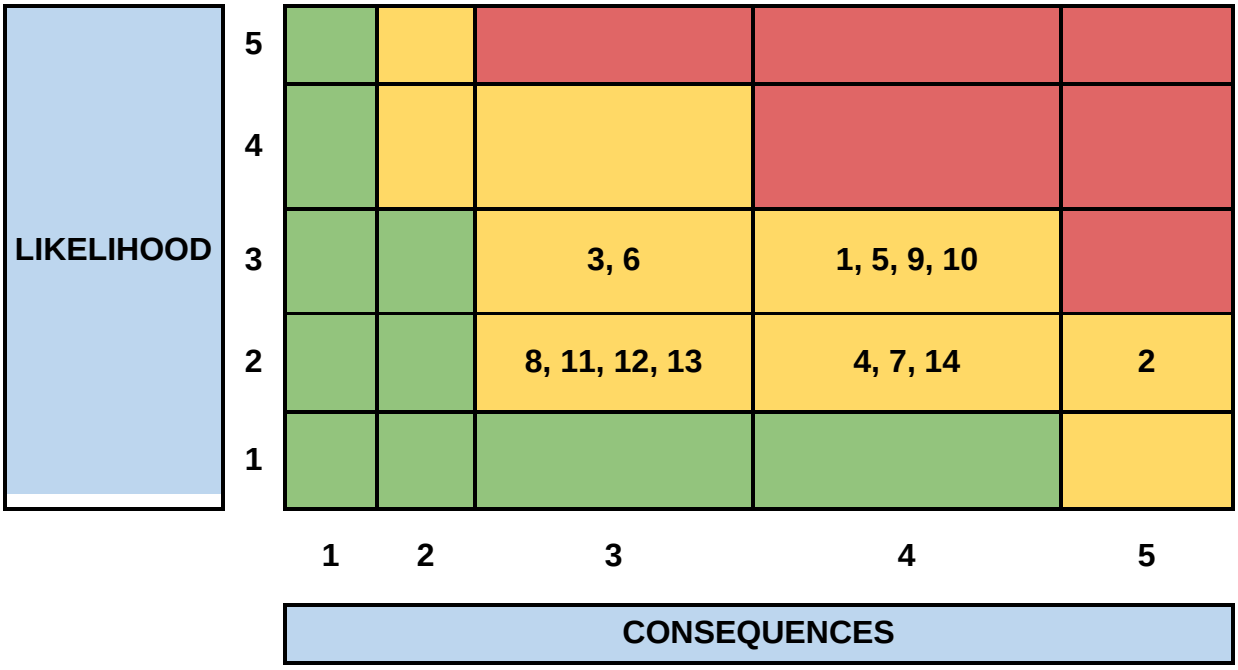
To ensure ongoing risk management, the team will continually monitor and update risks through project milestones, integrating testing data to refine mitigation strategies. Planetary protection protocols will remain a critical component of the mission, ensuring compliance with established standards and minimizing environmental impacts. This risk analysis will form the backbone of our strategy to maintain safety and achieve our objectives.

4.1.1. Risk Analysis

ID	Summary	L	C	Trend	Approach	Risk Statement	Status	Risk Category
1	Thermal	3	4	↓	R	Fluctuating temperatures risk failure in thermal insulation systems.	Active	Technical
2	Mechanical	2	5	↓	R	High radiation in pits could interfere with equipment functionality and harm crew members.	Active	Technical
3	Mechanical	3	3	↓	W	Unstable and uneven lunar terrain may impede rover mobility.	Active	Technical
4	CDH	2	4	↓	M	Maintaining reliable communications prone to issues, due to exploration sites being on the far side.	Active	Technical
5	Power	3	4	↓	M	Maintaining sufficient power for mission-critical tasks may be challenging.	Active	Technical
6	Mechanical	3	3	↓	R	Lunar regolith could cause long-term damage to equipment.	Active	Technical
7	Budget	2	4	↓	M	Mission costs could exceed the budget of $375M.	Active	Programmatic
8	Schedule	2	3	↓	M	Unexpected delays in assembling or manufacturing of complex components.	Active	Programmatic
9	Instrumentation	3	4	↓	M	Compromised mechanisms involving deployment and retractions due to payload issues.	Active	Technical
10	Instrumentation	3	4	↓	M	Failure to protect sensitive instruments from harsh environments and movements.	Active	Technical
11	Power	2	3	↓	M	Deployable solar panels or radiators may experience issues in unfolding or angling optimally.	Active	Technical
12	CDH	2	3	↓	M	Issues with extending or angling the communication antenna could restrict transmission reliability	Active	Technical

13	Thermal	2	3	↓	R	Risk of thermal expansion-Induced misalignment of instruments	Active	Technical
14	Protection	2	4	↓	M	Risks of lunar contamination and failure to meet planetary protection standards.	Active	Programmatic

Table 13: Risk Analysis Table



Thermal Risks

Fluctuating Temperatures and Thermal Insulation Failure

Given that temperatures on the lunar surface fluctuate drastically between day and night, there is a risk of failure in the thermal insulation systems. This would compromise the spacecraft's thermal management system. As a result, internal components could overheat or freeze, leading to critical mission failures. To address this, the team plans to use space-grade insulation materials with proven thermal durability and implement active thermal management systems. Insulation materials will undergo rigorous thermal cycling tests to confirm resilience in extreme conditions.

Thermal Expansion Leading to Instrument Misalignment

Given that lunar temperatures cause rapid expansion and contraction, thermal expansion may misalign sensitive instruments over time. This affects the rover's precision instruments and calibration stability. Consequently, data accuracy may degrade, compromising the mission's scientific objectives. The team intends to use materials with low thermal expansion coefficients and include software-based calibration routines to correct any misalignments caused by thermal stress.

Current Status: Risk likelihood reduced due to successful testing; continued monitoring during integration phase.

Mechanical Risks

High Radiation in Lunar Pits

Given the high radiation levels within lunar pits, there is a risk that radiation will interfere with equipment functionality and harm potential crew members. This impacts radiation-sensitive electronic components and human safety protocols. Radiation interference could lead to equipment malfunctions and increased health risks for crew members, limiting exploration viability. The team plans to shield critical components with radiation-resistant materials and conduct extensive testing in radiation-rich environments to validate component resilience.

Unstable Terrain Affecting Rover Mobility

Given the unstable and uneven terrain on the lunar surface, the rover may experience impaired mobility or even tipping. This could immobilize the rover, halting data collection and exploration capabilities. Failure to navigate the terrain effectively could result in the mission's inability to reach or explore key target areas. The rover will be equipped with gyroscopic stabilization and adaptive traversal algorithms to manage terrain challenges. Its wheels and suspension are designed with abrasion-resistant materials tested on simulated lunar surfaces.

Current Status: Risk likelihood moderately reduced; performance during extended analog testing will further inform final mitigation strategies.

Command and Data Handling (CDH) Risks

Communication Delays on the Lunar Far Side

Given that some exploration sites are on the far side of the Moon, maintaining reliable communication may be challenging. This affects the data handling and transmission systems, essential for real-time mission control. Delays or loss in communication could impede timely data transfers and disrupt mission guidance. The team will employ a high-gain antenna for stronger signals and utilize orbiting relay systems to minimize communication interruptions. Multiple communication tests will be performed to optimize antenna angling and

signal strength.

Antenna Deployment and Alignment Issues

Given the need for antenna deployment on-site, there is a risk that the communication antenna may not extend or align optimally. This directly affects the reliability of the communication link to Earth. Communication failures could result in data gaps, limited command transmission, and potential mission loss. The team is incorporating a robust multi-joint deployment system and testing antenna reliability under thermal and mechanical stress. This approach includes self-locking mechanisms to ensure optimal deployment.

Current Status: Risk mitigation strategies are robust; additional simulations planned to validate performance during obstruction scenarios.

Power Risks

Battery Efficiency in Extreme Cold

Given the extreme cold during lunar nights, batteries may lose efficiency, leading to rapid discharge. This threatens the spacecraft's power subsystem, which supports all mission-critical tasks. Power loss could lead to shutdowns of essential systems, compromising mission success. To counter this, the team will use low-temperature-rated batteries with integrated heating elements for nighttime operation. Extensive thermal testing will validate battery performance under simulated lunar night conditions.

Deployable Solar Panel Malfunction

Given that solar panels must deploy and align under lunar temperature shifts, there is a risk of solar panels not unfolding or aligning optimally. This impacts the spacecraft's power generation capability. Failure to generate sufficient power could shorten mission duration and limit power for instruments. The solar panels will be designed with self-locking joints and flexible deployment mechanisms, tested to withstand thermal cycling. Panel durability will be confirmed through repeated deployment tests.

Current Status: Risk likelihood significantly reduced following successful prototype testing.

Instrumentation Risks

Payload Deployment and Retraction Challenges

Given the precision required for payload mechanisms in harsh lunar

conditions, compromised mechanisms could hinder deployment and retraction. This would impact scientific instruments and their ability to gather data. Mechanical failure could result in incomplete data collection or inability to retrieve instruments safely. To reduce risk, the team will use materials tested for extreme conditions, ensuring that moving parts have appropriate clearance and lubrication to function in low gravity. Testing will also validate instrument stability.

Environmental Protection for Sensitive Instruments

Given exposure to lunar dust and temperature extremes, sensitive instruments may be inadequately protected from the environment. This could compromise the operational reliability of scientific instruments. Loss of instrument functionality would hinder data accuracy and completeness, impacting mission objectives. Instruments will be protected with dust-resistant coatings and temperature-insulated casings. The design includes a monitoring system for environmental exposure, enabling adjustments to mitigate impact.

Current Status: Risk likelihood stable but monitored closely; proactive management continues to mitigate impacts.

Programmatic and Planetary Protection Considerations

- **Programmatic Risks:** Monitoring mission costs against the $375M budget, tracking any unforeseen delays in component manufacturing, and ensuring that resources are allocated efficiently will be essential. Risks related to mission scheduling, such as unexpected component delays, will be actively tracked, and alternate plans will be prepared to mitigate any impacts.
- **Planetary Protection:** A key environmental consideration is ensuring compliance with planetary protection protocols to avoid contamination of the Moon or Earth. Procedures for spacecraft sterilization, tracking of contamination risks, and adherence to international guidelines will be integrated into the risk management plan.

  Current Status: Risks mitigated to acceptable levels; ongoing adherence to established standards ensures compliance.

Accepted Risks

- **Thermal Expansion of Non-Critical Components:** Minor alignment shifts in non-critical components were identified as low-severity risks. Accepted due to low impact on mission objectives and successful compensatory measures during testing.

- **Single-Point Mechanical Failure in Secondary Payload Systems:** Backup systems for secondary payloads were deemed non-essential, and risks are accepted due to mission cost constraints.

Risk Tracking and Updates

Risks that are not accepted, such as those related to primary systems or critical science objectives, are tracked rigorously through updated FMEA and subsystem reviews. Likelihood and severity ratings are re-evaluated during each project milestone, with mitigation strategies adapted based on new data. Testing results and operational insights from the PDR phase will inform risk management strategies in subsequent design and implementation phases, ensuring mission reliability and success.

4.1.2. Failure Mode and Effect Analysis (FMEA)

Function	Failure Mode	Effects	Sev	Cause	Occ	Prevention	Det	RPN	Actions
Mechanical	Equipment Functionality and Crew risks	Missed Timeline	10	Unknown and High Radiation	4	Using radiation analayzers before sending human and other equipment	3	120	Build another Detection system just for radiation
	Lunar Regolith	Long-Term Operation Damage	5	Lunar Regolith	8	Using High quality equipment and shielding from Lunar Regolith	4	160	Increase funding for better quality materials
Thermal	Insulation System	Missed Timeline	7	Fluctuating Temperatures	5	Changing and improving Insulation materials used	9	315	Change in material
Power	Sufficient Power	Lack of functioning of equipment	10	Electrical Failure	4	Checking if batteries and electrical equipment are met with the required standards	8	320	Improving Electrical equipment
Instrumentation	Payload	Compromised mechanisms involving deployment of and retractions due to payload	6	Mechanism Failure	2	Simulating environment for all mechanisms.	10	120	Change mechanical equipment used for paylaods
	Protection	Instrument failure due to harsh environments and movements.	7	Harsh Enviornments and movements	4	Testing Equipment in harsh and rigorous enviornments	8	224	Use high grade shielding equipment

Figure 13: FMEA Chart

A. Mechanical

One of the most serious concerns is the possibility of equipment failure due to the harsh radiation environment on the Moon. If we fail to account for high radiation levels, it could cause significant delays or force us to miss critical mission milestones, with a severe impact on both the safety of the crew and the functionality of the spacecraft. Radiation is a silent, invisible threat, but by using radiation detectors before sending humans and equipment to the lunar surface, we can minimize this risk. We also need to create specialized radiation detection systems to ensure that we can detect potential problems early. While we can't completely eliminate the risk, planning for it with the right equipment, including shielding, will help protect our team and equipment.

Lunar regolith, the fine, abrasive dust found on the Moon's surface, poses a significant risk to equipment during long-term operations. Over time, regolith can cause damage to mechanical systems, sensors, and other critical components, potentially leading to system failure or degradation. This is a moderate-severity risk (rated 5), but it is highly likely (occurrence of 8) due to the pervasive nature of regolith on the lunar surface. The best prevention for this is the use of high-quality, durable materials and shields designed to protect equipment from regolith abrasion. It is also important to use regolith-resistant coatings and filters on moving parts and sensitive instruments. Detection involves monitoring equipment for signs of wear and damage due to regolith exposure, and any issues should be addressed promptly. The risk can be significantly reduced by investing in advanced materials and technologies to shield the spacecraft and instruments, ensuring that the mission can continue safely and efficiently for extended periods.

B. Thermal

Thermal fluctuations on the Moon can be extreme, with temperatures ranging from over 100°C in direct sunlight to -150°C in the shadowed regions. Failure of the spacecraft's insulation system to properly regulate temperatures could lead to mission delays or even the failure of critical systems. This is a high-severity risk (rated 7) due to the significant impact on the mission timeline, as well as the potential to damage equipment or life-support systems. The cause of thermal failures often stems from fluctuating temperatures that the current insulation system cannot handle. To prevent such issues, insulation materials need to be regularly reviewed and improved to ensure they meet the demands of extreme lunar conditions. Using advanced thermal protection materials designed specifically for space exploration will help mitigate this risk. Detection of thermal failures can be done through monitoring temperature readings across the spacecraft, and any discrepancies would be addressed through corrective actions. Upgrading insulation materials is a vital step to reduce the likelihood of thermal failure and ensure the mission remains on schedule.

C. Power

Power failure, particularly in the form of electrical failures that cause key equipment to stop functioning, is one of the most critical risks faced by the mission. If the spacecraft loses power, either due to battery malfunction or issues with electrical components, it could result in non-functioning equipment, a breakdown in communication, and a failure to execute mission-critical tasks. The severity of this failure is rated 10 because it could halt the entire mission. Causes include improper electrical system checks or failure to meet the required power standards for mission operations. Preventative measures include rigorous testing of all electrical systems and ensuring that batteries and power supply systems meet the necessary specifications and safety standards. This includes inspecting wiring, battery health, and power output before and during the mission. Detection of power failures can be done through constant monitoring of the spacecraft’s power systems, with immediate corrective actions taken if needed. Ensuring power redundancy and more resilient electrical components will help minimize the occurrence of power failures, providing greater reliability for mission success.

D. Instrumentation

The payload mechanism’s deployment and retraction systems are integral to the mission’s success, as these systems ensure that scientific instruments and other

payloads are properly deployed and function as intended. A failure in these mechanisms could compromise the entire mission by preventing payloads from being deployed or retracted at the necessary times. While the likelihood of failure is low (rated 2), the severity (6) still warrants significant attention. The failure could occur if the payload mechanisms are not adequately tested for the extreme lunar environment, leading to mechanical failure when exposed to harsh conditions. Preventative measures include rigorous simulations and testing of the deployment and retraction systems in conditions that replicate the lunar environment. This would ensure that the mechanisms can withstand the stresses of the mission. Detection is relatively straightforward (rated 10) because any malfunction in the payload system would be immediately visible during operations. If failure does occur, switching to more robust mechanical equipment for the payload system could be considered to ensure continued mission success.

The protection of sensitive instrumentation against harsh lunar environments is a critical challenge. Failure of protection systems could lead to the lack of functioning of key instruments, which would significantly impact the mission's scientific objectives and data collection. The severity of this failure is rated 7, as it could cause mission delays or render certain experiments or instruments unusable. The primary cause is the failure to fully account for the harsh environmental conditions during the design phase, such as excessive heat, dust accumulation, or radiation exposure. Preventative strategies include extensive testing of instruments in simulated lunar environments to verify that protection systems can handle these conditions. Moreover, using high-grade shielding and robust materials for the instrumentation will ensure it is protected. Detection can be accomplished through continuous monitoring of the instruments, and the protection systems can be checked regularly for wear and tear. Upgrading to more advanced shielding and ensuring that all equipment meets the stringent demands of the lunar environment will help prevent failure and ensure the mission's scientific goals are achieved.

4.1.3. Personnel Hazards and Mitigations

During the manufacturing, integration, and testing phase of equipment used for lunar missions, personnel are exposed to various potential hazards, requiring them to follow safety standards to minimize risk. Some of these risks and their associated mitigation are shown below:

A. Machinery and Manufacturing Hazards

During the machinery and manufacturing process, there are several risks such as cuts, punctures, or crushing from using heavy machinery. To mitigate these risks, all machine operators must receive proper training and certification, ensuring they are familiar with safe machine operation. Workers should always wear personal protective equipment, including gloves, safety goggles, and steel-toed shoes, to protect against injuries. Additionally, safety barriers or shields should be installed around hazardous machinery to prevent accidental contact by any personnel. Inspections must be conducted as required for any machinery and equipment during the required period as it ensures safety and functionality. Strict protocols must be enforced, especially for high-powered machinery, to minimize risks. During manufacturing, personnel might be exposed to harmful fumes, dust, and particles that can occur during operations such as welding, soldering, or cutting these contaminants can pose significant health risks. Ventilated systems must be maintained inside the facility to remove fumes and dust particles from the building plant. Hazard assessments should also be performed routinely to identify and address any potential risks.

B. Thermal Hazards

Thermal hazards, such as burns from hot surfaces or equipment, are a risk during the testing of high-temperature components or when handling hot machinery. To mitigate these risks, workers should be equipped with heat-resistant gloves, aprons, and face shields to protect against burns and other injuries. Areas with elevated temperature risks must be marked to alert workers to potential hazards. Adequate training should be provided on how to safely handle hot equipment and conduct high-temperature tests. Thermal insulation and shields should be used around hot components or machinery to prevent accidental exposure to heat, ensuring that workers are not at risk of burns.

C. Electrical Hazards

Electrical hazards such as electric shock or arc flash can occur during the testing and integration of electrical components, particularly high-voltage systems. To

prevent electrical accidents, all electrical work must be performed by trained personnel who adhere to safety codes and industry standards. Workers should use insulated tools and wear appropriate PPE, including rubber gloves and dielectric boots when working with electrical systems. Lockout/tagout (LOTO) procedures must be implemented during testing to prevent accidental energizing of circuits, ensuring that equipment is properly de-energized before maintenance or testing begins. Ground-fault circuit interrupters (GFCIs) should be installed, and routine inspections of electrical systems must be carried out to identify and address any potential risks.

D. Radiation Exposure

During the testing of radiation-sensitive components, such as sensors, LIDARS, or power systems, there is a potential risk of exposure to harmful levels of radiation. To prevent exposure, strict protocols should be followed when handling and testing radiation-emitting materials and components. Shielding materials should be used around systems that emit radiation, and containment areas must be established for testing to prevent contamination. Personnel exposure should be minimized by creating restricted access zones during high-radiation tests, and radiation levels should be regularly monitored to ensure they remain within safe limits. Additionally, workers must wear appropriate PPE, such as lead aprons and radiation badges, to protect themselves from exposure.

E. Fire and Explosion Hazards

Combustible materials, flammable chemicals, or electrical systems in the manufacturing and testing areas can present fire and explosion risks. These hazards must be managed by storing flammable chemicals in fireproof cabinets and following strict fire safety regulations. Fire suppression systems, such as sprinklers or fire extinguishers, should be installed in critical areas where fire risks are highest. Regular fire drills and training must be conducted to ensure that all personnel can respond quickly and effectively in the event of a fire. Electrical systems must be installed and tested by qualified personnel to prevent shorts, overheating, or other electrical malfunctions that could lead to fires or explosions.

5. Activity Plan

5.1. Project Management Approach

Overview of Personnel Requirements
The success of the LSpace mission concept relies on a dedicated and highly specialized team of individuals with expertise across various domains. The personnel involved in the mission are allocated across different phases of the mission lifecycle, with each phase requiring a specific set of skills to address the needs of the mission. The following discusses the number of personnel needed per phase and fiscal year (FY), the rationale for supporting these numbers, and the specific roles each team member will play.

Pre-Mission Phase: During the pre-mission phase, the team focuses on mission design, initial research, system development, and risk mitigation strategies. A large portion of the team will be involved in this phase to establish the mission's foundation.

- **Personnel Count**: 30-35 individuals
- **Key Roles**: Project Manager, Chief Scientist, Systems Engineers, Mechanical Engineers, Thermal Engineers, Electrical Engineers, Scientists (Astrophysicists), Program Analysts, Mission Assurance Specialists, and Outreach Officers.
- **Rationale**: This phase requires significant technical expertise in system design, environmental analysis, and mission strategy. Program analysts and scientists will work closely to finalize the mission's research objectives, while engineers and technical staff will begin developing systems and subsystems.

Development and Testing Phase: This phase involves detailed design, system testing, and validation, as well as the integration of systems into the spacecraft. Specialized subteams will work on their respective components, such as propulsion, thermal management, and communication systems.

- **Personnel Count**: 45-55 individuals
- **Key Roles**: Systems Engineers, Mechanical Engineers, Electrical Engineers, Mission Assurance Specialists, and key scientists involved in subsystem testing.
- **Rationale**: The focus will shift to system integration, testing, and validation, requiring more specialized roles for each system. Teams working on individual subsystems will become more autonomous, and system-level engineers will lead the integration efforts.

Launch and Operations Phase: In this phase, the team focuses on final mission preparations, launch procedures, and the first months of operational monitoring and control. Personnel will be primarily responsible for real-time problem-solving and ongoing mission adjustments.

- **Personnel Count**: 30-35 individuals
- **Key Roles**: Systems Engineers, Project Manager, Chief Scientist, Mission Assurance Specialists, Outreach Officer.
- **Rationale**: The team size reduces during this phase as many systems and subsystems are established and ready. However, active monitoring and response to mission anomalies require critical operational staff.

The Rationale for Personnel Numbers

The rationale behind these numbers is based on the complexity of tasks in each phase. During early phases (pre-mission and development), the need for interdisciplinary work is highest, and thus, a larger team is required. However, as the mission progresses toward launch and operations, many systems are finalized, reducing the overall number of personnel required for real-time support and monitoring.

Importance of Personnel

Each member of the team plays a vital role in ensuring the success of the mission. The personnel requirements are not just based on skill sets but also on their ability to collaborate and lead specific mission aspects. The following outlines the significance of each role:

- **Project Manager (Ashi Poorey)**: The Project Manager oversees the entire mission, ensuring that all phases align with the mission's objectives and timelines. The Project Manager is also responsible for making critical decisions, managing the team, and ensuring that resources are allocated efficiently.

- **Chief Scientist (Jesus Garcia Garcia)**: The Chief Scientist provides the scientific vision and direction for the mission. They are responsible for setting research goals, ensuring that scientific objectives align with the mission's technical capabilities, and overseeing the scientific integrity of the mission.

- **Deputy Project Manager of Resources (Len Johns Shaji)**: This role focuses on managing resources, including personnel and materials. They help maintain the mission's schedule, budget, and overall coordination, ensuring that all systems function as planned.

- **Lead Systems Engineer (Jay K)**: The Lead Systems Engineer manages the technical aspects of the spacecraft's design, integration, and testing. They ensure that the spacecraft's various systems work together efficiently and oversee the overall engineering efforts of the team.

- **Scientists (Urva Vashi and Jesus Garcia Garcia)**: Scientists play a critical role in defining mission goals and providing technical input. Astrophysicists and other scientific experts are necessary for identifying potential research areas, testing equipment, and interpreting mission data.

- **Program Analyst (Rohaba Sajjad)**: The Program Analyst supports the mission by conducting data analysis, performing mission evaluations, and helping to optimize mission strategies through continuous feedback.

- **Mission Assurance Specialists (Sreeram and Raghav Ramesh)**: These individuals focus on ensuring that the mission adheres to high standards of safety and reliability. They are responsible for mitigating risks, performing quality assurance, and developing procedures to avoid mission failures.

- **Outreach Officer (Vanina Nguemsop)**: The Outreach Officer is responsible for managing communication between the mission team and external stakeholders, including the public, educational institutions, and media. Their work promotes the mission and educates the public about its importance.
- **Mechanical Engineers (Jay K)**: Mechanical engineers are essential for designing and testing the physical systems of the spacecraft, including propulsion, mobility, and structural integrity. They ensure that all systems meet the demands of the lunar environment.

- **Electrical Engineers (Vanina Nguemsop)**: Electrical engineers are responsible for designing the spacecraft's electrical systems, ensuring power generation, storage, and distribution are optimized. They play a key role in supporting communication and operational control.

- **Thermal Engineers (Raghav Ramesh)**: Thermal engineers are crucial for managing the spacecraft's heat exchange systems, ensuring that it operates within safe temperature limits across various environmental conditions on the Moon.

- **Computer Hardware Engineer (Benita Benjamin)**: This role focuses on designing and implementing the spacecraft's computing systems, ensuring that the hardware can handle the mission's computational requirements.

Subteams/Groups

To effectively manage the complexity of the mission, the team is divided into sub teams responsible for different technical and programmatic aspects. Each subteam has its specific tasks and levels of autonomy.

- **Systems Engineering Subteam**: Led by the Lead Systems Engineer, this group is responsible for the overall integration of the spacecraft systems. They ensure that each system (thermal, electrical, propulsion) is designed to meet mission specifications and function cohesively.
- **Mechanical Engineering Subteam**: This group is responsible for designing the physical systems and subsystems, including the rover and structural components. They work closely with the systems engineering team to ensure compatibility with other spacecraft components.
- **Electrical and Computer Hardware Engineering Subteam**: Led by Electrical Engineers and the Computer Hardware Engineer, this group designs and tests the spacecraft's electrical and computing systems. They have a significant amount of autonomy when testing hardware, but they will collaborate with the overall systems engineering group to ensure compatibility.
- **Science and Research Subteam**: Comprised of scientists, this group determines the scientific goals of the mission and ensures that instruments are designed to collect meaningful data. This team works closely with engineers to ensure scientific instruments are integrated and calibrated properly.
- **Programmatic and Risk Management Subteam**: This team focuses on managing mission risks, budget tracking, and ensuring the mission stays on schedule. They are responsible for assessing and mitigating risks and reporting to the Project Manager.
- **Mission Assurance Subteam**: Ensures that all systems and procedures meet mission standards and quality expectations. They play a critical role in maintaining mission integrity and avoiding failures.
- **Outreach and Communications Subteam**: This team handles public relations and external communications. Their goal is to ensure that stakeholders are kept informed and engaged throughout the mission's development and execution.

5.2. Mission Schedule

5.2.1. Schedule Basis of Estimate

Pre-Phase Considerations:

- Phases A and B provided a robust foundation of feasibility analysis, trade studies, and conceptual design, enabling the project to proceed confidently into Phase C-F. These earlier efforts included assessments of site selection, instrumentation feasibility, and risk analysis.
- As these phases are considered complete, the focus shifts to detailed design, hardware fabrication, and mission implementation.

Milestone-Oriented Structure:

- The schedule is built around key milestones, including the Preliminary Design Review (PDR), Critical Design Review (CDR), and Mission Readiness Review (MRR). Each milestone acts as a decision gate, ensuring the mission progresses only when technical and programmatic readiness is confirmed.
- A 2-week buffer has been included for each milestone to accommodate any review feedback or unforeseen delays.

Subsystem Development and Testing:

- The timeline incorporates subsystem-level activities such as thermal testing, payload integration, and communications validation. Testing will occur in simulated lunar environments, including analog testing at Black Point Lava Flow in Arizona and drop testing at NASA Glenn Research Center.
- Each subsystem undergoes iterative testing to validate performance in vacuum and temperature-controlled conditions before integration into the final spacecraft.

Phase-Specific Timelines:

- **Phase C (Design Completion)**: Focused on finalizing designs for all spacecraft components and conducting detailed design reviews. The estimated duration is 12 months, including multiple iterations and external peer reviews.
- **Phase D (Assembly, Integration, and Testing)**: Spanning 18 months, this phase includes hardware fabrication, system integration, and environmental testing to ensure the spacecraft can withstand launch and lunar conditions.
- **Phase E (Mission Operations)**: Planned for a 6-month duration post-landing, this phase involves data collection, analysis, and relay back to Earth. The operational timeline is designed around power availability and thermal constraints on the lunar surface.

Contingency and Risk Management:

- Buffer periods totaling 10% of the total schedule are embedded across all phases to mitigate risks associated with component delays, testing failures, or unforeseen environmental challenges during field testing.

Tool Utilization:

- Advanced project management tools, including Gantt charts and critical path analysis, were used to identify dependencies, optimize workflows, and maintain alignment with the March 2030 launch readiness date.

Operational Planning:

- Surface operations, constrained by power availability and communication bandwidth, are scheduled to maximize data collection efficiency within a limited timeframe. Power management systems ensure continuous operation during lunar days and nights.

5.2.2. Mission Schedule

The mission schedule is divided into four main phases once Phases A and B are completed, each with estimated start and end dates based on the mission's design and operational needs. Furthermore, our first phase after Phase B, Phase C, will start one month after Phase B ends, around January 25, 2024. If no delays occur, the shortest time to complete the mission is as described in the schedule estimation table and the Gantt Chart, and the timeline for these phases is as follows:

- **Phase C: Final Design and Fabrication:**

This phase is estimated to run from January 25, 2025, to June 24, 2027, spanning approximately 2.5 years. During this phase, the mission's design will be refined and finalized, with extensive collaboration between engineering teams to address technical challenges. The Critical Design Review (CDR) will serve as the pivotal milestone, ensuring that all design specifications meet the mission's objectives. This phase also involves the development of detailed plans for fabrication and testing, laying a solid foundation for the subsequent stages. By its conclusion, the mission's design will be finalized, and necessary resources will be allocated for the build phase. This phase Subsystem Design Completion will run from January 25, 2025, to March 24, 2025, followed by Material and Component Selection from March 25, 2025, to August 24, 2025. Simulation and Final Modeling will be carried out from August 25, 2025, to October 24, 2025, ensuring detailed design validation, and Fabrication Planning will take place from October 25, 2025, to December 23, 2025. The First Program Implementation Review (PIR) is scheduled for December 24, 2025, before the Critical Design Review (CDR) begins on December 25, 2025, and spans until June 24, 2026, with the CDR Submission Deadline on June 24, 2026. The phase concludes with Preliminary Assembly Planning from June 25, 2026, to January 24, 2027, preparing for the next stage. The Gantt chart below reflects this phase with Blue colored corresponding to tasks tailored for the Engineering (Engineering and/or Technician) Team(s), Green corresponding to the Science Team, Red corresponding to the Programmatics (Administration and Management) Team(s) and Orange corresponding to all Teams.

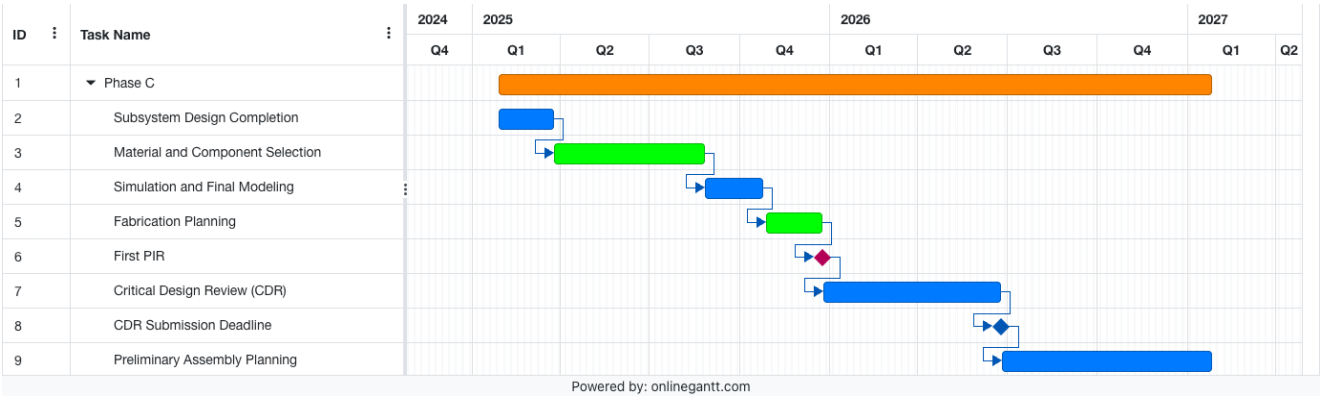


Figure 14: Phase C Gantt Chart

- **Phase D: System Assembly, Integration & Test, Launch & Checkout**

Beginning on January 25, 2027, and continuing until March 4, 2030, this phase will last approximately 3 years. During this time, the mission focuses on assembling and testing all components under simulated conditions to ensure functionality and reliability. The phase begins with the System Integration Review (SIR), running from January 25, 2027, to February 25, 2027, a critical checkpoint to verify readiness for full system integration. The SIR Submission Deadline is set for February 26, 2027, ensuring that the review's findings are documented and addressed.

Following the SIR, Instrumental Acquisitions will occur between March 1, 2027, and August 31, 2027, during which all necessary instruments and components are procured, ensuring they meet the mission's design specifications. Afterward, Integration Testing is conducted from September 1, 2027, to February 27, 2029, validating the performance of integrated systems under mission-simulated conditions. The Second Program Implementation Review (PIR) is scheduled for February 28, 2029, to assess progress and confirm readiness for the next steps.

The next stage involves Rover Assembly and Testing, running from March 1, 2029, to June 30, 2029, during which the rover is built and tested independently. The Rover Test, a critical milestone, is scheduled for June 28, 2029, ensuring its readiness for deployment. The Operational Readiness Review (ORR) follows, taking place between July 1, 2029, and July 31, 2029, confirming that all systems are operational and ready for final validation. The phase concludes with Final System Validation, which runs from August 1, 2029, to January 31, 2030, to ensure all mission systems function as intended before launch. The spacecraft is scheduled to Reach the Launch Location by February 25, 2030, and the Launch Window is planned for the entire day of March 1, 2030, with the launch expected on the same date. The phase culminates with a Successful Landing and Rover Deployment on March 4, 2030, marking the transition to the mission operations phase. Following the same color scheme, the gantt chart below represents the Phase D schedule.

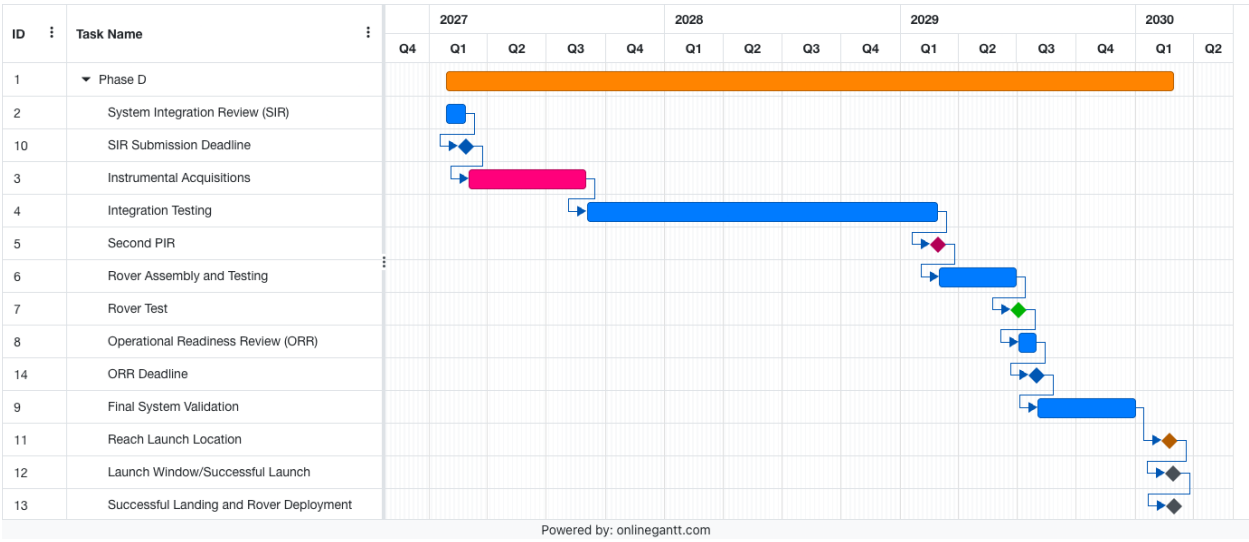


Figure 15: Phase D Gantt Chart

- **Phase E: Mission Operations**

Starting on March 5, 2030, and extending through January 24, 2032, this phase provides approximately 2 years for active mission operations. The primary milestones include the rover's successful traversal to and entry into the targeted pit. The Rover's Successful Traversal to the Pit will be completed between March 5, 2030, and March 7, 2030, with entry into the pit scheduled for March 7, 2030, to March 8, 2030. During this period, the mission will focus on Data Collection and Scientific Measurements, which will begin on March 9, 2030, and continue through December 31, 2031. The Final Program Implementation Review (PIR) is planned for February 1, 2031, to evaluate progress and address any remaining objectives. Preparations for the mission's end-of-life will occur between January 1, 2032, and January 24, 2032, ensuring a seamless transition to the closeout phase. Following the same color scheme, the gantt chart below represents the Phase E schedule.

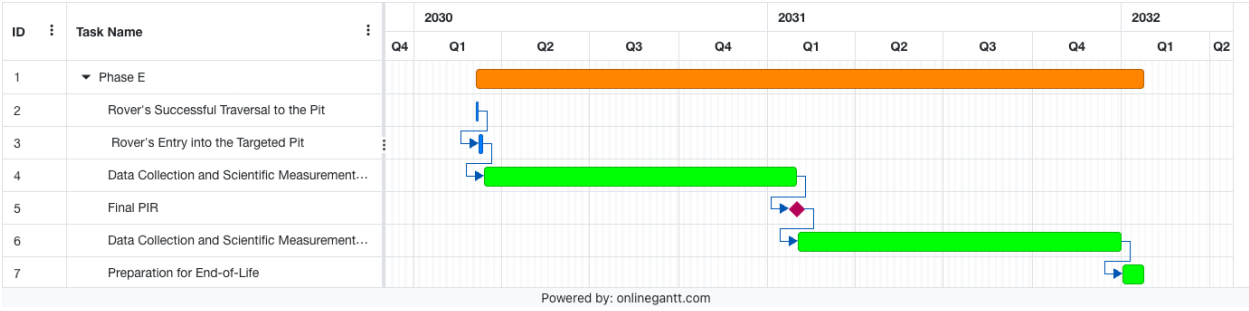


Figure 16: Phase E Gantt Chart

- **Phase F: Closeout**

From January 25, 2032, to July 31, 2032, this 6-month phase focuses on wrapping up mission activities. Milestones include comprehensive Data Archiving, which will take place between January 25, 2032, and March 31, 2032, ensuring that all collected data is systematically stored and accessible for future research. System Decommissioning will be conducted from April 1, 2032, to May 31, 2032, involving the careful shutdown of operational systems and safe disposal of hazardous components. The mission team will then work on the Preparation of Final Reports, scheduled for June 1, 2032, to July 15, 2032, summarizing the mission's achievements, results, challenges, and lessons learned. The final milestone, Submission and Dissemination of Reports, will occur between July 16, 2032, and July 30, 2032, with the mission officially ending on July 31, 2031. Following the same color scheme, the Gantt chart below represents the Phase F schedule.

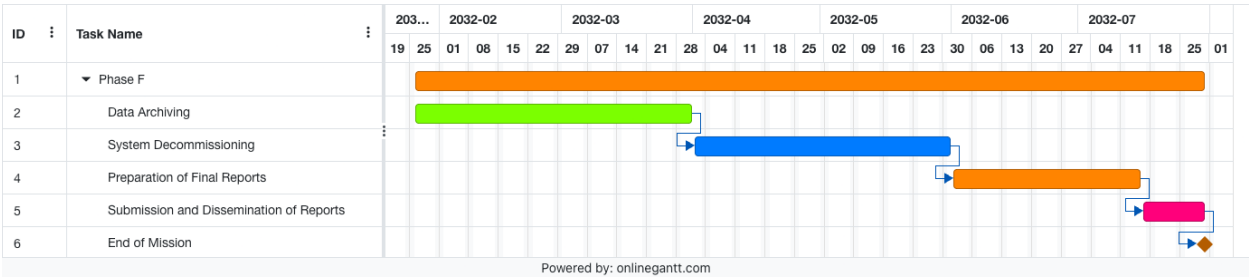


Figure 17: Phase F Gantt Chart

The total mission duration is estimated at 7.5 years. While these estimates provide an estimated timeline, the project's dynamic nature means schedules and resource allocations will change as new challenges and opportunities occur. Detailed plans will be continuously refined, ensuring flexibility and responsiveness to the mission's requirements. By actively adapting to unexpected circumstances, the mission team will maintain the execution standards. Through careful planning and responsive management, the mission aims to achieve its ambitious objectives while laying the groundwork for future explorations.

- Schedule Estimation Table:

Phase	Systems Engineering Timeline Deliverables and Milestones	Start date	End Date	Duration Estimation
Phase C: Final Design and Fabrication	Subsystem Design Completion	January 25, 2025	March 24, 2025	~ 2 years
	Material and Component Selection	March 25, 2025	August 24, 2025	
	Simulation and Final Modeling	August 25, 2025	October 24, 2025	
	Fabrication Planning	October 25, 2025	December 23, 2025	
	First Program Implementation Reviews (PIR)	December 24, 2025		
	Critical Design Review (CDR)	December 25, 2025	June 24, 2026	
	CDR Submission Deadline	June 24, 2026		
	Preliminary Assembly Planning	June 25, 2026	January 24, 2027	
	System Integration Review (SIR)	January 25, 2027	February 25, 2027	

Phase D: System Assembly, Integration & Test, Launch & Checkout	SIR Submission Deadline	February 26, 2027		~ 3 years
	Instrumental Acquisitions	March 1, 2027	August 31, 2027	
	Integration Testing	September 1, 2027	February 27, 2029	
	Second PIR	February 28, 2029		
	Rover Assembly and Testing	March 1, 2029	June 30, 2029	
	Rover Test	June 28, 2029		
	Operational Readiness Review (ORR)	July 1, 2029	July 31, 2029	
	Final System Validation	August 1, 2029	January 31, 2030	
	Reach Launch Location	February 25, 2030		
	Launch Window	March 1, 2030		
	Successful Landing and Rover Deployment	March 4, 2030		
Phase E: Mission Operations	Rover's Successful Traversal to the Pit	March 5, 2030	March 7, 2030	~ 2 years
	Rover's Entry into the Targeted Pit	March 7, 2030	March 8, 2030	

	Data Collection and Scientific Measurements	March 9, 2030	December 31, 2031	
	Final PIR	February 1, 2031		
	Preparation for End-of-Life	January 1, 2032	January 24, 2032	
Phase F: Closeout	Data Archiving	January 25, 2032	March 31, 2032	0.5 years
	System Decommissioning	April 1, 2032	May 31, 2032	
	Preparation of Final Reports	June 1, 2032	July 15, 2032	
	Submission and Dissemination of Reports	July 16, 2032	July 30, 2032	
	End of Mission	July 31, 2032		
Total Mission Duration	-	-	-	**~ 7.5 years**

Table 14: Schedule Estimation Table

5.3. Budget

5.3.1. Budget Basis of Estimate

Personnel Costs:

- Salaries were standardized based on industry benchmarks for roles including engineers, scientists, technicians, and administrative staff. The total personnel cost reflects phased team scaling, with larger teams during development and reduced numbers in operational phases.

Testing and Simulation:

- This allocation covers hardware testing in lunar-like environments and software simulations for mission-critical scenarios. A split of $5 million for initial testing and $45 million for extended environmental simulations ensures robust validation of mission hardware and systems.

Subsystem Costs:

- Costs for subsystems such as mechanical, power, thermal, and instrumentation were refined from SRR budgets, incorporating adjustments (e.g., -$5M for Mechanical and Thermal Management, -$25M for Payload) to align with mission priorities and constraints.
- Estimations relied on trade studies and NASA's MCCET tools for subsystem-level cost analysis.

Travel and Outreach:

- Travel budgets accommodate hardware testing, review boards, and outreach activities to engage the scientific community and the public. Outreach funding supports public education on the mission's objectives and findings.

Contingency Allocation:

- A 15% contingency of the total budget addresses potential risks such as schedule delays, unforeseen technical challenges, or inflationary impacts.

5.3.2. Total Mission Cost

The total mission cost shows an overview of the costs of the Personnel Budget, Travel Budget, Outreach Budget, and Direct costs. The total mission cost for the Mare Tranquillitatis Pit exploration is $325 million, as indicated in the detailed breakdown of direct costs. Here's a summary of the major cost components:

Objective	Cost
Personnel Costs	$113 million
Travel Budget	$89,500
Outreach Budget	$290,000
Direct Cost	$261.62 million
Total Cost	$375 million

Table 15: Total Mission Cost Summary

Each category is carefully allocated and accounted for, ensuring the mission remains within budget while addressing the technical, operational, and outreach needs.

5.3.3. Personnel Budget

The personnel budget for Mission Phases C through F is crucial, totaling approximately $113 million, which represents about 30% of the total mission budget of $375 million. This allocation is strategically designed to ensure the mission progresses efficiently within financial constraints, while providing the necessary expertise and support at each stage of the mission lifecycle.

The personnel costs for the Mare Tranquillitatis Pit exploration mission are meticulously planned to guarantee the availability of skilled personnel throughout phases C to F. Salaries are set in line with industry standards for engineers, scientists, technicians, and administrative staff. The budget accounts for team adjustments at different phases: a larger workforce during development and testing, and a smaller, more focused team during operational stages. Personnel costs are calculated based on the number of required staff for each role and the duration of their involvement in the mission. A detailed breakdown of roles, staffing levels, and the reasoning behind these decisions is outlined below.

1. **Pre-Mission Phase (2 years)**: The mission will employ 35 individuals, including both full-time and part-time staff. This team will consist of 3 Project Managers, 15 engineers, 12 Scientists, and 5 administration personnel. This personnel will assist the engineers in their task. who will work on mission design, initial research, system development, and risk mitigation strategies
2. **Development and Testing Phase (3 years):** As we advance in Phase D, about 55 employees since this phase has a greater emphasis on scientific and engineering expertise. Therefore, an increase of 6 technicians to assist the engineers in their task, and 6 scientists, 6 engineers and 2 administrators will be added to ensure detailed design, system testing, and validation, as well as the integration of systems into the spacecraft are efficiently conducted.
3. **Launch, Operations Phase and Closeout (2.5 years):** The same number of individuals in as in the development and testing phase scientists will be maintained to ensure the team conducts final mission preparations, launch procedures, and the first months of operational monitoring and control.

Table 6 below outlines salary for each personnel throughout the mission of the Mare Tranquillitatis pit, encompassing health insurance, Medicare taxes, and social security benefits, from Phase C to Phase F. The personnel costs are calculated based on the

number of people required in each role and the duration of their involvement in the mission.

The estimated cost of the personnel is as follow:

- **Project Manager**: $120,000
- **Administrators:** $60,000
- **Engineers:** $80,000
- **Scientists:** $80,000
- **Technicians:** $80,000

These costs are essential to ensure that the mission has the necessary expertise and support at each phase, from initial planning through to operational execution.

	Phase C	Phase C	Phase D	Phase D	Phase D	Phase E	Phase E	Phase F
Year	FY 1	FY 2	FY 3	FY 4	FY 5	FY 6	FY 7	FY 7.5
Science Personnel:	12	18	18	18	18	18	18	18
Engineering Personnel:	15	21	21	21	21	21	21	21
Technicians:	0	6	6	6	6	6	6	6
Administration Personnel:	5	7	7	7	7	7	7	7
Management Personnel:	3	3	3	3	3	3	3	3

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Mission Phase	Phase C	Phase C	Phase D	Phase D	Phase D	Phase E	Phase E	Phase F	
Year	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 7.5	Cumulative Total
PERSONNEL									
Science Personnel	$ 960,000	$ 1,477,440	$ 1,514,880	$ 1,552,320	$ 1,589,760	$ 1,627,200	$ 1,664,640	$ 1,702,080	$ 10,386,240
Engineering Personnel	$ 1,200,000	$ 1,723,680	$ 1,767,360	$ 1,811,040	$ 1,854,720	$ 1,898,400	$ 1,942,080	$ 1,985,760	$ 12,197,280
Technicians	$ -	$ 492,480	$ 504,960	$ 517,440	$ 529,920	$ 542,400	$ 554,880	$ 567,360	$ 3,142,080
Administration Personnel	$ 300,000	$ 430,920	$ 441,840	$ 452,760	$ 463,680	$ 474,600	$ 485,520	$ 496,440	$ 3,049,320
Project Management	$ 360,000	$ 369,360	$ 378,720	$ 388,080	$ 397,440	$ 406,800	$ 416,160	$ 425,520	$ 2,716,560
Total Salaries	$ 2,820,000	$ 4,493,880	$ 4,607,760	$ 4,721,640	$ 4,835,520	$ 4,949,400	$ 5,063,280	$ 5,177,160	**$ 31,491,480**
Total ERE	$ 787,062	$ 1,254,242	$ 1,286,026	$ 1,317,810	$ 1,349,594	$ 1,381,378	$ 1,413,161	$ 1,444,945	**$ 8,789,272**
TOTAL PERSONNEL	$ 3,607,062	$ 5,897,573	$ 6,200,263	$ 6,510,527	$ 6,828,365	$ 7,153,779	$ 7,486,766	$ 7,827,329	$ 43,684,335

Table 16: Summary of Personal Cost

The estimated total personnel cost amounts to **$43.7 million**, successfully adhering to the stipulated budget limit of **$113 million** for the entire mission. These costs reflect the required staffing for each phase and provide a comprehensive view of the personnel budget over the course of the mission lifecycle.

5.3.4. Travel Budget

The following outlines the estimated costs for various travel-related activities associated with the mission, including analog testing, key decision points (KDPs), pre-launch activities, international collaboration, and contingency travel.

Analog Testing Travel

- **Destination**: Black Point Lava Flow, Arizona
- **Purpose**: To simulate lunar conditions for rover mobility and lander deployment, using the site's volcanic terrain as an Earth analog for lunar regolith.

Expense	Cost
Airfare for 10 team members	$4,000
Lodging for 4 nights	$3,000
Meals and per diem	$2,500
Equipment rental and transport	$3,500
Miscellaneous testing supplies	$2,000
Total Estimated Cost	**$15,000**

Table 17: Testing travel

Key Decision Points (KDPs) Travel

- **Destination**: NASA Headquarters, Washington, D.C.
- **Purpose**: To present mission readiness and transition reviews at KDP-C, KDP-D, and KDP-E phases.

Expense	Cost
Airfare for 6 key personnel (per review)	$9,000
Lodging for 2 nights per review	$6,000
Meals and per diem	$4,500

Local transportation	$3,000
Total Estimated Cost	**$22,500**

Table 18: Key Decision point travel

Pre-Launch Travel

- **Destination**: Cape Canaveral, Florida
- **Purpose**: Final spacecraft integration, testing, and launch preparations.

Expense	Cost
Airfare for 8 team members	$8,000
Lodging for 5 nights	$6,000
Meals, per diem, and transport	$5,000
Team celebration and media event	$2,000
Total Estimated Cost	**$21,000**

Table 19: Pre-Launch travel

International Collaboration Travel

- **Destination**: European Space Agency (ESA) Facility, Netherlands
- **Purpose**: Collaborative development of spectrometer payload systems and testing of ISRU instrumentation.

Expense	Cost
Airfare for 3 technical team members	$9,000
Lodging for 7 nights	$7,000
Meals, per diem, and transport	$3,500
Cultural exchange and team integration event	$2,500

Total Estimated Cost	**$22,000**

Table 20: International travel

Contingency Travel

- **Purpose**: Reserved funds for unexpected reviews, troubleshooting, or late-stage collaboration needs.

Expense	Cost
Reserved funds for unexpected airfare and lodging	$7,000
Miscellaneous expenses	$2,000
Total Estimated Cost	**$9,000**

Table 21: Contingency Travel

Grand Total Travel Budget: **$89,500**

5.3.5. Outreach Budget

The outreach program for the Mare Tranquillitatis Pit exploration is designed to engage a diverse audience through educational initiatives, community involvement, digital outreach, and public engagement efforts. The goal is to raise awareness, inspire interest in space exploration, foster partnerships, and support STEM education. Below is a detailed budget estimate, explaining the cost breakdowns and explanations according to Team 3.

Educational programs will comprehend structured programs to integrate the mission into STEM education. A yearly allocation of $35,000 has been earmarked for Team 3's curriculum development and teachers' training workshops. The goal for the curriculum development will be to enhance STEM learning by connecting it with real-world space exploration challenges by preparing lesson plans, activity kits, and multimedia resources for K-12 and college levels, centered around the Mare Tranquillitatis Pit-all for 15,000. The teacher training workshops are for conducting professional development sessions for educators to prepare them to teach mission-related content, preparing educators to incorporate mission data into their curriculum and promote student engagement- all for a total of $20,000.

Team 3's community Outreach plan will emphasize on engaging local communities and underrepresented groups. Public lectures and panel discussions, budgeted at $22,000 will bring mission scientists and engineers to discuss exploration challenges and potential discoveries, facilitating direct dialogue with diverse audiences. Collaborations with local organizations will cost $25,000, supporting co-hosted events like "Space Nights" and community science exhibitions. An additional $40,000 will be dedicated per year to outreach programs tailored for underrepresented populations, offering mentorship opportunities and resources to inspire participation in space-related fields. This reflects Team 3's commitment to inclusivity.

Digital Outreach will leverage online platforms to maximize the mission's global reach. Website development and maintenance, annually allocated $15,000, will create a centralized hub for interactive features providing a centralized source for mission updates and educational content on the Mare Tranquillitatis Pit. Team 3 will allocate a yearly budget of $50,000 to collaborate with social media influencers for Professional content creation, targeted advertising, in order to engage with a wide audience and share real-time mission updates on social media platforms. A budget of $8,000 will be allocated for online sessions, including Q&A events with mission scientists to reach a broader audience and allow for live interaction, making mission information accessible globally.

Public Engagement events will be designed to inspire widespread interest and participation. Interactive exhibits at science centers, costing $25,000, will allow visitors to explore the mission's exploration process through hands-on displays. The exhibits will showcase mission equipment and planned scientific processes for exploring the pit. Media coverage and press releases, budgeted at $15,000, will ensure continuous media presence and broad public awareness of key mission milestones. This includes media outreach, press conferences, media kits, and strategic engagement with media outlets to ensure mission coverage. Collaboration with news channels, news agencies, TV networks, and online media platforms.

An additional $17,500 is added as margin to cover potential variations or unforeseen expenses in the outreach budget.

OUTREACH									
Total Outreach Materials	$ 2,000	$ 2,000	$ 2,000	$ 1,500	$ 1,500	$ 1,500	$ 1,500	$ 1,500	**$ 12,000**
Total Outreach Venue Costs	$ 5,000	$ 5,000	$ 6,000	$ 4,000	$ 6,000	$ 6,000	$ 6,000	$ 6,000	**$ 38,000**
Total Outreach Travel Costs	$ 8,000	$ 9,000	$ 10,000	$ 9,000	$ 7,000	$ 7,000	$ 7,000	$ 7,000	**$ 57,000**
Total Outreach Services Costs	$ 12,000	$ 12,000	$ 12,000	$ 10,000	$ 8,000	$ 8,000	$ 6,000	$ 6,000	**$ 68,000**
Total Outreach Personnel Costs	$ 14,000	$ 14,000	$ 12,000	$ 12,500	$ 10,000	$ 10,000	$ 8,000	$ 8,000	**$ 80,500**
Outreach Margin	$ 2,500	$ 2,500	$ 2,500	$ 2,500	$ 2,500	$ 2,500	$ 2,500	$ 2,500	**$ 17,500**
Total Outreach Costs	$ 43,500	$ 45,657	$ 46,814	$ 42,581	$ 38,640	$ 39,550	$ 35,836	$ 36,642	**$ 292,578**

Table 22: Summary of Estimated Outreach Budget

This budget ensures comprehensive outreach that not only educates and engages but also builds sustainable public interest, creates and supports an inclusive, informative, and participatory outreach strategy that underlines the importance of the Mare Tranquillitatis Pit exploration and its potential implications for future lunar habitability research.

5.3.6. Direct Costs

Direct Costs Analysis

This section provides a detailed breakdown of the direct costs associated with our mission, which are projected to total approximately $211.62 million. These costs encompass a wide range of expenses, from the mission vehicle and related systems to critical subsystems, personnel expenses, testing and simulation costs, and a contingency fund for unexpected developments.

Personnel Costs

Personnel costs are a significant portion of the mission budget, totaling an estimated $113.4 million. This figure accounts for the salaries of our team members over the course of the project. The breakdown of personnel expenses is as follows:

Role	Annual Salary
Scientist	$80,000/year
Engineers	$80,000/year
Technicians	$80,000/year
Administration	$60,000/year
Managers	$120,000/year

Table 23: Summary of Personnel Budget

These salary estimates reflect industry-standard pay rates for each role, adjusted to account for the project's specific needs and the skills required throughout the mission's lifecycle. The personnel costs cover the full span of the mission, including pre-launch development, testing, and post-launch operations.

Scientist: Salaries for scientists who will develop the mission's scientific objectives and data analysis frameworks.

Engineers: Compensation for engineers who will design, develop, and test the spacecraft and its subsystems.

Technicians: Wages for technicians who will assist in assembling and testing the spacecraft components.

Administration: Pay for administrative staff who will support the project through clerical work, procurement, and coordination.

Managers: Salaries for project managers who will oversee the mission operations, ensuring project milestones are met within budget and on schedule.

These figures are derived from industry-standard rates and are adjusted for the project's duration and specific skill requirements.

Subsystem Costs

The subsystem costs are calculated to be $160 million, distributed among the following key areas:

Subsystem	Estimated Cost
Mechanical	$30 million
Command and Data Handling	$25 million
Power	$20 million
Thermal Management	$35 million
Payload (Science Instruments)	$50 million

Table 24: Subsystems Estimated Costs

Mechanical System: The mechanical subsystem includes the spacecraft's frame, housing, and internal components that must endure extreme environmental conditions. A $5 million reduction was achieved by optimizing design and reducing material costs without affecting system integrity.

Command and Data Handling (CDH): This critical subsystem ensures spacecraft operations and manages the collection and transmission of scientific data. The CDH systems play an essential role in controlling spacecraft functions and enabling communication with mission control.

Power System: The spacecraft's energy sources, such as solar panels and energy storage devices, are key to its operation. Innovations in energy storage and solar power technology allow for cost-effective solutions without compromising energy efficiency.

Thermal Management: The spacecraft must endure extreme temperature variations, making the thermal management system vital to maintaining its operational

integrity. This system's cost was optimized by improving design specifications, resulting in a $5 million reduction.

Payload (Science Instruments): The science payload represents the instruments responsible for gathering mission data. These include high-tech sensors, cameras, and other scientific equipment designed for the specific objectives of the mission. A $25 million reduction was achieved by refining the design and production process.

These costs have been derived from detailed cost models, including the NASA Instrument Cost Model (NICM) and the Mission Concept Cost Estimating Tool (MCCET). These models use industry-standard cost-estimating relationships (CERs) to ensure the accuracy and reliability of the estimates.

Testing and Simulation Costs

Testing and simulation are a vital part of the mission to ensure that the spacecraft performs optimally in the harsh environment it will encounter. The testing and simulation budget totals $50 million, allocated as follows:

- **Pre-Launch Testing**: $5 million will be dedicated to rigorous testing of the spacecraft to ensure its durability and functionality under simulated launch conditions. This phase includes vibration tests, thermal testing, and vacuum testing to replicate space conditions.
- **Simulation**: $45 million is allocated to testing the spacecraft's hardware and software in a controlled environment designed to replicate the conditions of the lunar surface. This includes simulating extreme temperatures, terrain challenges, and space radiation to ensure the spacecraft's systems will function under operational conditions.

These testing and simulation activities are crucial to mitigating risks and ensuring the mission's success. This portion of the budget reflects the high level of preparation required to deal with the complexities of a lunar mission.

Contingency Fund

The contingency fund is an essential part of the project's financial planning, allowing us to manage unforeseen expenses. A fund of $60 million, or approximately 15% of the total budget, is set aside to address potential challenges that may arise

during development, testing, or operations. This fund is designed to cover unexpected issues such as:

- Unanticipated delays in development or testing.
- Cost overruns due to unforeseen technical challenges or supply chain disruptions.
- Emergent needs for additional resources or equipment to ensure mission success.

Each aspect of the mission's budget has been reviewed and thoroughly justified to ensure effective financial management and mitigate the risks of cost overruns. The contingency fund allows for flexibility in managing unexpected costs without compromising the mission's overall success.

The direct costs of the mission have been carefully estimated to ensure that the project remains within budget while meeting all necessary requirements. This analysis covers all major cost categories, from personnel and subsystems to testing, simulation, and contingency funds. By utilizing advanced cost models and applying industry best practices, we are confident in the accuracy and reliability of these cost estimates. The allocation of resources is designed to ensure that the mission is successful, and the spacecraft can achieve its objectives within the projected timeframe and budget.

5.4. Scope Management

5.4.1. Change Control Management

Our team uses Discord and Google Docs to track all proposed and completed changes for the project. These tools make it easy for us to stay organized and ensure everyone is on the same page. Anytime a change is proposed by a team member, we fill out a Change Request form, which is basically our way of explaining what the change is, why it's needed, and what we expect it to accomplish. This form is sent to the main stakeholder, L'SPACE, for approval before we move forward. Using this system, we can keep track of the status of each request and whether it's approved, rejected, or needs revision. If a request is rejected, we take the feedback we receive, make improvements, and submit a revised version. Once a change is approved, we implement it and document it in the appendix of our major deliverables so we have a clear record of what was updated.

When it comes to requesting and implementing changes, the process usually starts with input from team members, stakeholders, or anyone else involved in the project. Each request includes a detailed description of what needs to change, why it's important, how it could benefit the mission, and any potential risks involved. We also make sure to explain how the change could impact the overall goals or science of the project. If it gets approved, we go ahead and implement it. If it doesn't, we revise it based on the comments we get from stakeholders and try again. This process helps us make sure that every change we make is well-thought-out and aligns with the project's goals.

For stakeholder feedback, we handle Requests for Action (RFAs) and Advisory (ADVs) as part of our change management process. RFAs are usually more urgent, so we prioritize them and review their impact on the project. Once we know what needs to be done, we make the necessary changes to the deliverables, focusing on fixing the specific issues or sections that the stakeholders highlighted. On the other hand, ADVs tend to have a lower priority, but we still take them seriously. We review the feedback, update the relevant sections, and make sure everything meets the stakeholder's expectations. Even though ADVs aren't as urgent as RFAs, we make it a point to address them on time because they're still important for improving the mission overall.

Our process for change control is built on communication and feedback. By using tools like Discord and Google Docs, we're able to stay organized, respond quickly to feedback, and make sure that changes are properly documented. This approach helps us keep track of everything while making sure that the project stays on course and aligns with what the stakeholders want.

5.4.2. Scope Control Management

Team 3 has developed comprehensive plans to manage scope changes, whether it involves downscoping to address overruns or increasing scope to leverage additional resources. They may vary depending on the phase of the mission to ensure alignment with mission objectives, requirements and constraints.

Downscope strategies:

1. **Schedule delays and cost overrun**
    - **Manufacturing and Procurement Plans:** Due to the budget cut, team 3 has first looked at altering manufacturing and procurement methods. This involves switching to more cost-effective suppliers, using off-the-shelf components instead of custom-built processes to reduce lead times and costs.
    - **Reducing the Personnel cost:** team 3 has reduced the number of personnel in non-critical roles to ensure the new budget constraint is met and mission objectives are not altered.
    - **Crash of Schedule:** More personnel may also be added to critical tasks, to accelerate progress. This approach will be carefully managed to avoid diminishing returns and ensure that quality is maintained.

In case of an increasing scope:

1. **Utilize additional budget and Schedule Margin:** If extra budget or schedule margin are available, team 3 will explore opportunities to enhance mission capabilities, such as upgrading instruments and manufacturing plans, extending the mission duration or incorporating additional outreach initiatives.
2. **Reallocate resources:** if this case were to happen, team 3 will leverage the additional resources to address any existing risks, emphasize on the mission constraints, further enhancing mission success rates.

Mission life cycle considerations:

1. **Development, and Testing (Phase C and D):** Downscoping options will focus on manufacturing and procurement adjustments. Increasing scope in this phase may involve using additional funds for system upgrades or risk reduction efforts. The focus here will be on optimizing designs and ensuring that any changes do not significantly impact the overall mission timeline.
2. **Launch & Mission operations (Phase D & E):** Team 3 will prioritize maintaining mission stability and will only make critical adjustments that are necessary for mission success. If any additional scope, it will emphasize on enhancing data analysis to ensure mission longevity.

Team 3’s scope control management plans are designed to be flexible and responsive, ensuring that the mission can adapt to both constraints and opportunities. By having clear strategies for downsizing and increasing scope, the team can effectively manage resources and maintain focus on achieving the mission’s objectives.

5.5. Outreach Summary

To increase public awareness and appreciation of the team's mission, the team will use different strategies, emphasizing the team's mission impact and community engagement, to enhance public awareness and appreciation about it. The team will reach out to K-12 schools, universities and colleges, and community centers. The team will also reach out to underrepresented populations, initiate public engagement and awareness campaigns, and collaborate with other organizations if they can.

In K-12 schools, the team will host interactive workshops and presentations that can connect their mission to real-world problems related to STEM education. Maybe they can use online platforms like zoom to have the workshops. But on the other hand, in-person workshops and meetings can help to connect more easily. The team could have activities or projects tailored to various schools like middle school students can work on different projects or maybe papers identifying different relevant parts of the mission that match with their school work or science class. For highschool students, they can explore technology or engineering aspects of the mission or even can explore different roles in the mission that are connected to their careers they aspire. From the team's side, they can also develop interactive toolkits or information papers about their mission so the students can use it to learn more about the mission and its purpose, maybe the teachers can integrate it in to their lessons or curriculum and the school can use it to have a deep career exploration on space and career in space related fields.

For universities and colleges, the team will try to contact the university departments that align with their mission, maybe focusing on engineering, science and technology departments. Also, the team can try to contact the Aerospace Department in specific colleges. They can also try to select particular colleges that give more importance or weigh a lot for space related careers and missions. The team also thinks it's a good idea to contact colleges that are partnered with NASA and opportunities at NASA, it can make it easier to connect and to hold events or public awareness campaigns. If the team can, then they can also contact or connect with colleges that the members of the team are part of.

For community colleges, public and private universities, the team can have interactive workshops just like in schools, or maybe informative workshops too. The workshop is about the team's mission ultimately, but can also be broad on the aspect of engineering, science and technology. While sharing and making awareness about their mission, the team can also touch base about different internships and opportunities at

NASA and can give slight insights on careers with NASA and space to get more people into the workshops and meetings. They can hold both in person and online workshops and meetings. They can also have follow up meetings to have deep driven conversations about their mission to students and even to faculty interested. The team can also hold review or experience chats or meetings to share their experience working on the mission and the things they learned from it, to motivate and to inform students. If the team can and are allowed, then they can create “unpaid” research collaborations and student ambassador opportunities to increase the public awareness, engagement and appreciation for the team’s mission. The team can open the research collaborative opportunities to anyone in any major but preferably STEM majors to learn about the team’s mission. They can read and learn from the team’s mission and project and can create research on it or maybe seek mentorship on research on something similar from the team. For the student ambassador positions, basically students helping the team to promote the team’s mission using their own strategies and ideas. This can be for anyone in any major but preferably someone in business or marketing major and interested in space, cause it aligns more with their career. These strategies can help to reach students and faculties at college and universities about the team's mission.

For community centers, the team can organize events such as information sessions where they can explain their mission, its purpose and impact and maybe its relevance to communities. The team can also share their experience like how they knew about this opportunity, their application process, their motivation or goal in doing it. They can also share about how the experience was, the different things they learned, the different skills they enhanced, the challenges they faced and the things they liked the most. These things are generally interesting aspects to focus on community talks as it includes a variety of diverse people with different perspectives. Maybe the team can pass out flyers about their mission and do interactive games or quizzes and question and answer sections to have engagement during the talks. These strategies can help to make awareness about the team’s mission in the community.

To outreach to the underrepresented population, the team will try to focus on certain groups or communities of underrepresented populations and will design specific programs that address the needs of the underrepresented populations and will connect it with the team’s mission. One of the biggest factors in the underrepresented population is lack of resources and the team can gather enough information about space exploration and the different programs and opportunities for them. Can also offer opportunities for mentioning and supporting them. A good strategy would be to have chats with them about different opportunities while concentrating and sharing information about the team’s mission. It is also important to share about the team members' experience with other members and mentors and within the academy to

motivate and inspire those from the underrepresented communities to learn about the program and about the team's mission. The team members can also share their experience if they are part of any underrepresented communities, it can help to connect with them easily. Ensuring direct involvement is essential to make awareness about the team's mission.

For initiating Public engagement and awareness campaigns, it is important to use Social Media. Even though the team can use Social Media for other outreach programs too, it is important to use it for public engagement. The team members can use Instagram, Twitter, LinkedIn or make youtube videos under an account for the team's mission if allowed, to share about it with the World around. Maybe they can do blogs and or Instagram reels or youtube shorts like 60 seconds videos to share one thing they learned from each part they did in each deliverable or maybe one challenge they faced while doing it. It should be interesting and informative content which can help the team to share about the mission. They could write a little note about some interesting facts and post it. The team could also do interviews of the team members sharing their insights or experience about the mission. They could do like a daily vlog or mission vlog sharing their whole mission experience and making awareness about their mission and its purpose. The team could also do little social media polls or quizzes, or go live for engagement and to hear people's thoughts and insights and answer questions while sharing about the mission in every opportunity. It would be better for different members to do the appearance in each video or shorts, especially if there are members from diverse populations, so different people can connect and can also establish the theme of equality and collaboration. If it is permitted, the members can also share within their own personal social media about the mission and their progres, role and tasks, to make public awareness.

To collaborate with other organizations, the team could partner with other schools, universities or community organizations focusing on underrepresented populations or preferably STEM education. In schools or universities, they could partner with other organizations that focus on STEM or space to share about the team's mission. The team could visit the school or can do something like a virtual meeting during the organization meeting to share. Preferably, it is better to offer something in return like maybe mentorship, exclusive resources, resume check, help to connect with mentors, give career advice or something that is interesting for the group, so they can share and learn about the team's mission.

Maybe they could also seek help from their friends and family to make awareness about the team mission. The team could make little note cards or flyers about their mission and share with the people around them to share about the mission.

The team members could also use open spaces like noticeboards in their part time job office or in their college to share about the mission.

The team could maybe do like open competitions with some simple prizes, to give an opportunity for people to learn about the mission in an interesting way and use their knowledge to compete for a prize. This can be done using social media or even in-person events. But social media can help to reach more people and can help to elevate the public engagement. These can ensure to reach many more people about the team's missions and can help people to connect and learn.

Furthermore, press releases highlighting key milestones of the mission and space-focused publications will be created and sent to major news outlets. This will give access to a global audience by engaging people beyond immediate educational and scientific communities and inspire students and professionals to pursue space exploration careers. Team members will feature their personal stories and their journey so far in the mission lifecycle to foster connection with the public about exploration and investment.

All these strategies can help to increase the public awareness and appreciation of the team's mission and can help to enhance the mission's reach and ensure meaningful, long-term impact within diverse communities.

6. Conclusion

The MDR successfully established the foundational capabilities, constraints, and design framework required to meet the mission's science objectives. These requirements address the unique challenges posed by the lunar environment, including extreme temperature fluctuations, abrasive regolith, and rugged terrain. Key developments include a robust mechanical subsystem, featuring a titanium alloy chassis and motorized wheel drive for the rover, as well as a stable, shock-absorbing lander frame designed to ensure structural integrity and operational resilience. Significant progress has been made, with most subsystems currently achieving Technology Readiness Level (TRL) 5, while targeted innovations in deployment, mobility, and environmental adaptations continue to advance subsystem readiness.

The Preliminary Design Review (PDR) builds upon this foundation, marking a pivotal step toward detailed system integration and implementation. Leading up to the PDR, extensive design refinements and subsystem validation efforts have been conducted under simulated lunar conditions. These include rigorous testing of the rover's mobility and deployment systems, enhanced thermal management strategies, and recovery protocols to ensure system resilience. The focus has also been on optimizing payload performance to maximize scientific output and operational efficiency within the mission's constraints.

If additional time or resources were available, extended prototype testing would be prioritized to further enhance subsystem reliability and refine power distribution and management systems for extended mission durations. The PDR reflects the culmination of these efforts, presenting updated design specifications, refined risk assessments, and comprehensive contingency plans. All subsystems have been assessed to meet or exceed TRL 5, demonstrating their readiness for subsequent design phases.

This progress solidifies the mission's trajectory toward achieving its goals of sustainable lunar exploration and habitation. By addressing critical design and operational challenges at this stage, the mission is well-positioned to advance scientific understanding of lunar environments, evaluate potential habitation sites, and contribute to the development of long-term human exploration capabilities on the Moon and beyond.

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APPENDIX

Table of Changes	
Section	Changes Made
SRR - 1.2 (MDR - 1.2)	● STM is now formatted on a landscape formatted page for better view ● STM is now one table; it was separated into two tables per goal, but now it is all in one table. ● Science goals and requirements are introduced at a higher level of detail ● The STM included this instrument: "Bruker TRACER 5i Portable XRF Analyzer" which was a handheld device is now changed to Planetary Instrument for X-ray Lithochemistry (PIXL - XRF Analyzer) which was on Mars rover ● The LiDAR and GPR are now specified that we are using for the mission -- WISDOM for Stepped Frequency Continuous Wave (SFCW) Ground-Penetrating Radar (GPR) & Space Qualified Rover Lidar (SQRLi) for Light Detection and Ranging (LiDAR) ● STM tables are now not introduced with bullet points and there just the table captions. I deleted the bullet points. ● It is now avoiding the use of language such as "this table shows" in scientific writing. Instead of introducing the table, it is now introducing the material itself. ● Now it does not explain what an STM is. Instead, the narrative is introducing the key things from within the STM that need to be highlighted.
SRR - 1.7.3 (MDR - 1.9.2)	● The mission schedule high level overview table now includes key dates for each of the milestones within the mission. ● The schedule now provides specific key dates for milestones instead of indicating the months in which these events will occur. ● The launch date (per the Mission Task Document) is now specified in the mission schedule. ● The approximate arrival date on the lunar surface is now included in the mission schedule. ● The total mission duration is now not as quite long and is within the suggested span of 7-8 years. ● The start of Phase C is now within 30 days after the academy ends.

MDR - 1.9.2 (PDR 5.2.)	● Included the launch date provided by customer constraints. ● Tried to include more LCRs, SRBs, and KDPs.
SRR- 1.5.3.1 (PDR 2.1.2.1)	● Child Requirements Included ● Relevant subsystem requirement ID ● Specific parameters on the requirements' description
SRR- 1.5.4.1 (PDR 2.1.3.1)	● Child Requirements Included ● Relevant subsystem requirement ID ● Specific parameters on the requirements' description
MDR- 1.7 (PDR 2.1.2.4)	● Relevant Contractor and Clear justification provided ● Lead times based on the primary supplier's estimation and customization plus verification plan
MDR- 1.7 (PDR 2.1.3.4)	● Specific components included instead of just a general knowledge of the equipment ● Relevant contractor and clear justification provided ● Lead times based on the primary supplier's estimation and customization plus verification plan